Wheelchair trajectory research

Accuracy, clock synchronization, route geometry and visual failure audit
Research continuation: 4 October 2026. All 71 accepted validation trajectories remain visible. Plot revision: 5 October 2026; IMU red, C3D black.

Latest signal-arbitrated experiment:94.835mm ownclock /94.662mm commonnative. It fixes a training clockcyclealias, but still FAILS the5%training-participant guard:No.8regresses5.80%. It is not promoted. Retained candidate remains100.702mm. All clock choices use signal evidence, never position RMSE.

Corrected-clock refit experiment:96.329mm on its own clock grid. On identical native observed timestamps and one shared initial pose, oldblend100.593mm becomes96.172mm; supported-route RMSE49.681 becomes46.630mm. **It fails the training-participant regression guard and is not promoted.** A repeated-route clock alias was found in training. See the corrected-clock section. The table below preserves the retained v34 benchmark rather than mixing clock/support definitions.

Metric / model	Original v21	Tilt motion blend .75
Synchronized 2D RMSE	102.585 mm	100.702 mm
Heading RMSE	1.362 degrees	1.261 degrees
Supported symmetric route RMSE	50.283 mm	49.726 mm
Mean sampled Hausdorff	148.641 mm	149.285 mm
Max-run discrete Frechet, mean	165.580 mm	165.919 mm

The new correction improves synchronized mean RMSE by 1.883 mm (1.84%). It is a **modest conditional development candidate**. Route-tail and ordering metrics slightly worsen. Forty-five trials improve and 26 worsen; No.14 contributes most of the gain. A descriptive four-participant bootstrap gain interval is -0.104 to +4.380 mm and includes zero. This is not a prospective, selection-adjusted confidence interval.

No.18 PS2 and No.5 AR2/AR3 remain unresolved. Millimeter accuracy is not established. Reference absolute uncertainty, firmware conversion and complete mounting rotations are still missing.

The work resumed existing artifacts without restarting. Three independent specialists audited synchronization, route metrics, and the candidate. All models, failed experiments and existing source acquisitions are preserved. No arbitrary post-hoc spatial transform, heading flip, scale, reflection, endpoint snap or C3D reset was applied.

Data, initialization and inference

Partition	Coverage
Training	473 accepted recordings: 266 earlier +207 target training
Target training people	No.1/3/4/6/7/8/13/15/16/17/19/20
Validation	No.5/10/14/18; 71 accepted recordings
Test	No.2/9/11/12; signals not accessed in this window

Every target participant stays in one partition across dates. Earlier trial-to-person mapping is incomplete and uses inherited non-overlap assumptions/date grouping. Historical validation exposure and retrospective test inspection prevent a new blind-test claim.

The physics/TCN/planar/ridge base and empirical per-setup calibration are inherited. v21 adds a bounded forward/yaw correction from 75 contextual linear features and 40 physical products, fitted using observed rate labels and relative endpoint equations. Dynamic tilt adds 30 contextual empirical acceleration-direction/gyro-projection features, with 0.3 Hz full-record forward/backward lowpass. It is not a verified attitude estimator.

Three target-training participant folds selected motion blend .75 before validation. Clipping, normalization and correction coefficients exclude held people within each fold. Inherited calibration/base previously saw those people, so this is **conditional correction selection**, not a full independent pipeline fold. Full model fit uses all473 training recordings.

Prediction uses IMU arrays, acquisition setup and original native startup IMU history only. One optical initial position/heading fixes the coordinate gauge, as previously authorized. No later optical updates occur. Trapezoid body velocities and midpoint heading integrate the motion. Invalid IMU support stops usable reconstruction. Filtering and future context are offline and noncausal.

Fresh native startup and the independent inference loader reproduced 100.702119424 mm. Maximum cached trajectory difference was 5.88e-09 m; startup gravity/bias matched exactly. Deliberately poisoned optical fields did not change motion output. These checks establish implementation reproduction, not generalization.

Metrics: time and geometric route

Primary synchronized localization compares prediction(t) against directly observed supported C3D(t). Trial RMSE is calculated over valid frames; trials are averaged within person, then people equally. This is not pooled-frame RMSE. Uneven maneuver coverage is retained; OS has only3 recordings /2 people.

Primary geometric metrics restrict both trajectories to the same observed support. Invalid frames disconnect the curves: no segment bridges a gap. Exact query-to-polyline distance uses the union of supported segments and isolated points. Spatial query samples have <=10 mm spacing, retaining piece endpoints. Direction means each receive weight one-half. Symmetric route RMSE is sqrt((mean d(pred,ref)^2 + mean d(ref,pred)^2)/2). P95 uses equally weighted directional distributions.

Hausdorff here is a sampled-point-to-exact-polyline maximum. With <=10 mm query spacing, continuous Hausdorff lies between this maximum and maximum+5 mm. A 20 mm spacing repeat measures discretization sensitivity. Summary P95 and Hausdorff are equal-person means of trial statistics, not global maxima or pooled percentiles.

Observed gaps prevent honest full-trial continuous Frechet comparison. Discrete Frechet is calculated separately inside supported temporal runs at25 mm spatial spacing (50 mm convergence repeat), taking the maximum run value. This preserves ordering inside runs only. Arc-length correspondence independently normalizes paired runs by their traveled distance; it is diagnostic and never corrects a model or resets an origin.

Endpoint means last-supported endpoint:8/71 actual recording endpoints are unsupported. Path length sums supported adjacent edges; unsupported distance is not invented. Observed coverage ranges43.94-100%, median98.75%, with1-71 runs (median2). Partial-route metrics cannot establish unseen route accuracy.

Metric (mm)	Original	Blend
Synchronized RMSE	102.585	100.702
Symmetric route RMSE	50.283	49.726
Mean trial P95	97.842	96.769
Mean sampled Hausdorff	148.641	149.285
Last-supported endpoint	133.760	127.772
Absolute supported length error	108.714	104.732
Max-run discrete Frechet	165.580	165.919
Within-run arc RMSE	92.248	91.060

Visual failures and error mechanisms

Case	Evidence	Interpretation
No.18 PS2	416.3mm sync;72.4mm route;515.9mm endpoint;521.1mm length deficit	Overlapping straight route hides a severe braking displacement deficit. Not evidence for timing alone.
No.5 AR2 / AR3	318.7/278.6mm sync;184.8/159.3mm route	Genuine opposed-route failure; physical-front/180-degree ambiguity unresolved. No evaluated-trajectory sign flip.
No.18 SL1	198.9mm sync;128.5mm route;301.0mm P95	Late yaw/cross-track drift persists.
No.10 PS2	Predominantly lateral displacement after early yaw excursion	Yaw later recovers but accumulated position offset remains.
ST ascent	Early yaw excursion changes sign by setup/day	Pitch/mounting coupling plausible, not independently verified body attitude.
No.18 long10x5	Residual signal alignment45-70ms	Clock offset and yaw/travel errors both matter.

The independent visual specialist reviewed all71 historical v34 six-panel plots, contact sheets and enlarged failures. Legends and straight-route Y-axis padding were repaired while retaining equal physical axis scale. Stored processed optical positions are dotted display-only in that historical atlas. The current report atlas instead contains all71 two-panel common-native comparisons: trajectory and yaw versus synchronized time. IMU is red and observed C3D is black; reference gaps stay disconnected. Yaw is shown relative to the shared initial heading and wrapped to -180/+180 degrees; wrap crossings are disconnected. No fitted heading correction is applied.

The three worst original cases contribute15.306mm to the102.585mm mean objective; even perfect correction of those three would leave87.279mm. Removing difficult trajectories would hide a broad residual error problem. The improvement does not justify dropping any accepted recording.

Clock synchronization evidence

Native timing audit includes544 accepted acquisitions (473train+71validation). C3D nominal frame rate is100Hz; IMU median rate is102.396Hz. The pipeline already robustly estimates each device-to-receiver affine clock. About90% receiver timestamps repeat because rows arrive in packet batches; median receiver-fit residual4.78ms, maximum82.65ms. Similar nominal rates do not prove alignment.

Residual offset/affine skew were evaluated using signed yaw and wheel-speed versus observed C3D motion in nonoverlapping8s blocks. A +/-100ms lag search and +/-2000ppm skew cap were fixed. Alternating blocks provide a training holdout correlation check. Training196 identifiable recordings selected offset over affine/guarded affine; validation only24 were identifiable. Short/quiet recordings cannot identify skew reliably.

Signal correlation does not establish absolute timing accuracy: slip, mounting, derivative filters and shape mismatch can mimic changing lag. Widths of the correlation objective are flatness diagnostics, not statistical confidence intervals. C3D-assisted alignment is offline evaluation synchronization; C3D is unavailable to a deployed predictor.

The separate signal-only timestamp counterfactual retains already-integrated paths and performs no spatial fitting. Its common-mask blend score changes100.707814 to99.506231mm (production remains100.702119mm). Originalv21 changes102.585825 to101.946985mm. All71 remain;24 receive shifts and28 observed frames are trimmed at boundaries. This is a timing diagnostic, not a promoted/retrained estimator.

No.18long10x5 gains roughly27-42mm from45-70ms residual offsets, while No.14trial1 worsens76.9 to98.5mm. Existing observed-syncv9 independently reports nearly identical offsets (+70.5,+46.5,+44.9ms) for those No.18 trials. New correction training on the corrected clocks is warranted, with explicit fixed-coverage fallbacks: v9 reaccepted465/473training recordings.

Preferred acquisition evidence: a shared trigger/timecode or periodic timebase spanning the trial, device timestamps and clock uncertainty. Qualisys documents a minimum20ms externally triggered start delay and recommends Sync out for synchronization. Motion-event alignment remains useful when hardware timing evidence is absent; do not choose lag from position RMSE.

Corrected-clock refit and independent route audit

Benchmark / motion model	Old blend	New clock, frozen	New clock, refit
own_clock_full	100.702mm	99.630mm	96.329mm
common_native	100.593mm	99.491mm	96.172mm

All473originaltrainingIDs remain:465use observed-only v9 corrected clocks and8retain explicit original-clock fallback. All71validation recordings use corrected clocks. Fixed v21regularization.1,endpointweight1,gain.5;fixedtilt.3Hz andblend.75. No validation retuning. Correction clipping, normalization, caps and coefficients are refitted on training data; the base and physical calibration remain inherited.

The common-native comparison interpolates each motion through its own clock mapping onto the identical native C3D timestamps. Allsix streams integrate from ONE common directly observed initial pose, with no internal reset or geometric transform. Each preparation own-clock score uses its own mask and initialpose, so those numbers are not interchangeable with nativecommon.

Common-native metric (mm)	Old blend	Clock refit
Synchronized RMSE	100.593	96.172
Symmetric route RMSE	49.681	46.630
Mean trial P95	96.769	90.051
Mean sampled Hausdorff	149.329	141.124
Last-supported endpoint	127.388	126.032
Absolute supported length error	105.714	112.104
Max-run discrete Frechet	165.876	159.650
Within-run arc RMSE	90.912	87.256

Independent audit reproduced213cached motion integrations and localization/heading trial scores exactly, with hashes and sharedinitialpose/masks verified.41trials improve synchronized error and30worsen;32route RMSEs worsen. Allfour participant means improve, but AR/CR/ST synchronized maneuver means worsen. Absolute supported path-length error worsens, especially10x5:125.834 to247.595mm. No.14long10x5trial1 worsens50.62mm;No.18longtrial1 improves87.91mm. These regressions remain in the atlas.

Do not promote: training guard fails. Three same-fit-ID conditional folds give old132.076 to new126.514mm mean, but worstperson ratio1.559583. No.13 worsens55.96% andNo.8 worsens5.80%. Heading2.081 to2.136degrees worsens. Full-training validation gains cannot override the predeclared5%participant guard.

Root cause evidence:No.13long10x5trial3 changes clock by+4.8517s, approximately one repeated route cycle. The independent bounded local-window diagnostic around the old clock finds medianlag-5ms over4identifiablewindows, supporting the old basin. A large clock correction can also be valid:No.15PS1 changes+4.1598s andimproves markedly. Thus a universal shift cap is inadequate; multichannel/event-order/common-support evidence must adjudicate basins without selecting from position error.

Reference continuity and calibration audit

Stored-processed positions provide a separately labeled weak secondary reference;32,441additionalmodel-gridframes restore continuous geometry-accepted support in61/71recordings. Weak-source blend synchronized RMSE103.878mm androute49.484mm do not replace primary observed-only results. Absolute supported length error becomes182.785mm rather than104.732mm: observed fragmentation hides distance comparisons, while processed paths introduce uncertainty.

Among61continuous recordings, full discreteFrechet152.924 to151.925mm andarc97.581 to96.359mm improve slightly. No.18long10x5trial1 has an apparent1.710m length deficit, but dottedprocessedturnexcursions and processed step speeds3.153m/s versus observed2.458m/s make it reference-dependent. It is not established physical undertravel. No.18PS2endpoint515.926mm remains unaffected by this support change.

A training-only calibration audit on207target recordings/12people gives freshall473centerphysics222.645mm versus held-person-excluded229.879mm. Axes/signs remain stable andscale shifts below.9%, yetNo.20worsens54.211mm. This illustrates calibration optimism and sensitivity; it cannot numerically adjust the96.3/100.7mm validation models.

Inherited v2calibration used477acceptedIDs and differentreference semantics; its231.330mm comparison is not an exact pairedfold control. Empirical wheel terms fit midpoint velocity jointly. A direct wheel-yaw interpretation gives1297.990mm and is invalid as a well-calibrated physicalbaseline. Radius metadata changes alone have exactlyzero effect on the current empirical velocity function.

Restoring individualwheel coefficients against rigidwheel targets improves purewheel-yaw training error to979.634mm but remains unusable. Training selectszero wheelyawfusion; individualmidpointspeed alone gives1.205mm conditionaltraining gain and.053mm validation regression. Small differential scale/intercept errors are amplified bydivisionby.52m. This experiment is rejected as a trajectory improvement.

Per-trialwheel startupbias testing exposes a missing assumption:only1/207heldtraining and1/71validation acquisitions satisfy the strict IMU-only initial5s quietgate, matching the originalv14 initialization. Most startupgravity/bias states are datefallbacks. Nonzero wheelyawblends remain rejected; zero-intercept refit yields only.091mm validationgain. Sparse eligibility does not prove continuous movement, but it invalidates an assumption of reliably measured quietstartup calibration.

Controlled experiments, including failures

Experiment	Validation RMSE mm	Decision / limitation
Original v21	102.585	Established control
Processed endpoints / +rates v32	105.172 /110.932	Reject; uncertain labels degrade performance
Confidence-weighted endpoints v33	.5selected:103.212	Reject; tiny gain at another validation weight is not selectable
Undamped dynamic tilt v33	100.565	Reject: >5% training-person regression guard
Training-selected tilt blend v34	100.702	Modest candidate;45 improve26 worsen
Long observed intervals v34	102.257	Small gain; slight heading regression; not combined
Braking bridge v34	114.483 /136.825	Reject; ordinary motion and tilt trigger false corrections
Joint planar intervals v35	102.610 selected	No useful gain;10x5 regression offsets small gains
Wheel acceleration phase v37	102.829 selected	Reject as improvement; training gain0.264mm fails to generalize

Every serious experiment preserves protocol, training/validation IDs, parameters, selection, full candidate tables, participant/maneuver outcomes and predictions. Selection uses training folds; validation audit cannot replace the selected candidate with a better grid member. No test acquisitions were opened this window.

Wheel phase used4Hz filtered acceleration-plane angle and a51sample derivative. Training-only phase-to-gyro-wheel mapping has correlations about.96-.99 and radius-like magnitudes.275-.317m, but those fitted gyro terms come from joint midpoint regression and are not guaranteed physical individual wheel speeds. Selected phaseblend.75 passes the conditional guard yet worsens validation and heading. Gyro axis is not a measured3D wheel normal; acceleration norms cannot eliminate propulsion/centrifugal contamination. A failed learned correction does not invalidate the physical rotation proxy, but it does reject this accuracy claim.

Calibration, limits and next experiments

Actual acquisition ranges are researcher-confirmed +/-2g and2000degrees/s. Integer CSV conversion remains unverified against firmware/receiver packing. Outward mirrored off-axis wheel sensors and a backward-facing center sensor do not specify full rotation matrices. Off-axis radius4cm, wheel radius.30m, hub width.52m and outboard reflector offset.066m are documented but their measurement uncertainty remains unverified.

Two axle markers establish midpoint and axle direction, not full body pitch. Gravity does not establish absolute yaw. Camera reprojection residuals are not absolute millimeter position uncertainty. The empirical raw-to-speed coefficients absorb configured radius; changing a radius metadata field alone may leave the estimator unchanged. Full mounting/lever-arm compensation needs calibrated axes, phase and body angular motion.

Current best hypothesis: heterogeneous errors from travel/braking deficits, yaw-induced lateral accumulation, residual clock offset, pitch/mounting coupling and unresolved reference-front semantics. A calibrated wheel/body estimator with wheelchair-trained interval displacement/heading corrections and uncertainty-aware SE(2) smoothing is defensible. Large networks or RL alone do not supply missing calibration or observable absolute position.

50mm: plausible as a research target, but not demonstrated as synchronized mean accuracy; current geometric mean is near50mm and still hides endpoint failures. 20/10mm: require substantial travel/yaw/timing improvements and independent reference/calibration evidence. 1mm: unsupported for this dataset and sensor system. No experiment here establishes attainability.

Illustrative isolated sensitivity: e_cross approximately distance times heading error in radians. At5m,0.1degree gives8.7mm;1mm would require about.0115degree under that simplified model. At2m/s,5ms timing error gives10mm localization error. These are scale calculations, not complete uncertainty bounds or achieved performance.

Next priorities: investigate No.8 after the completed multichannel clock arbitration and refit; complete full participant-fold calibration/base/normalization refits; stricter slip/bias confidence from allthreeIMUs; calibrated attitude/leverarms; uncertainty-aware interval fusion; native reference validation and a new independent acquisition. The first corrected-clock refit is completed and failed its participant guard. Preserve route-tail/endpoints alongside synchronized RMSE.

Signal-arbitrated clocks: latest unpromoted result

Training-only old/new clockbasins were compared on identical observed/sensor support with the same empirical calibration. Signed yaw, speed and distributed events reject No.13longT3wrongcycle andearlierSlopeHAR9badshift;No.19OS1isambiguousandkeepsold. No.15PS1/OS3supportedrepairsremain.462trainingrecordingsusecorrectedclocks;11oldfallbacks comprise8QCcases+3arbitrationcases. All473train/71validationIDsremain; fixed originalmodelparameters; no validationretuning.

Benchmark	Old blend	Arbitrated refit
Own-clock RMSE	100.702mm	94.835mm
Native-common RMSE	100.593mm	94.662mm
Conditional training mean	132.076mm	116.305mm
Worst-person training ratio	1.000	1.058005 - FAIL

No.13regression disappears aftersignal-basedaliascorrection (ratio.938);No.15stronglyimproves(.419). No.8stillworsens5.80%, abovefrozen5%guard. Do not loosen theguard to promote this result. Fullcalibration/base refits and investigation ofNo.8 remain nextsteps. The mean gain doesnot establish universalaccuracy or millimeterlocalization.

IMU-onlyloaderaudit verifiesall473/71IDs,11fallbacks,parenthashes,trainlabelcounts,and freshnative startup. All71modelmotionstreams reproduceexactly(maxdifference0). No referenceinputs enterprediction. Offlinefilteringand inheritedcalibrationremain limitations.

The latest atlas below uses thissignal-arbitratedrefit if thev44audit is available. It is diagnostic, not a promoted estimator. Retained100.702mm remains the currentguardedcandidate. Externalworkspacecreditfailure ended independentagentwork; unfinishedresearch andexactresumecommands are saved inresearch_state/NEXT_SESSION.md.

Latest native-common geometric diagnostics: symmetric route RMSE46.124mm; mean trial P9589.032mm; sampled Hausdorff139.983mm; maximum-run discrete Frechet155.162mm; last-supported endpoint122.308mm. Absolute supported length error worsens to113.306mm from105.714mm. These geometric means do not remove the failed participant guard or braking deficits.

Sources, reproducibility and trajectory atlas

Wheel-INS2, multi-IMU constraints
https://arxiv.org/html/2012.10593

TLIO, displacement and uncertainty fusion
https://arxiv.org/html/2007.01867

Wheelchair skidding study, institutional repository
https://repository.tudelft.nl/file/File_a9e38cd0-ca20-48ae-ab3a-f8056b64a4ee

VQF documentation, attitude and bias
https://vqf.readthedocs.io/en/stable/

C3D POINT:RATE
https://www.c3d.org/HTML/Documents/pointrate1.htm

C3D residual semantics
https://www.c3d.org/HTML/Documents/gloss_residual.htm

Qualisys external timebase
https://docs.qualisys.com/qtm/content/system_hardware/using_external_timebase_for_synchronization.htm

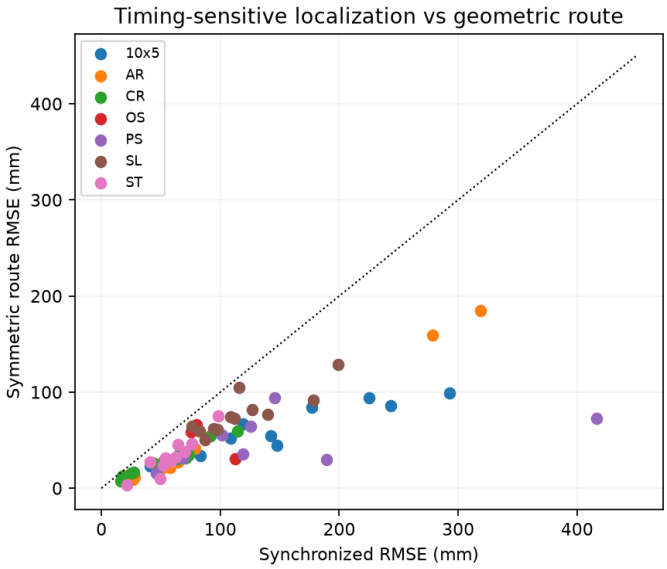
Qualisys external trigger
https://docs.qualisys.com/qtm/content/system_hardware/how_to_use_external_trigger.htm

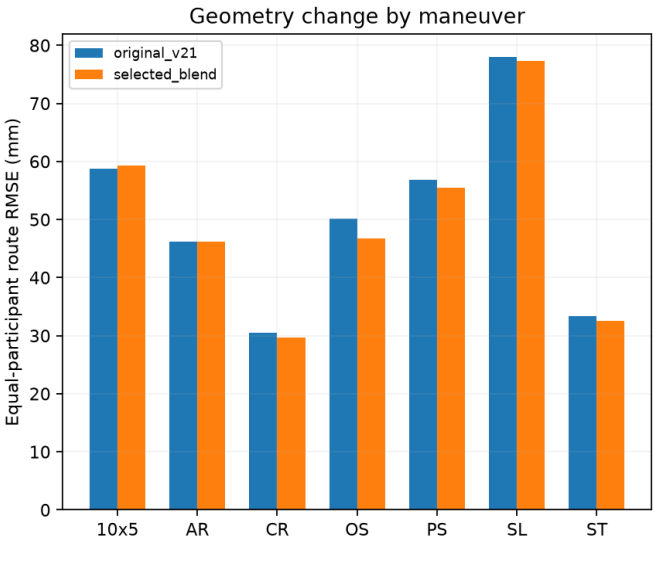
Inertial observability analysis
https://arxiv.org/abs/1805.05876

Primary-source mechanism notes: research_docs/trajectory_failure_research_v33.md. Experiment evidence: outputs_dynamic_blend_v34, outputs_dynamic_loader_v36, outputs_blend_audit_v36, outputs_route_metrics_v36, outputs_affine_clock_v36, outputs_signal_offset_v37, outputs_wheel_phase_v37, outputs_clock_refit_v38, outputs_clock_routes_v39, outputs_clock_folds_v39, outputs_clock_loader_v39, outputs_clock_alias_v40, outputs_clock_basin_v41, outputs_clock_arbitrated_v42, outputs_clock_arbitrated_folds_v43, outputs_clock_arbitrated_loader_v44, outputs_clock_routes_v44, outputs_stored_route_v37 and calibration/wheel gate audits. Persistent continuation files: research_state/STATUS, TODO, EXPERIMENTS, BEST_RESULTS, FAILED_IDEAS and NEXT_SESSION.md.

Following the metric overview are all71 accepted validation trajectories, one recording per landscape page. The current nativecommon atlas shows the same signal-arbitrated refit in red and native observed C3D in black. Each page has two panels: trajectory and yaw versus time. No storedprocessedpositions are shown. The refit is diagnostic and failed the trainingguard; retainedbest remains100.702mm. Physical trajectory scale is equal. Yaw is relative to the shared original initial heading; no post-hoc heading correction or spatial registration is applied.

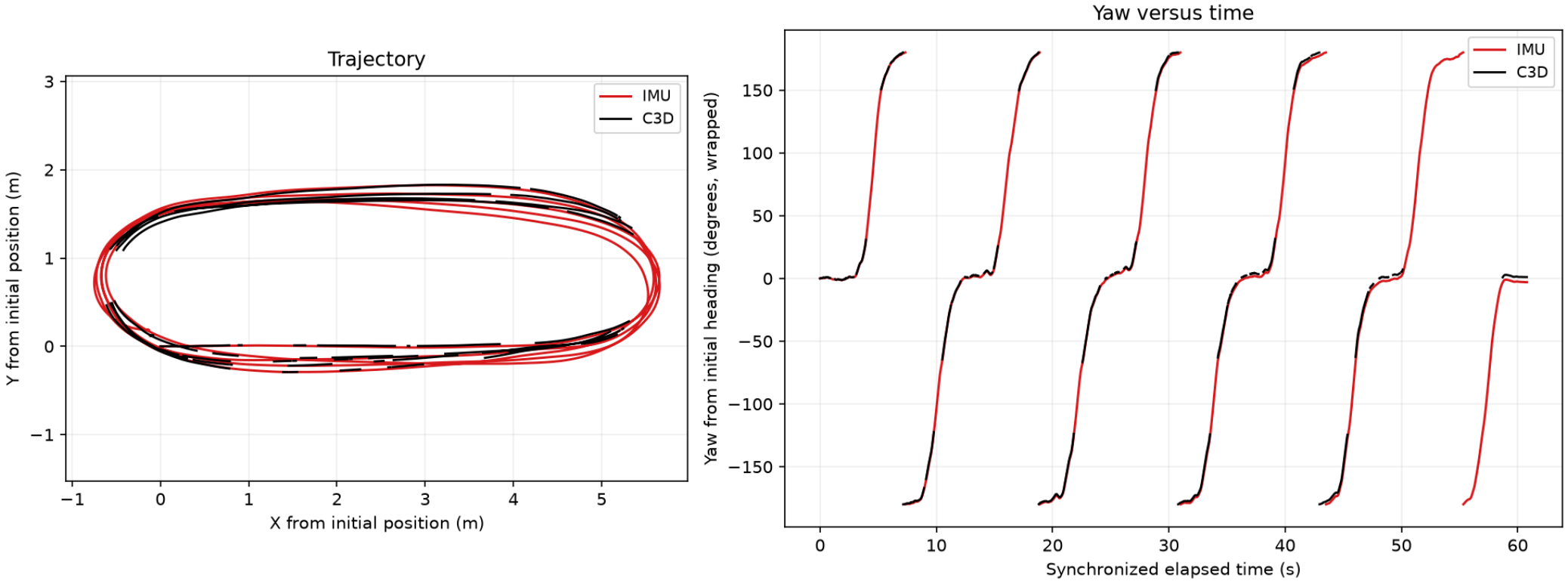
Metric overview





The gap between synchronized and nearest-route RMSE includes timing, tangential displacement deficits and route-order ambiguity. It cannot be assigned to clock error without a separate signal-based experiment. Slightly better means can coexist with worse tail metrics.

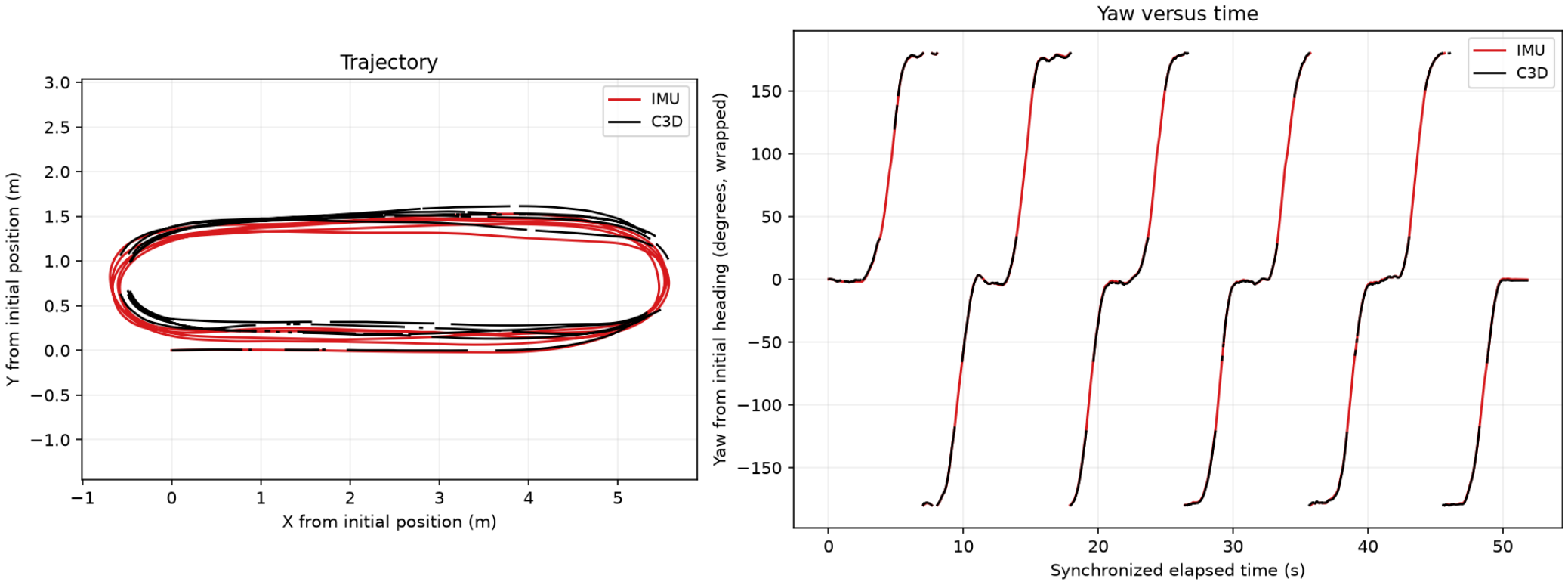
2026-09-24 | 10x5 | 10x5_Trial1_No.10
Synchronized 2D RMSE: 66.7 mm



Supported-route metrics (mm): route RMSE 31.9; P95 76.1; sampled Hausdorff 158.3; max-run discrete Frechet 188.1; last-supported endpoint 80.8; absolute length error 203.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

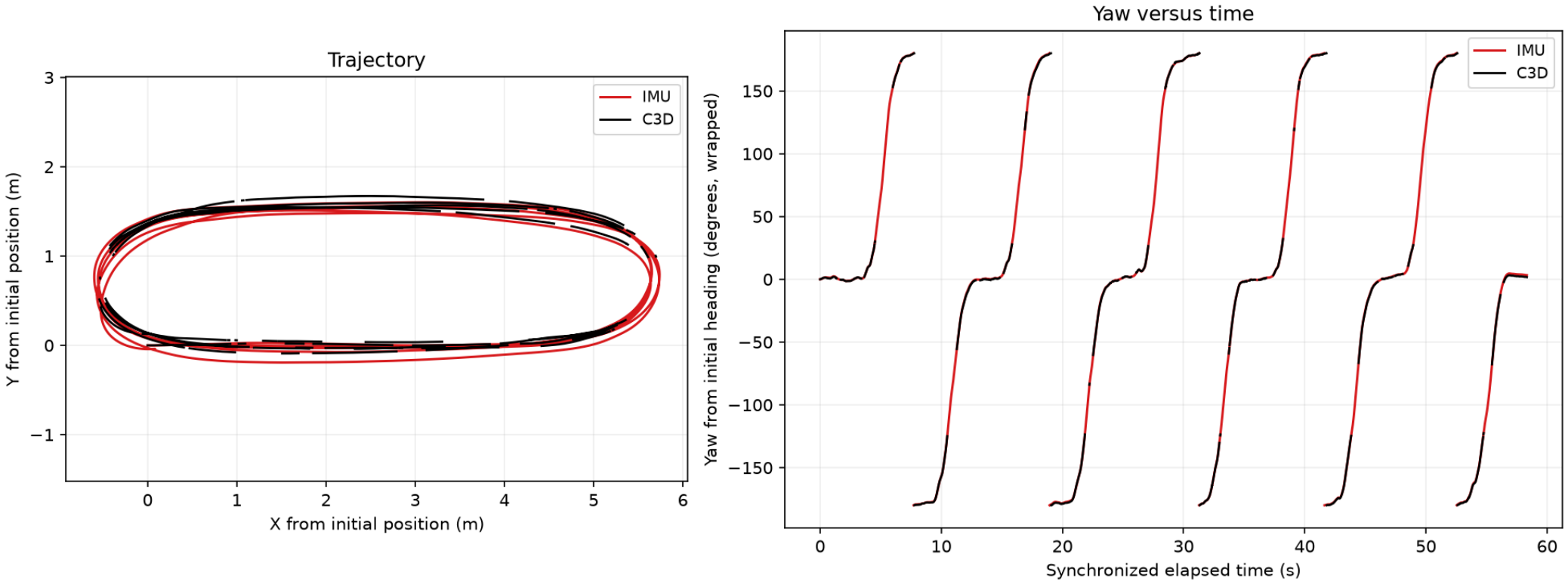
2026-09-24 | 10x5 | 10x5_Trial1_No.5
Synchronized 2D RMSE: 137.5 mm



Supported-route metrics (mm): route RMSE 54.2; P95 108.1; sampled Hausdorff 193.9; max-run discrete Frechet 220.6; last-supported endpoint 140.2; absolute length error 14.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

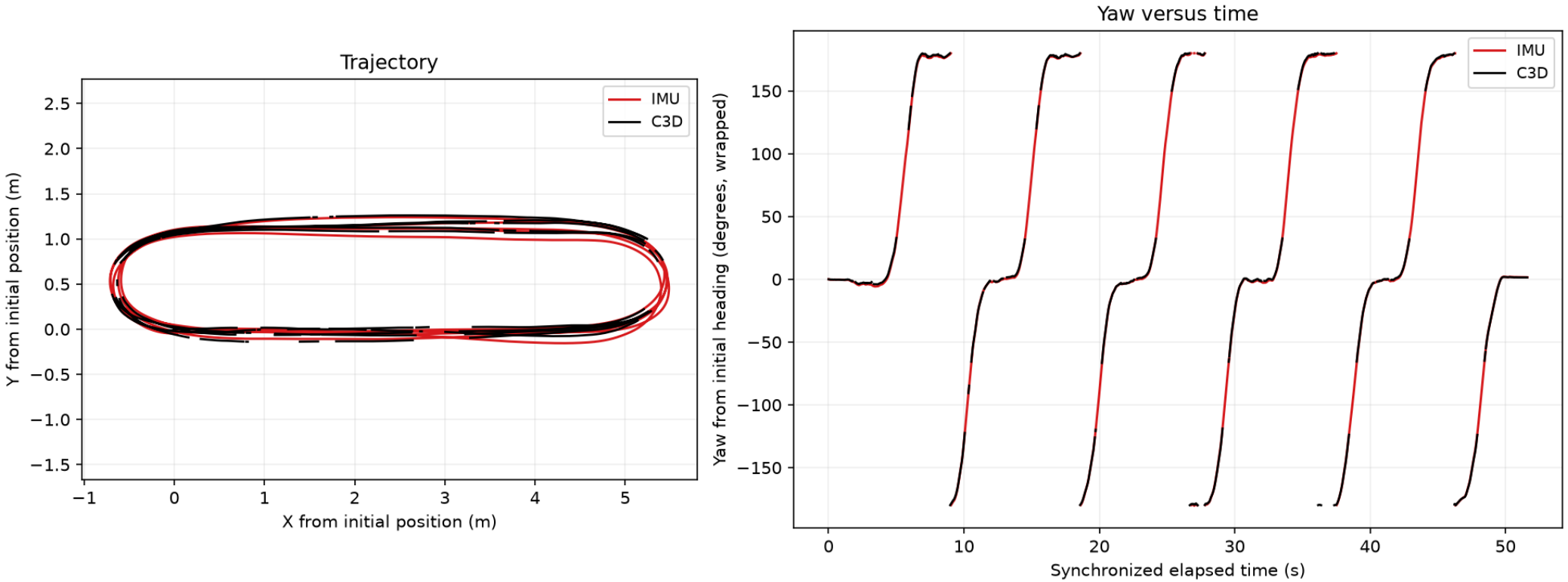
2026-09-24 | 10x5 | 10x5_Trial2_No.10
Synchronized 2D RMSE: 82.7 mm



Supported-route metrics (mm): route RMSE 34.8; P95 79.9; sampled Hausdorff 166.7; max-run discrete Frechet 186.5; last-supported endpoint 181.3; absolute length error 263.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

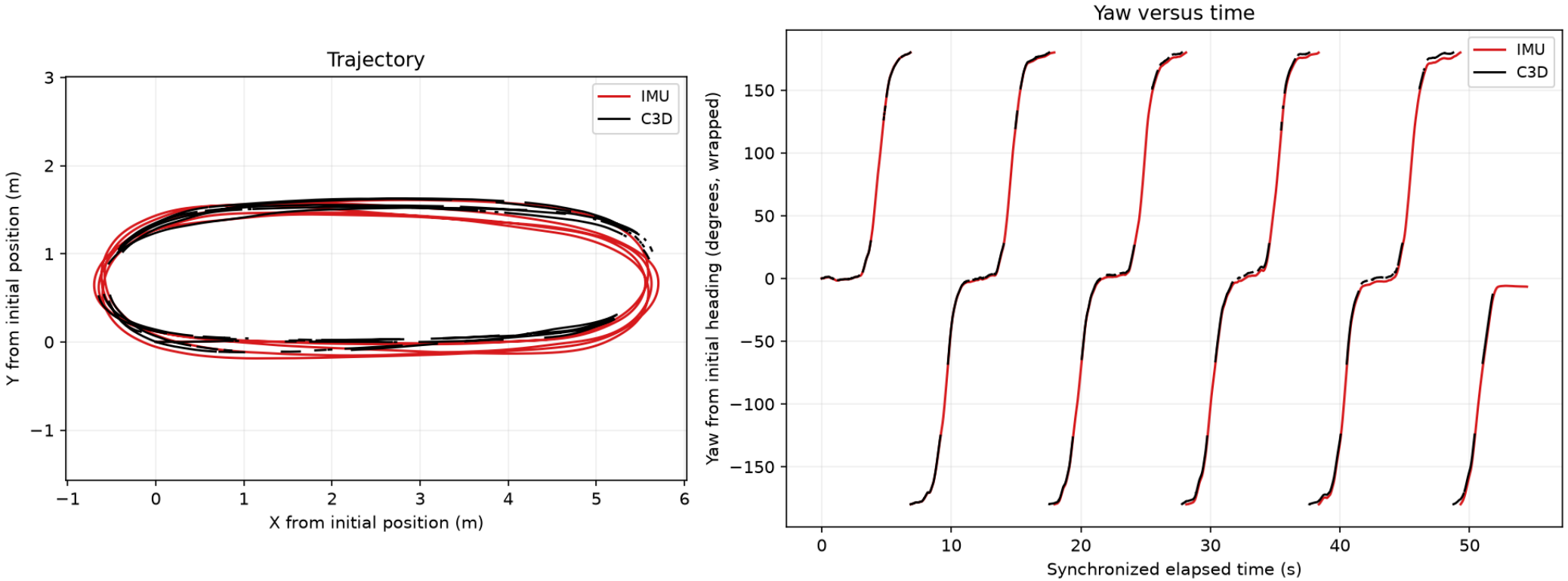
2026-09-24 | 10x5 | 10x5_Trial2_No.5
Synchronized 2D RMSE: 40.0 mm



Supported-route metrics (mm): route RMSE 22.3; P95 51.1; sampled Hausdorff 98.4; max-run discrete Frechet 98.4; last-supported endpoint 56.9; absolute length error 9.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

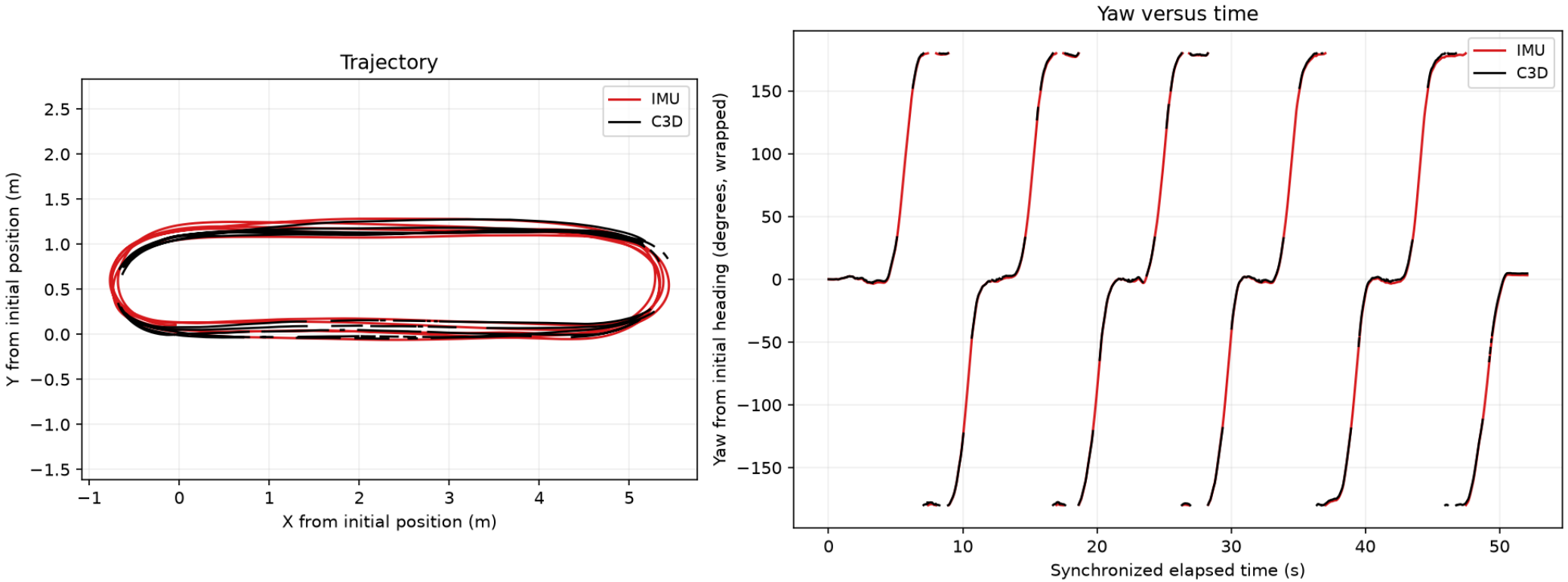
2026-09-24 | 10x5 | 10x5_Trial3_No.10
Synchronized 2D RMSE: 105.7 mm



Supported-route metrics (mm): route RMSE 57.0; P95 130.3; sampled Hausdorff 219.5; max-run discrete Frechet 265.1; last-supported endpoint 87.2; absolute length error 134.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

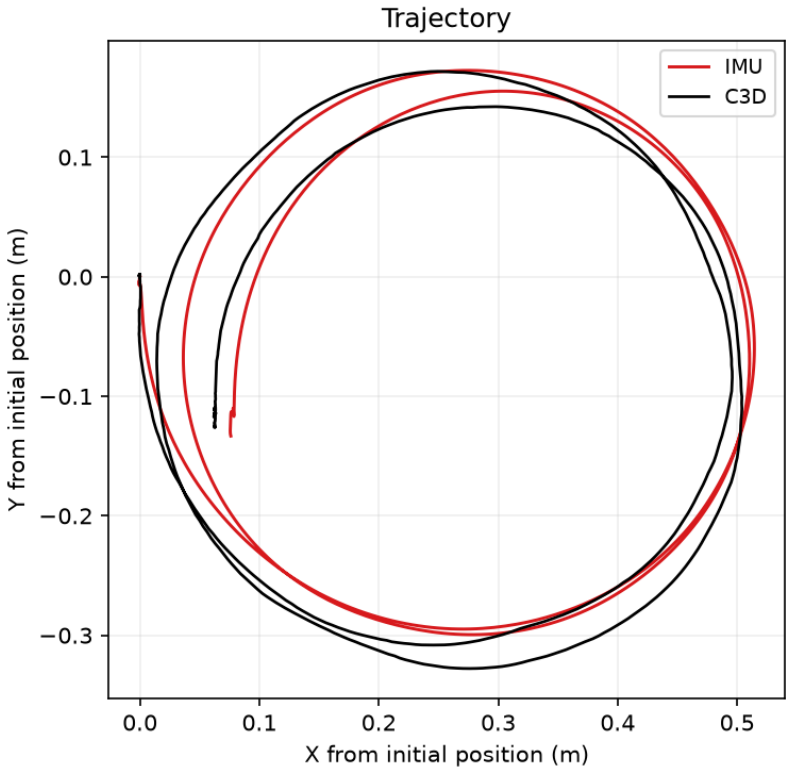
2026-09-24 | 10x5 | 10x5_Trial3_No.5
Synchronized 2D RMSE: 75.4 mm

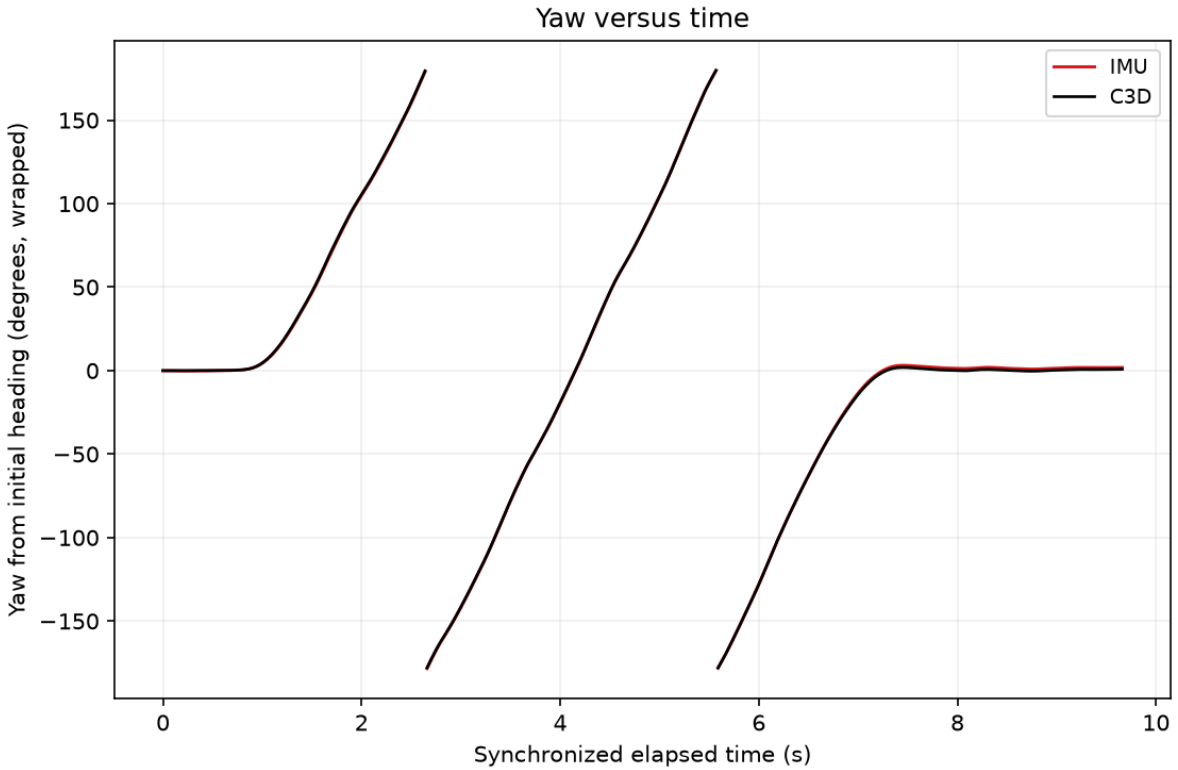


Supported-route metrics (mm): route RMSE 27.7; P95 57.5; sampled Hausdorff 116.8; max-run discrete Frechet 123.1; last-supported endpoint 107.8; absolute length error 161.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-24 | AR | AR_Trial1_No.10
Synchronized 2D RMSE: 21.4 mm

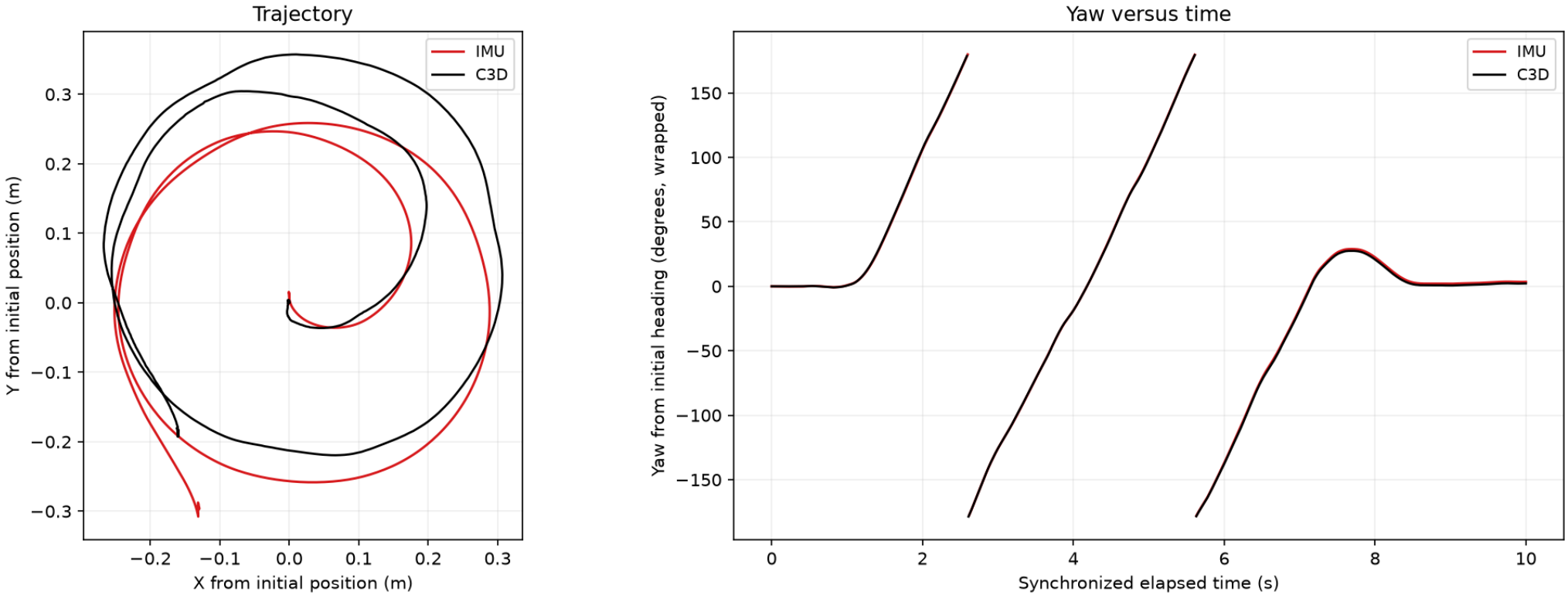




Supported-route metrics (mm): route RMSE 13.1; P95 23.4; sampled Hausdorff 28.4; max-run discrete Frechet 30.5; last-supported endpoint 15.6; absolute length error 64.5.

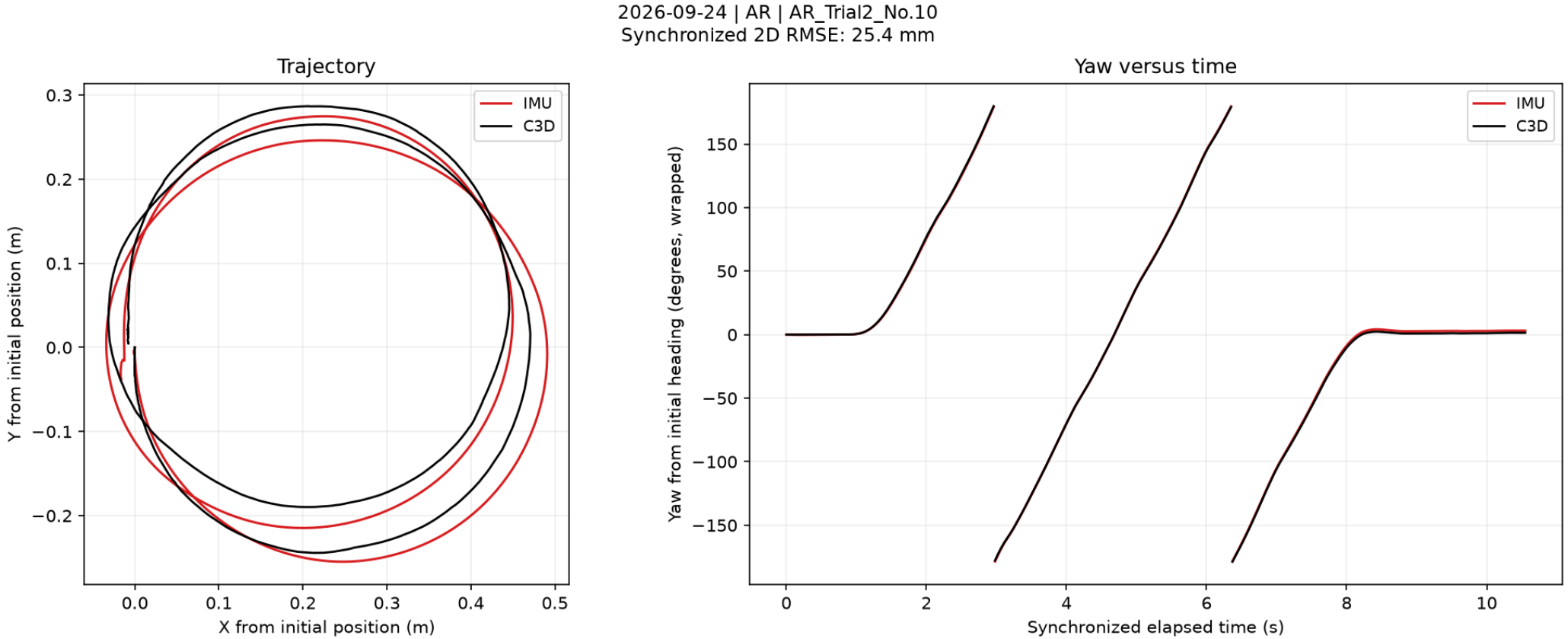
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-24 | AR | AR_Trial1_No.5
Synchronized 2D RMSE: 77.6 mm



Supported-route metrics (mm): route RMSE 41.6; P95 90.2; sampled Hausdorff 118.6; max-run discrete Frechet 119.2; last-supported endpoint 115.8; absolute length error 169.8.

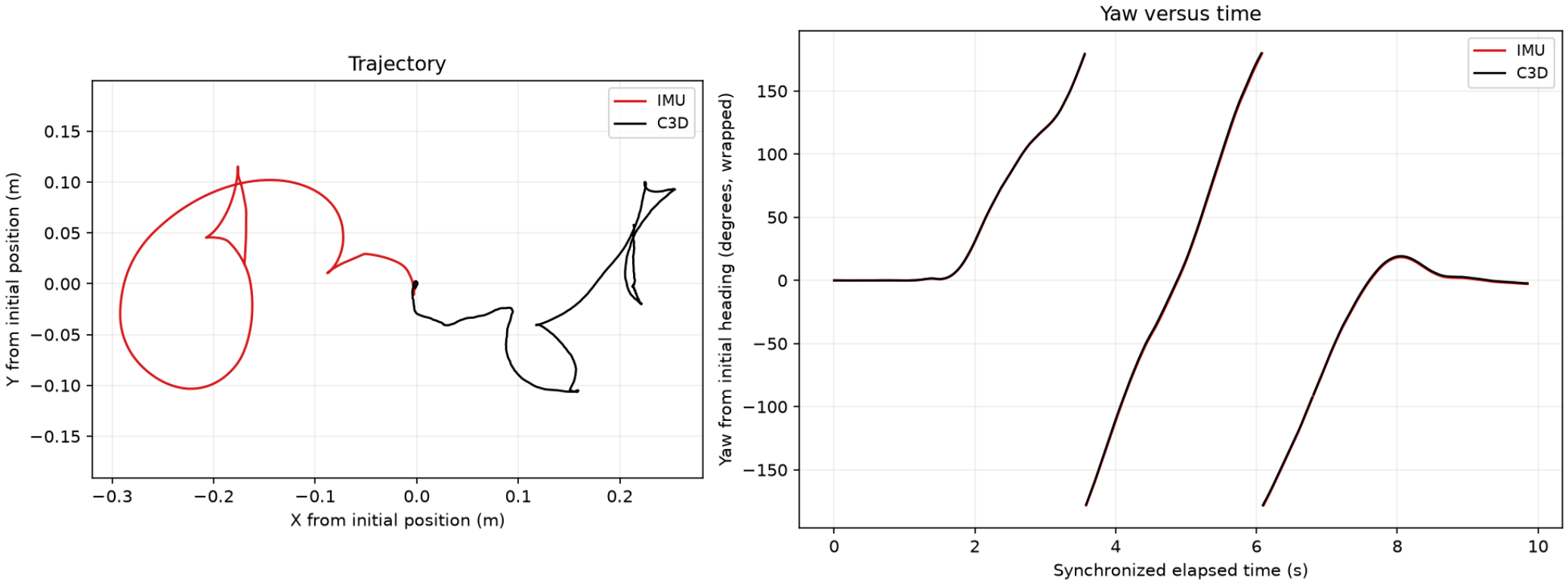
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 13.2; P95 23.3; sampled Hausdorff 28.9; max-run discrete Frechet 43.7; last-supported endpoint 43.7; absolute length error 78.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

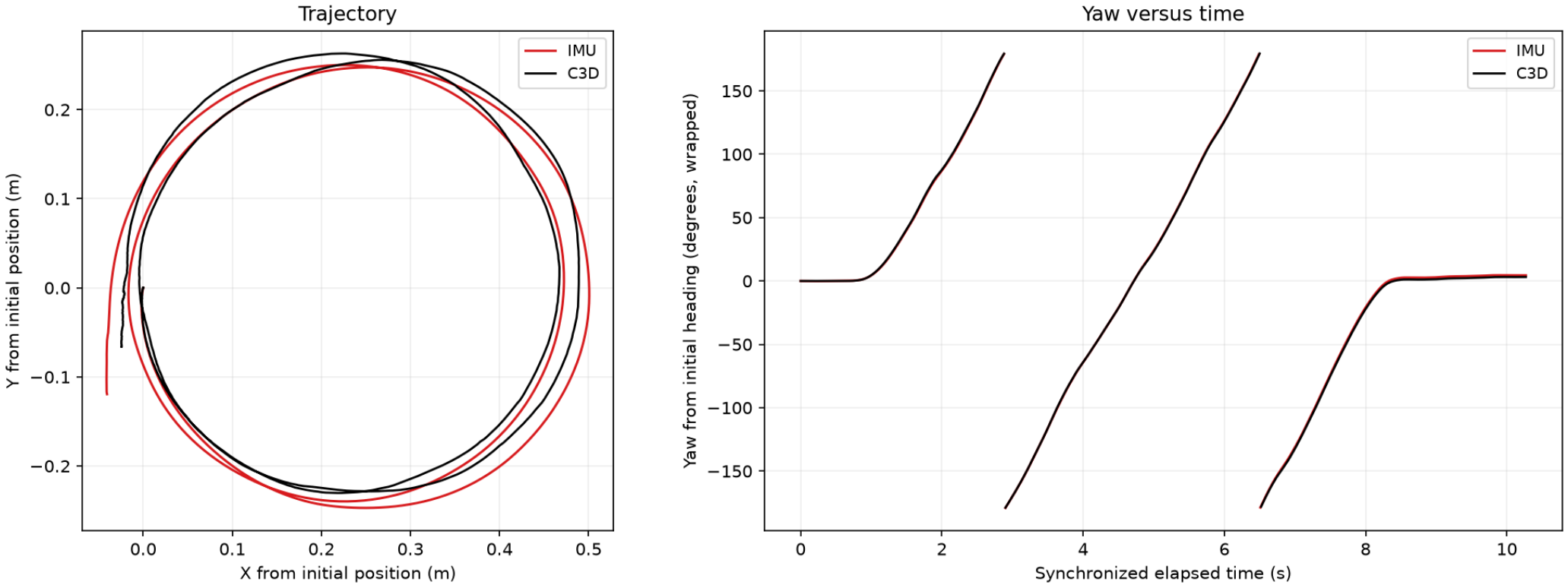
2026-09-24 | AR | AR_Trial2_No.5
Synchronized 2D RMSE: 318.7 mm



Supported-route metrics (mm): route RMSE 185.1; P95 277.8; sampled Hausdorff 288.5; max-run discrete Frechet 451.5; last-supported endpoint 385.2; absolute length error 153.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

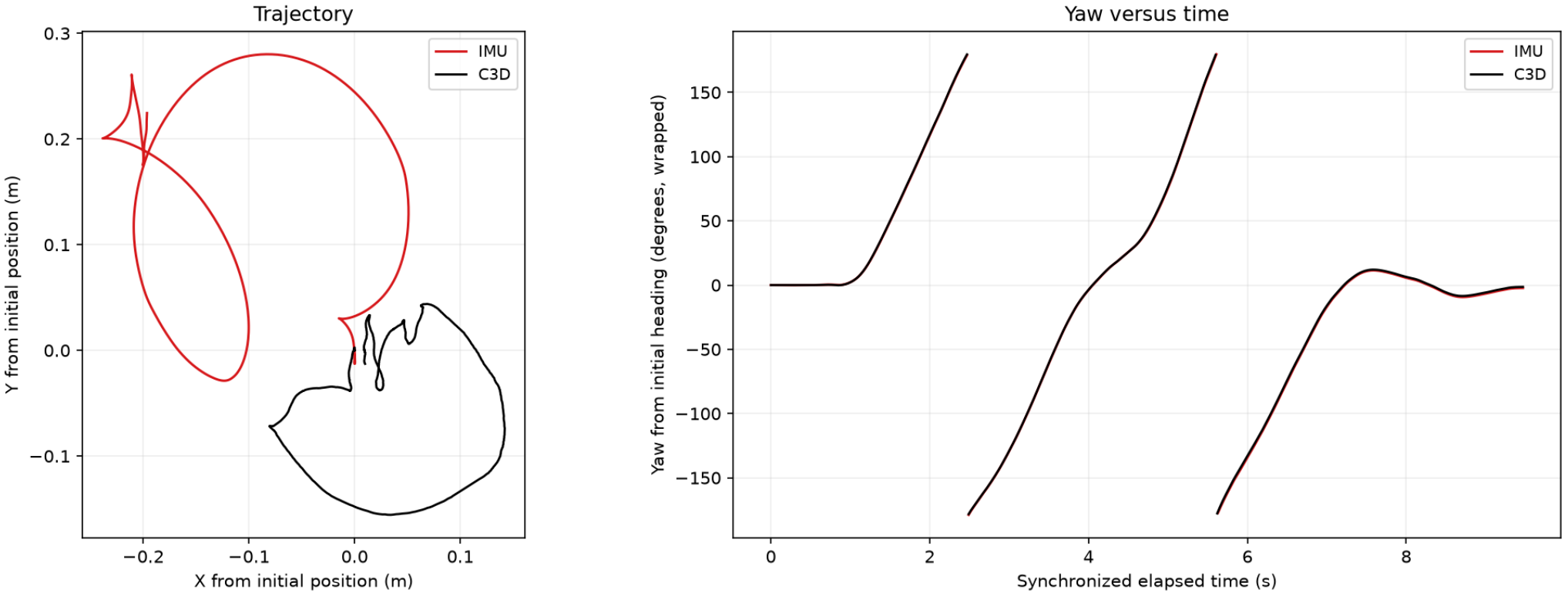
2026-09-24 | AR | AR_Trial3_No.10
Synchronized 2D RMSE: 25.9 mm



Supported-route metrics (mm): route RMSE 9.3; P95 16.2; sampled Hausdorff 55.5; max-run discrete Frechet 55.8; last-supported endpoint 55.8; absolute length error 97.0.

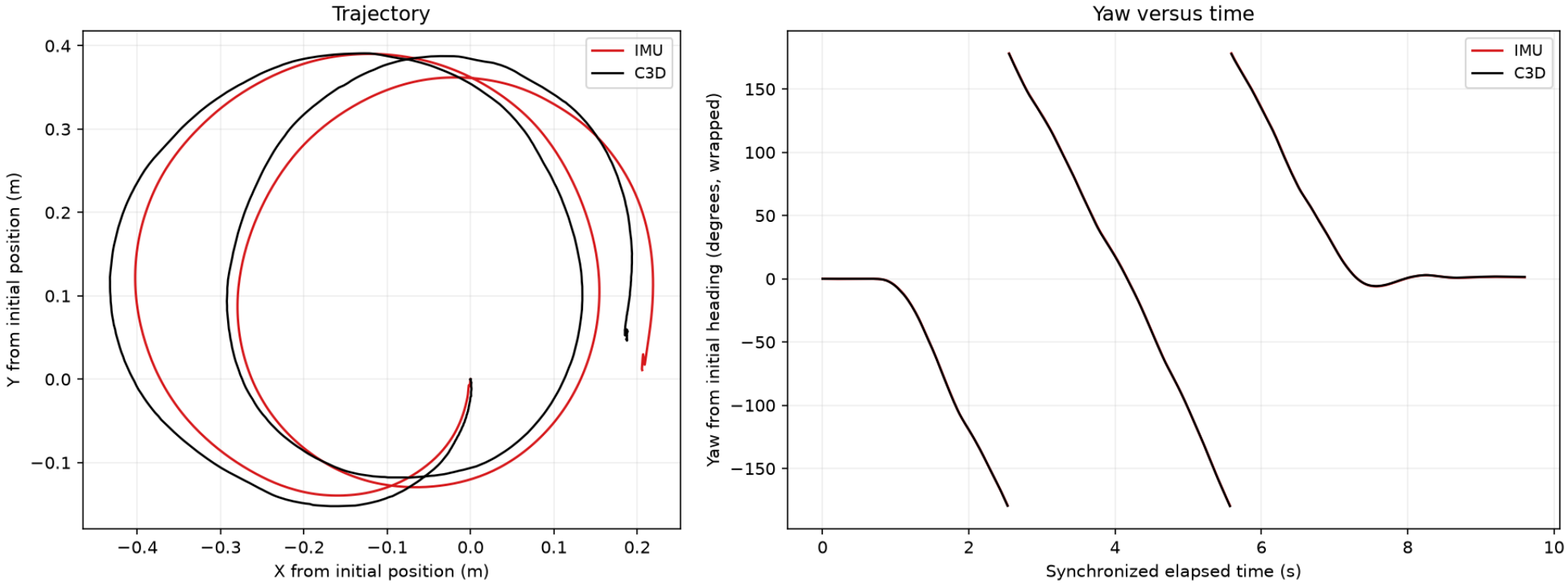
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-24 | AR | AR_Trial3_No.5
Synchronized 2D RMSE: 278.8 mm



Supported-route metrics (mm): route RMSE 159.4; P95 284.2; sampled Hausdorff 317.7; max-run discrete Frechet 314.5; last-supported endpoint 314.5; absolute length error 467.3.

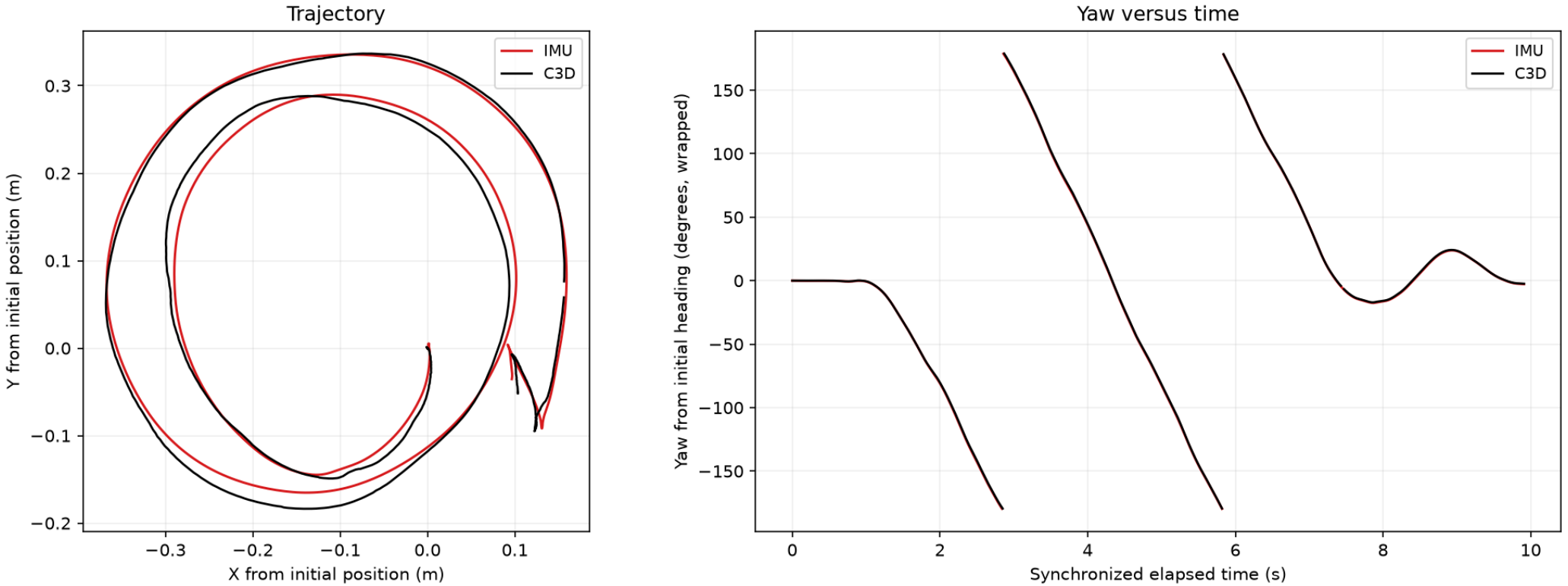
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 17.2; P95 29.5; sampled Hausdorff 39.9; max-run discrete Frechet 41.9; last-supported endpoint 41.9; absolute length error 40.2.

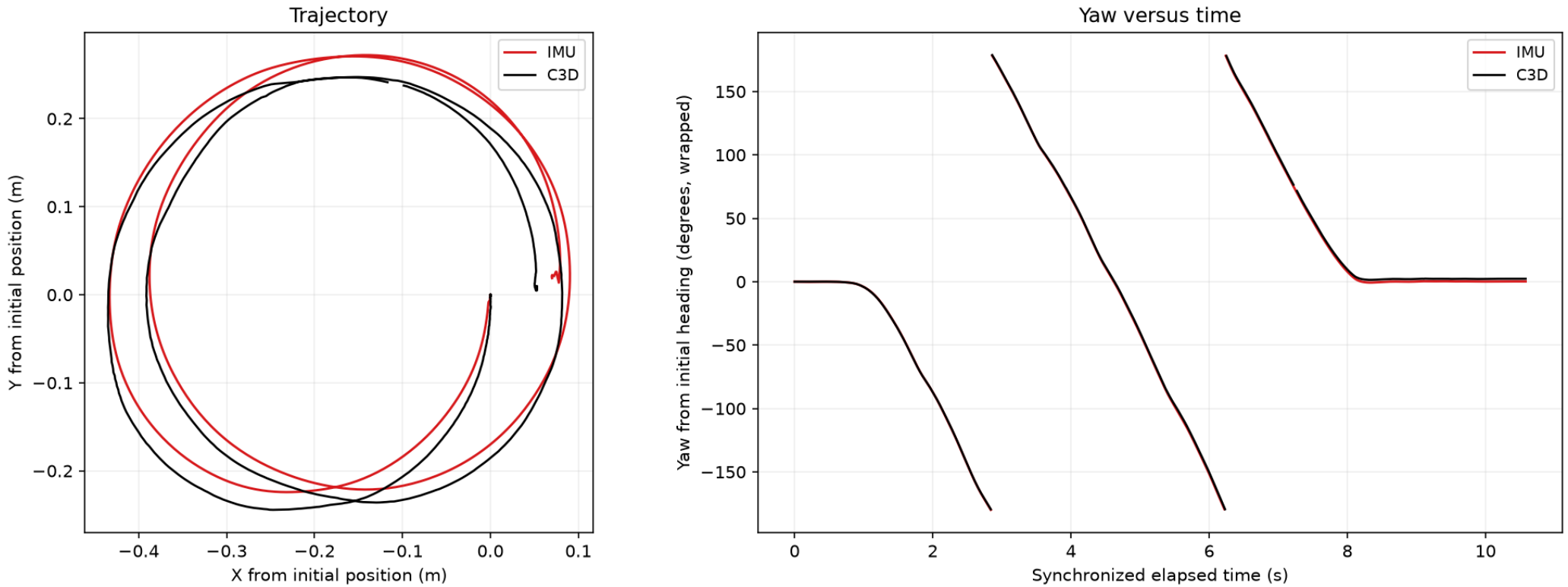
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-24 | CR | CR_Trial1_No.5
Synchronized 2D RMSE: 15.8 mm



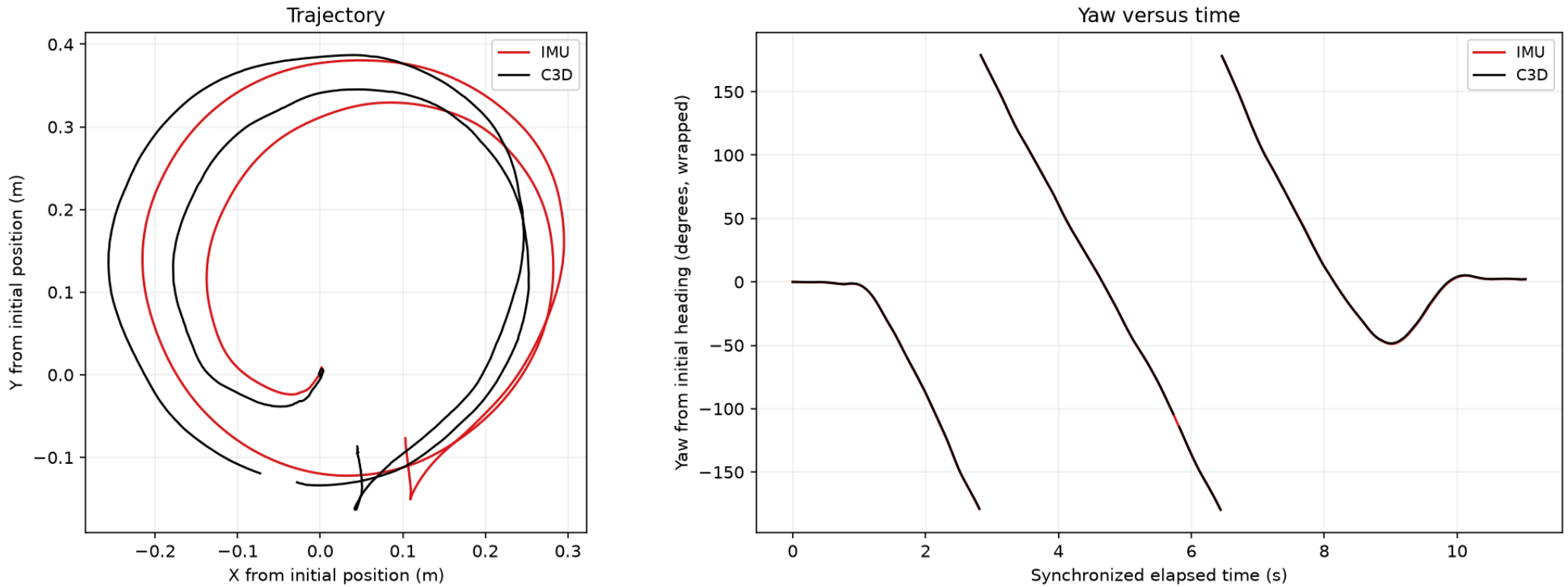
Supported-route metrics (mm): route RMSE 7.6; P95 15.5; sampled Hausdorff 18.5; max-run discrete Frechet 20.6; last-supported endpoint 17.7; absolute length error 47.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 15.2; P95 26.5; sampled Hausdorff 34.6; max-run discrete Frechet 38.0; last-supported endpoint 21.1; absolute length error 14.9.

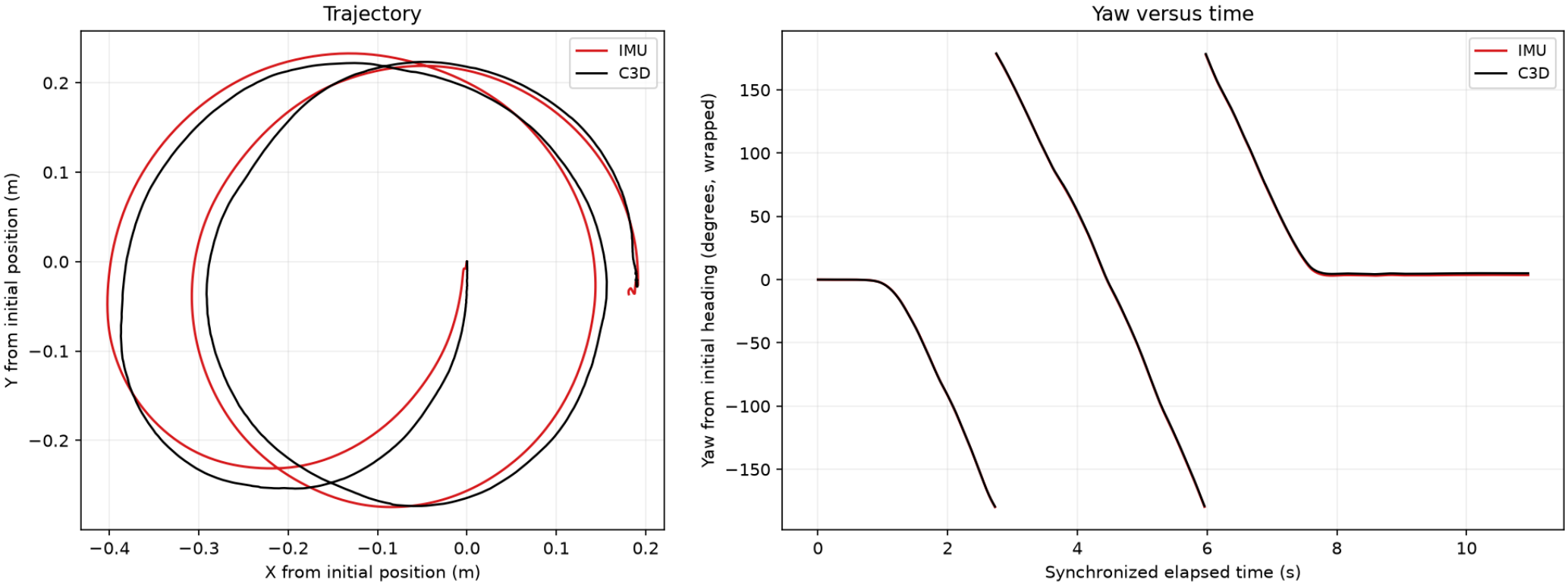
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 26.5; P95 41.2; sampled Hausdorff 47.6; max-run discrete Frechet 62.9; last-supported endpoint 59.2; absolute length error 157.8.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

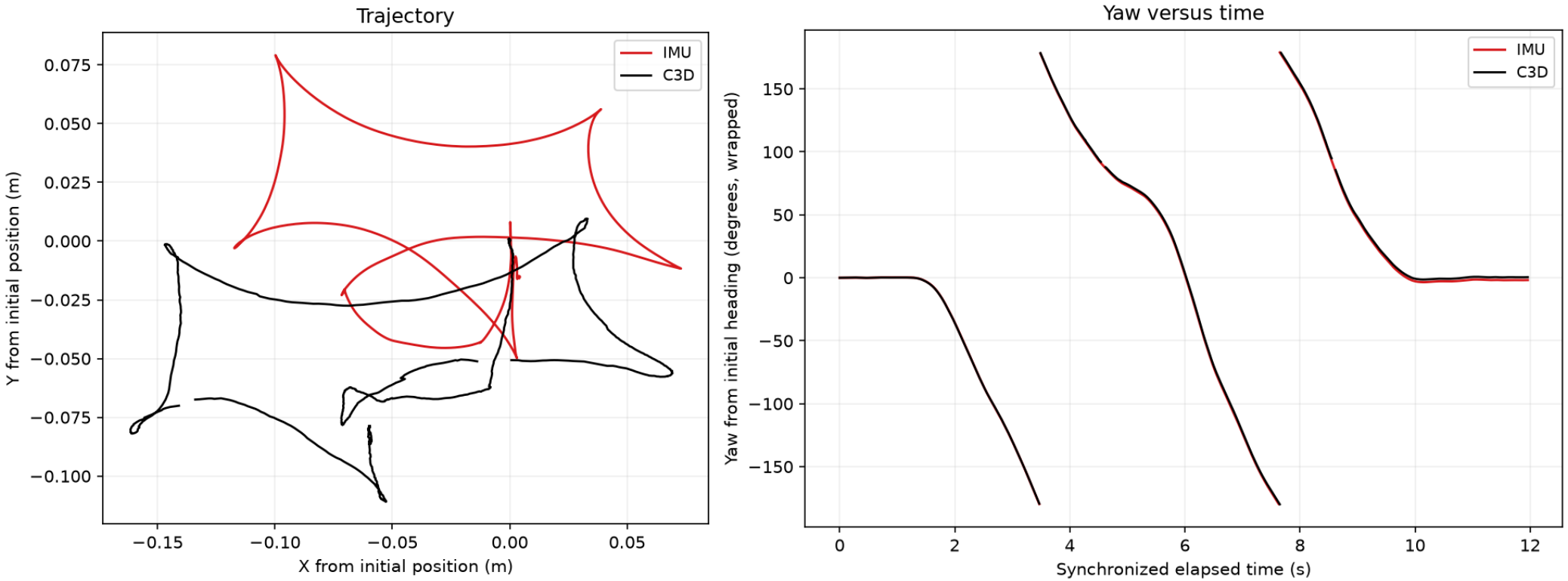
2026-09-24 | CR | CR_Trial3_No.10
Synchronized 2D RMSE: 17.9 mm



Supported-route metrics (mm): route RMSE 12.3; P95 19.9; sampled Hausdorff 27.2; max-run discrete Frechet 27.4; last-supported endpoint 17.7; absolute length error 5.4.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

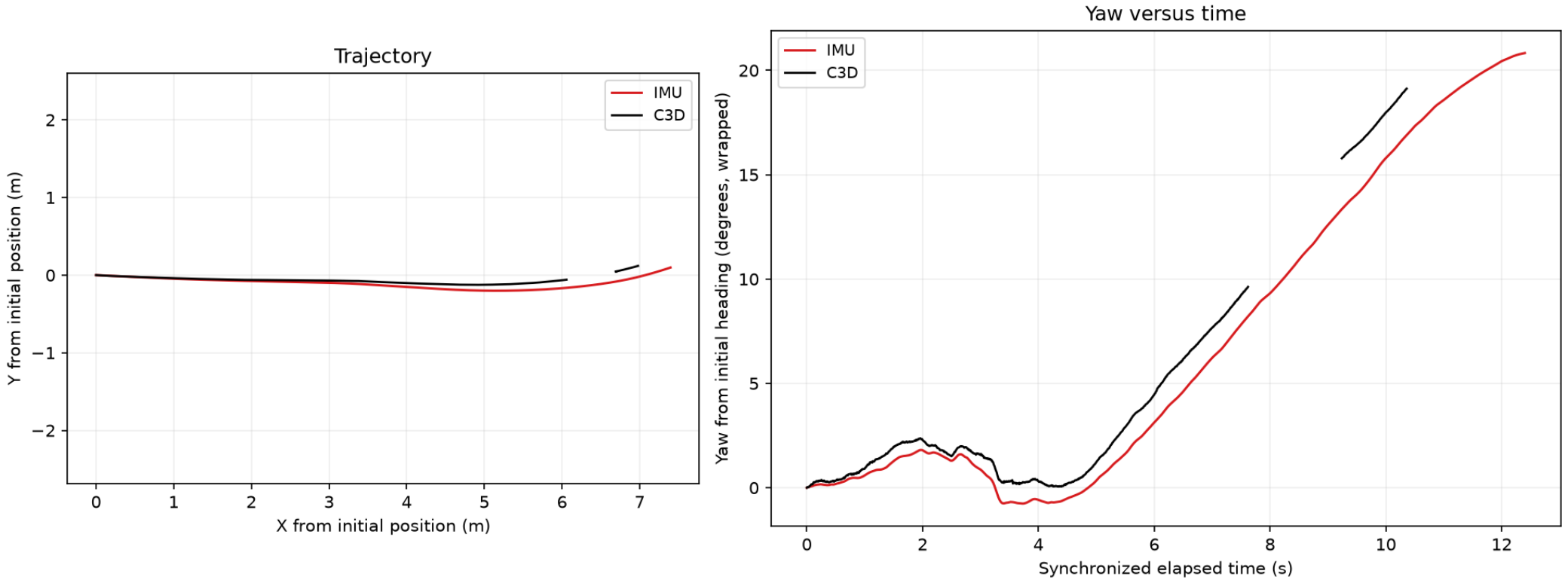
2026-09-24 | CR | CR_Trial3_No.5
Synchronized 2D RMSE: 66.7 mm



Supported-route metrics (mm): route RMSE 37.4; P95 79.0; sampled Hausdorff 90.5; max-run discrete Frechet 94.1; last-supported endpoint 94.1; absolute length error 69.4.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

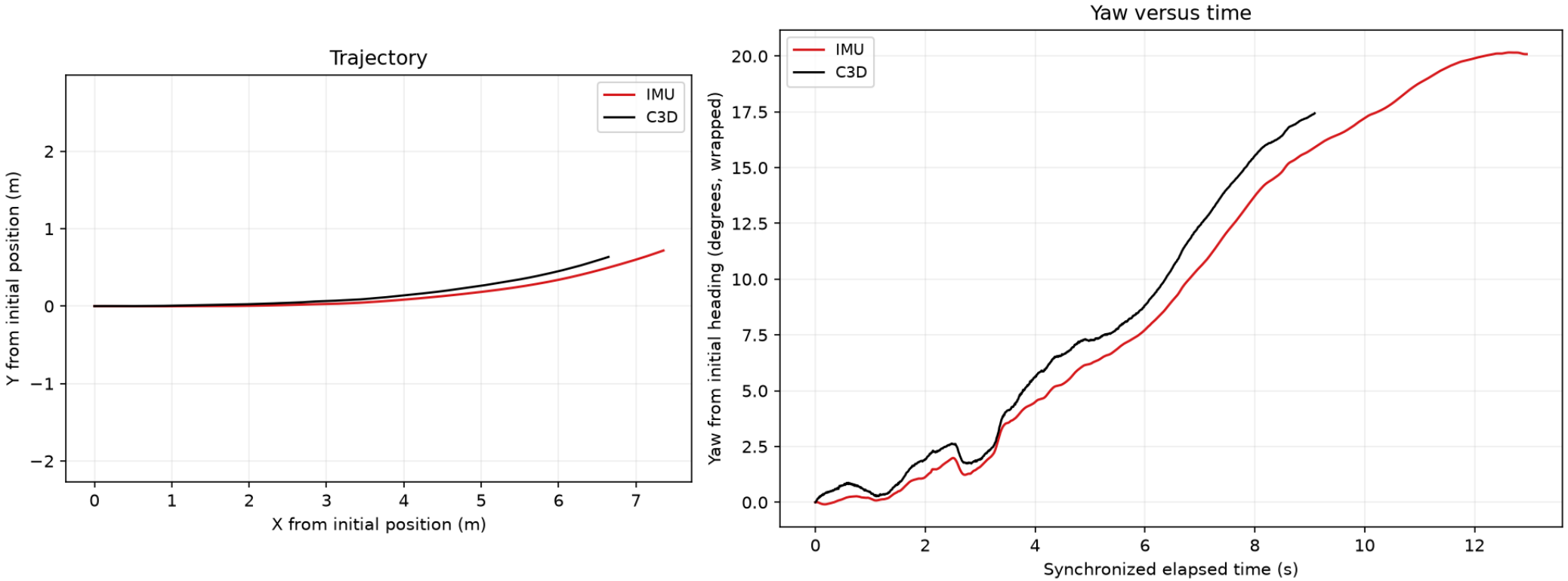
2026-09-24 | OS | OS_Trial1_No.10
Synchronized 2D RMSE: 71.8 mm



Supported-route metrics (mm): route RMSE 56.7; P95 127.4; sampled Hausdorff 139.3; max-run discrete Frechet 139.3; last-supported endpoint 139.3; absolute length error 31.8.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

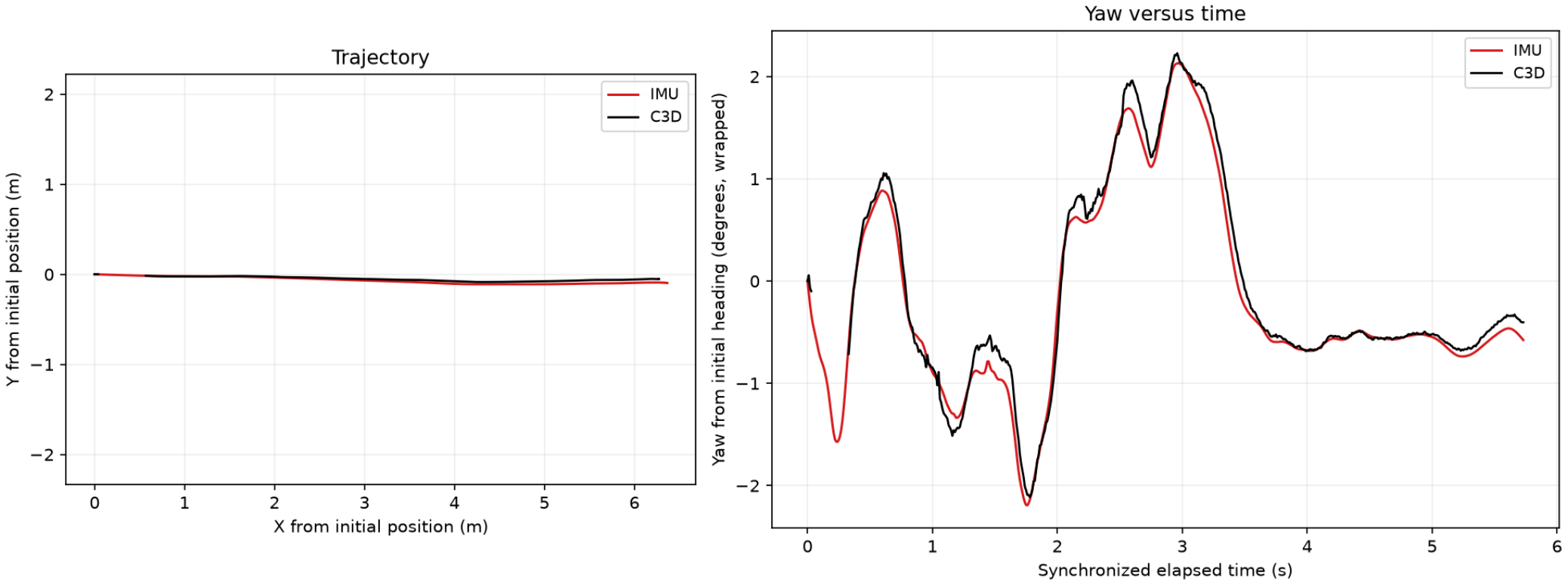
2026-09-24 | OS | OS_Trial3_No.10
Synchronized 2D RMSE: 74.6 mm



Supported-route metrics (mm): route RMSE 62.8; P95 119.5; sampled Hausdorff 132.9; max-run discrete Frechet 132.9; last-supported endpoint 132.9; absolute length error 16.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

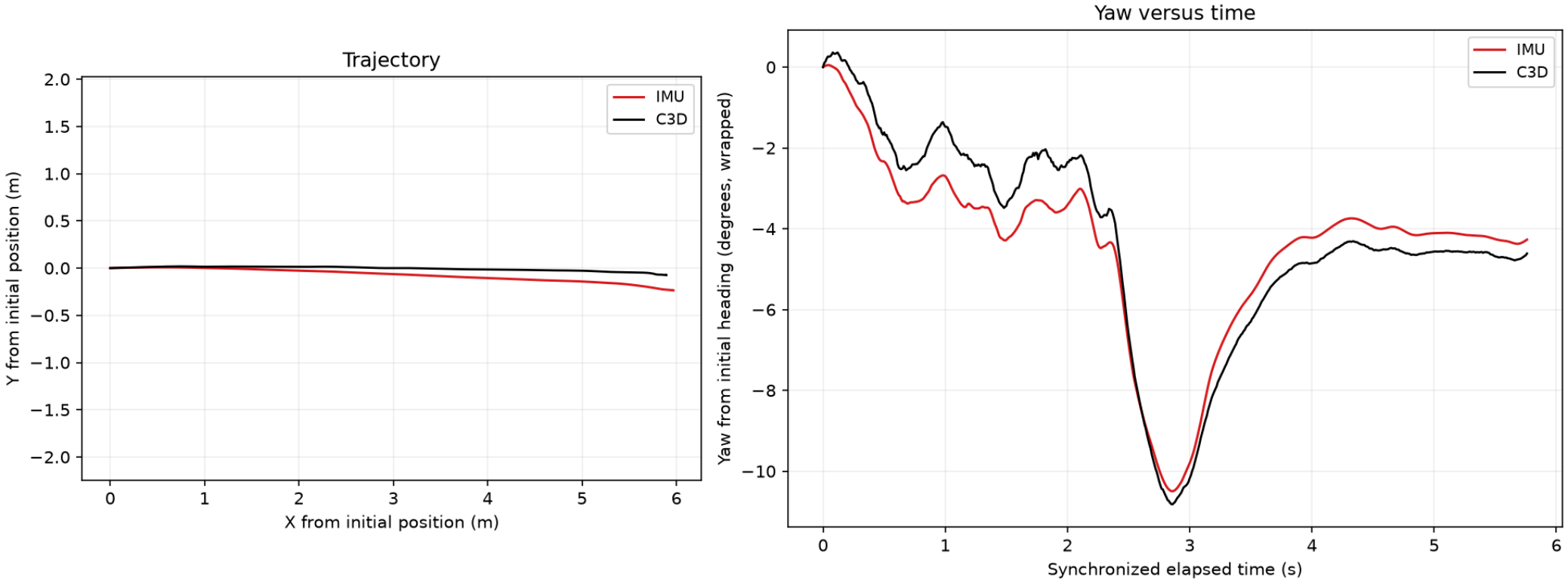
2026-09-24 | PS | PS_Trial1_No.10
Synchronized 2D RMSE: 66.5 mm



Supported-route metrics (mm): route RMSE 24.5; P95 38.8; sampled Hausdorff 101.0; max-run discrete Frechet 102.3; last-supported endpoint 102.3; absolute length error 109.8.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

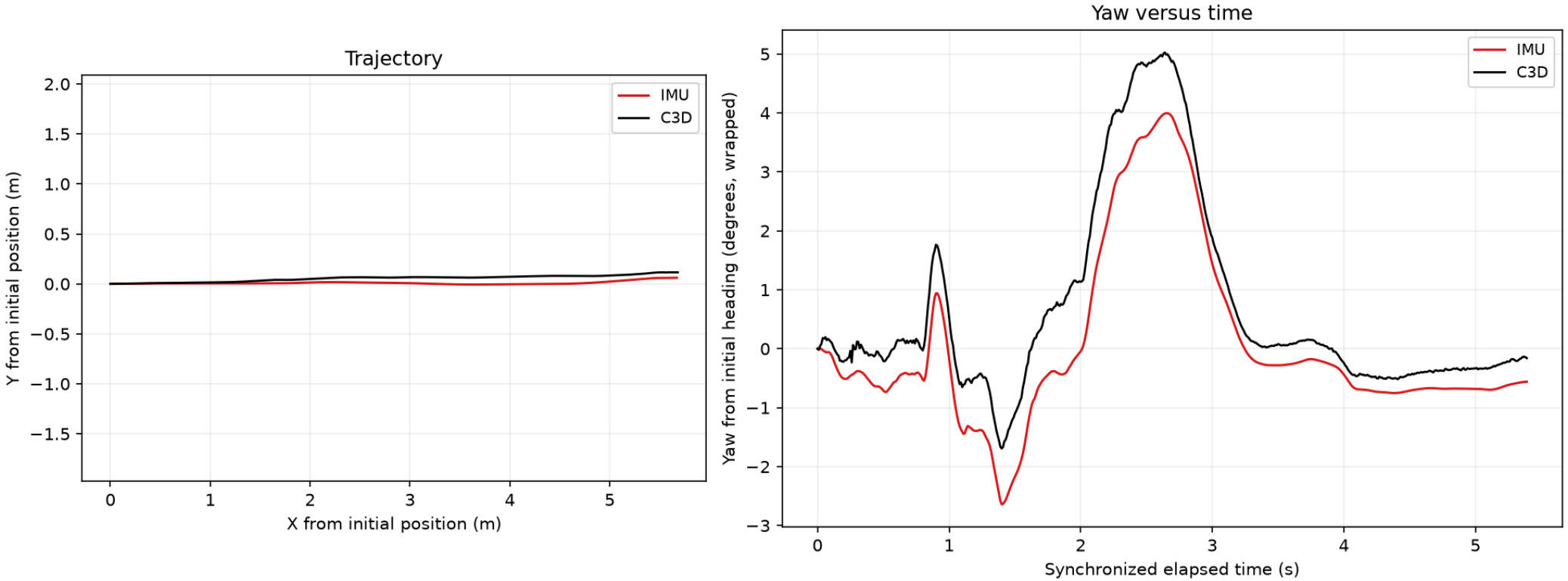
2026-09-24 | PS | PS_Trial2_No.10
Synchronized 2D RMSE: 135.9 mm



Supported-route metrics (mm): route RMSE 79.0; P95 143.6; sampled Hausdorff 178.1; max-run discrete Frechet 180.4; last-supported endpoint 180.4; absolute length error 66.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

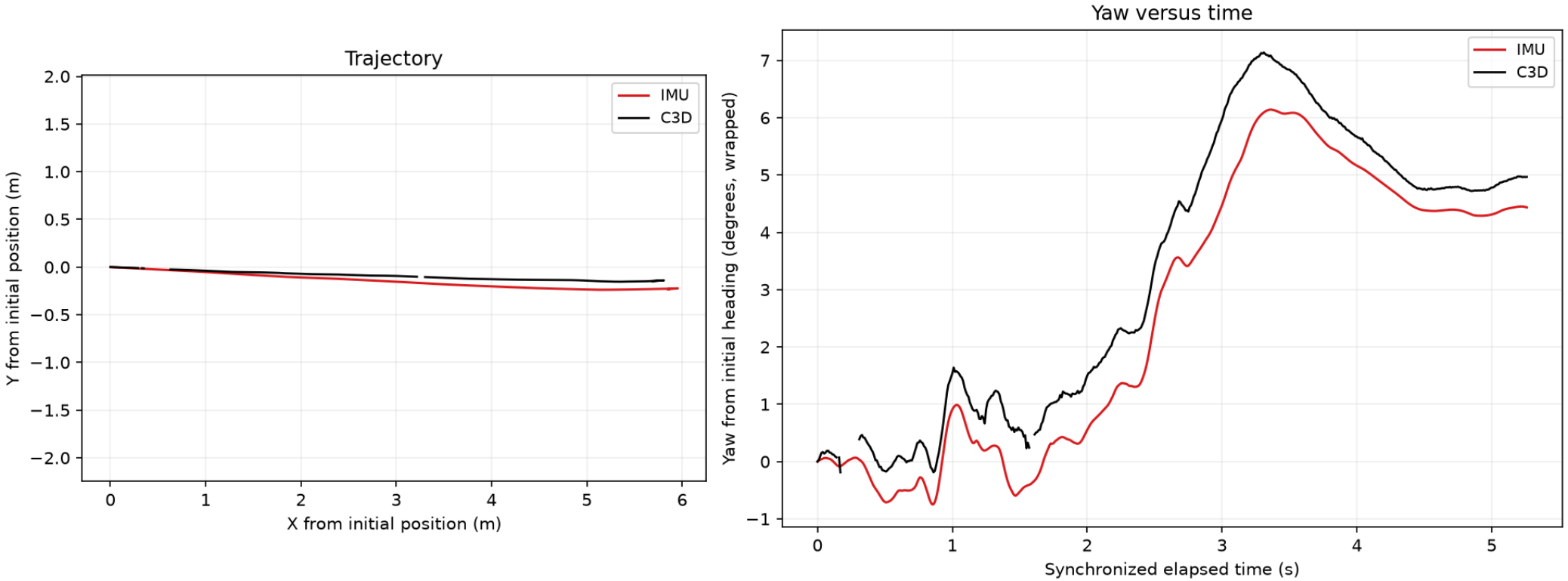
2026-09-24 | PS | PS_Trial2_No.5
Synchronized 2D RMSE: 60.0 mm



Supported-route metrics (mm): route RMSE 52.3; P95 79.5; sampled Hausdorff 81.4; max-run discrete Frechet 81.7; last-supported endpoint 54.5; absolute length error 8.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

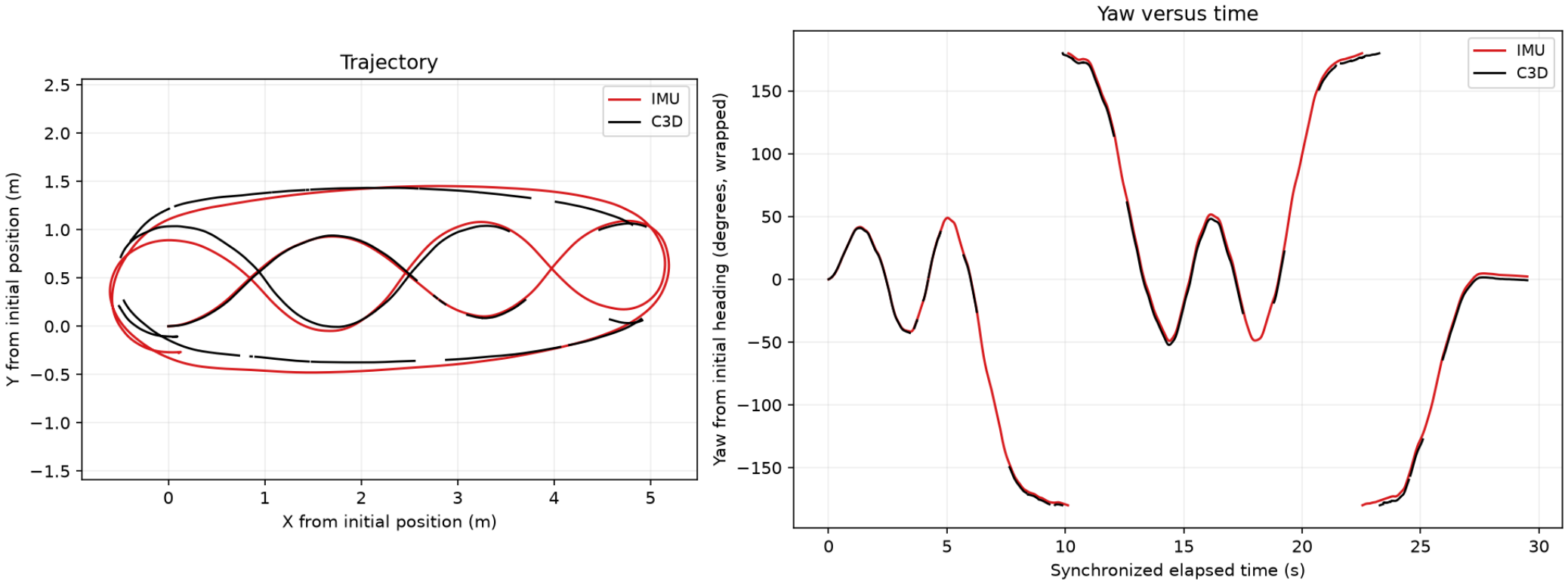
2026-09-24 | PS | PS_Trial3_No.5
Synchronized 2D RMSE: 125.2 mm



Supported-route metrics (mm): route RMSE 64.4; P95 94.2; sampled Hausdorff 166.3; max-run discrete Frechet 187.0; last-supported endpoint 187.0; absolute length error 145.6.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

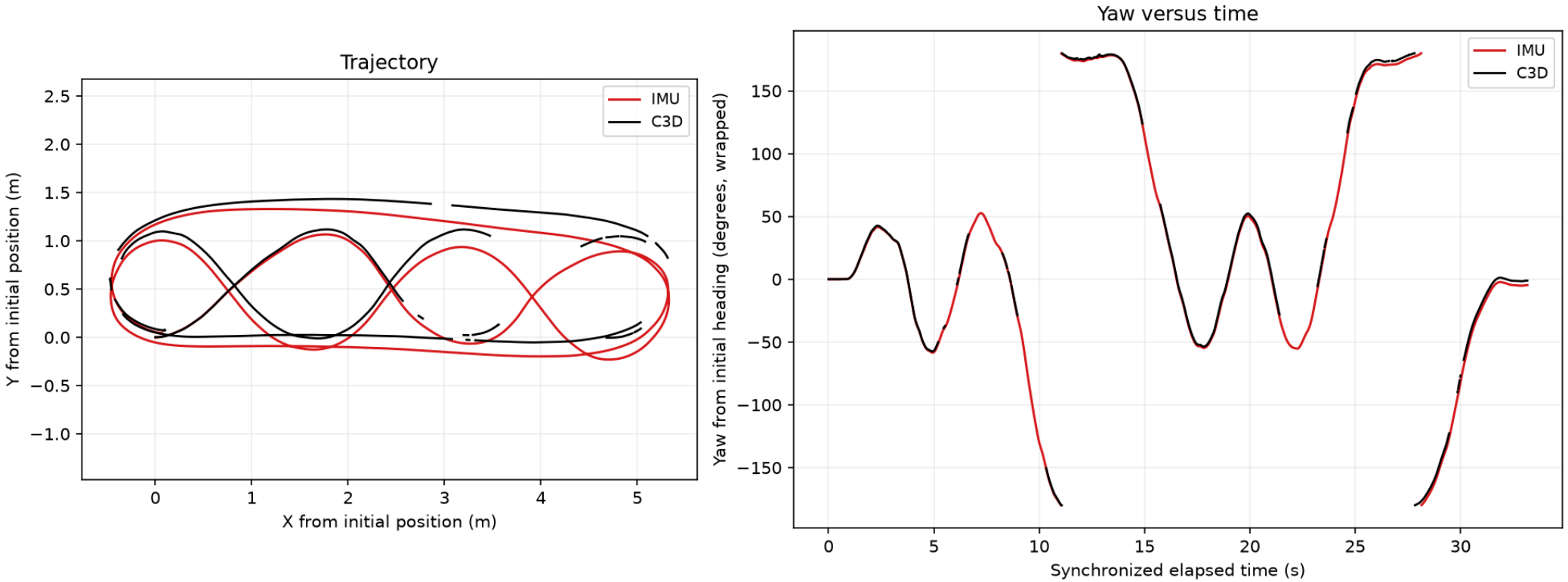
2026-09-24 | SL | SL_Trial1_No.10
Synchronized 2D RMSE: 119.0 mm



Supported-route metrics (mm): route RMSE 76.8; P95 151.7; sampled Hausdorff 184.3; max-run discrete Frechet 195.3; last-supported endpoint 174.8; absolute length error 4.3.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

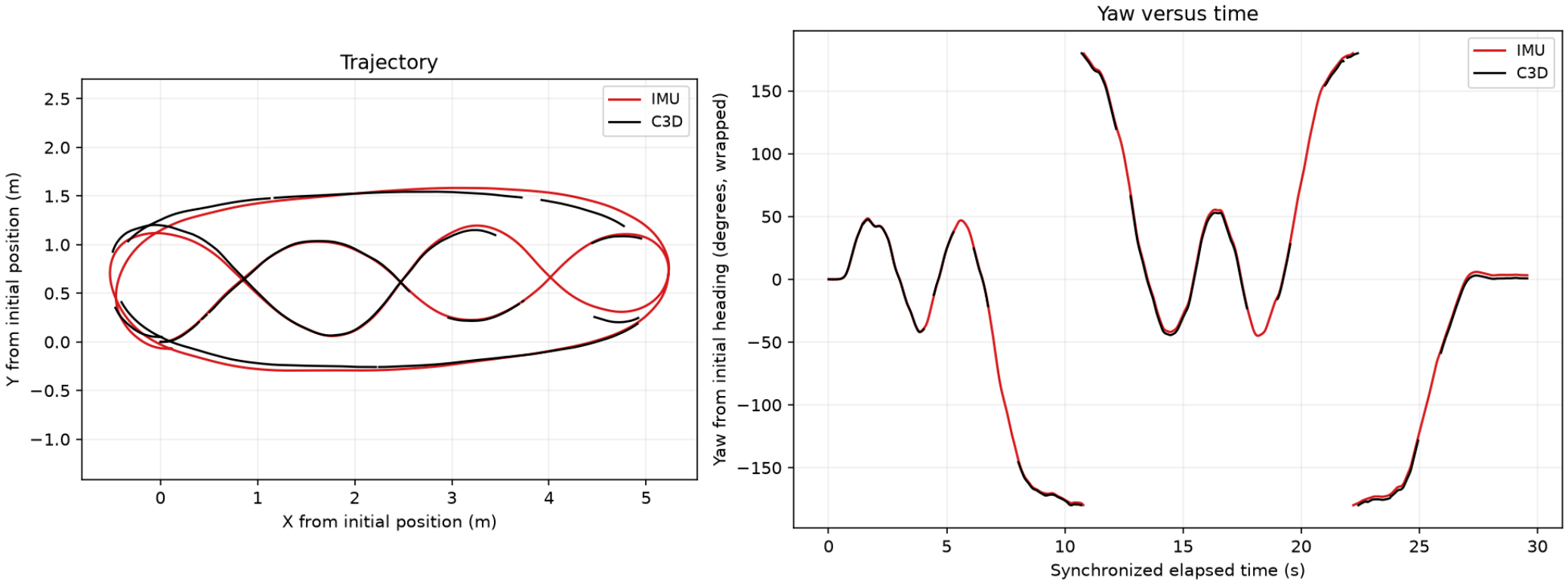
2026-09-24 | SL | SL_Trial1_No.5
Synchronized 2D RMSE: 117.6 mm



Supported-route metrics (mm): route RMSE 106.9; P95 201.7; sampled Hausdorff 248.5; max-run discrete Frechet 248.5; last-supported endpoint 42.3; absolute length error 52.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

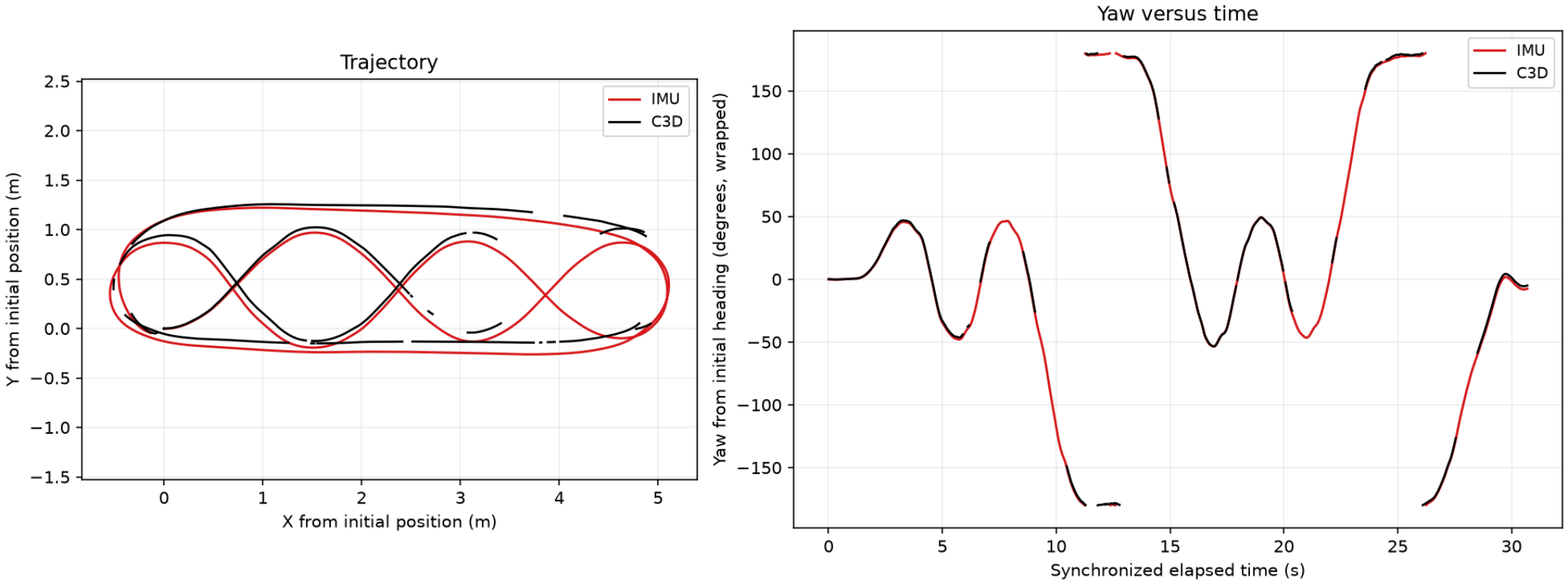
2026-09-24 | SL | SL_Trial2_No.10
Synchronized 2D RMSE: 83.5 mm



Supported-route metrics (mm): route RMSE 44.7; P95 97.9; sampled Hausdorff 139.1; max-run discrete Frechet 169.2; last-supported endpoint 165.2; absolute length error 19.4.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

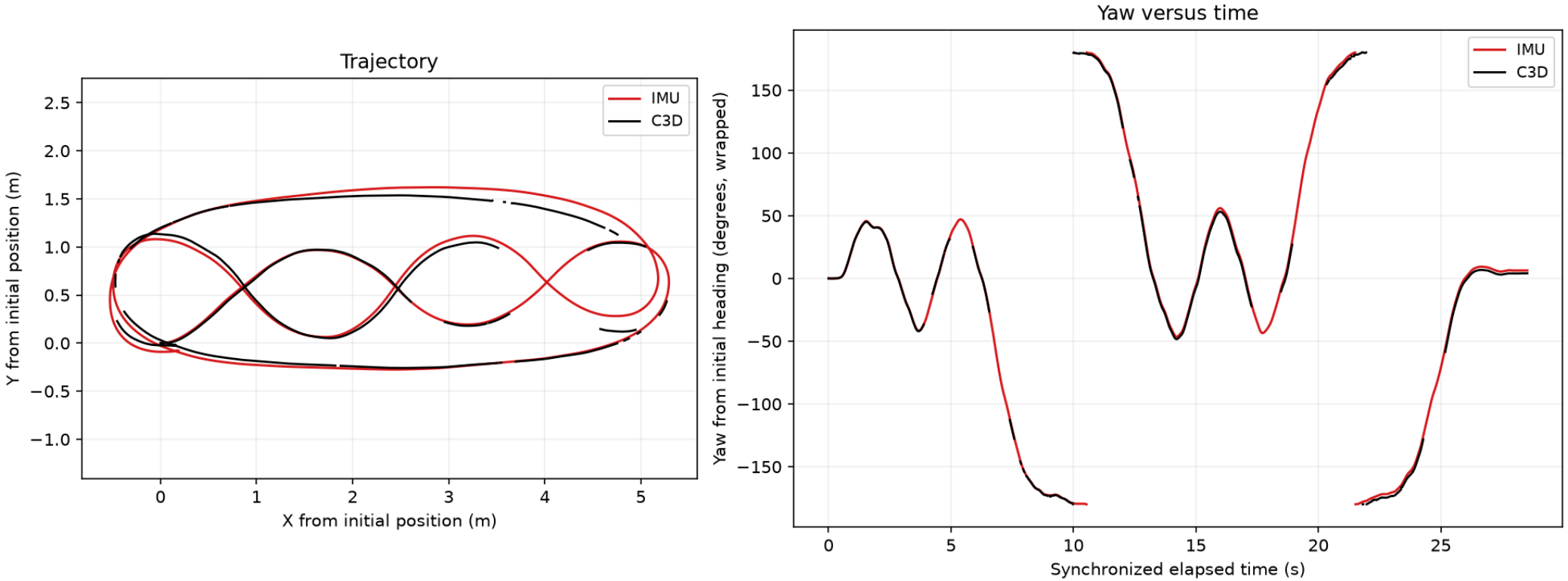
2026-09-24 | SL | SL_Trial2_No.5
Synchronized 2D RMSE: 79.9 mm



Supported-route metrics (mm): route RMSE 69.0; P95 117.8; sampled Hausdorff 143.2; max-run discrete Frechet 153.5; last-supported endpoint 58.4; absolute length error 141.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

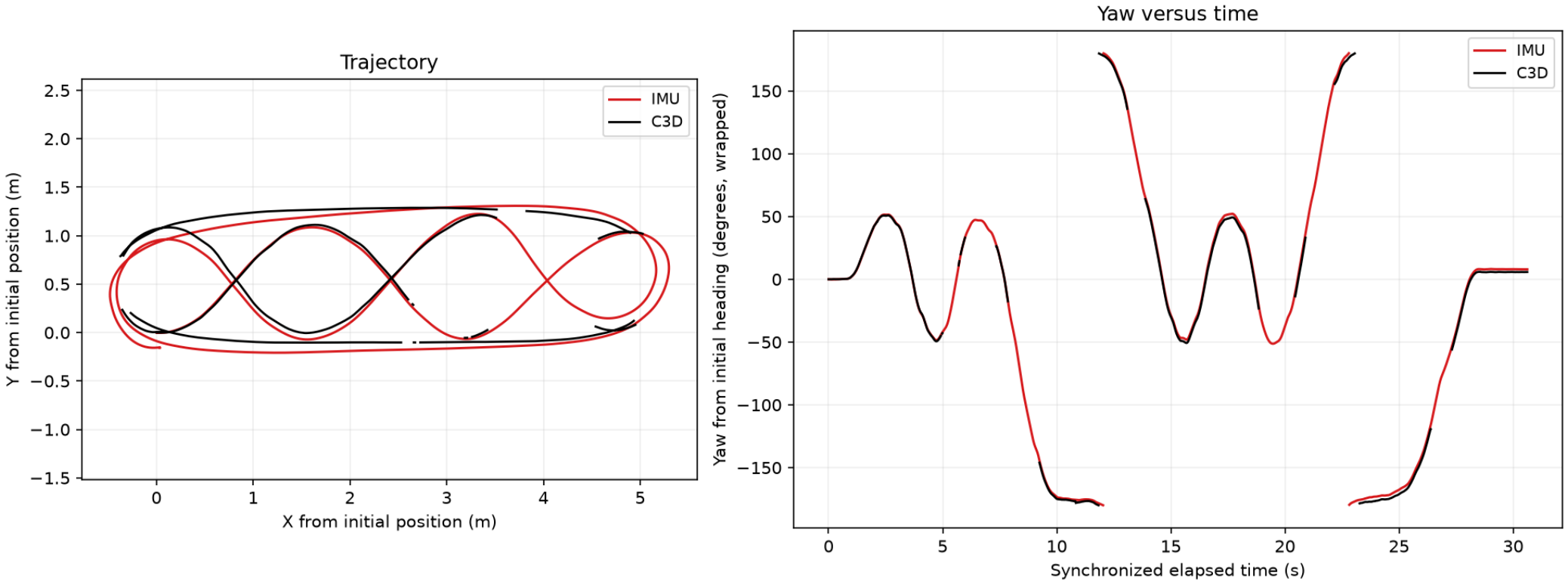
2026-09-24 | SL | SL_Trial3_No.10
Synchronized 2D RMSE: 77.6 mm



Supported-route metrics (mm): route RMSE 52.9; P95 131.9; sampled Hausdorff 183.5; max-run discrete Frechet 219.9; last-supported endpoint 67.5; absolute length error 19.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

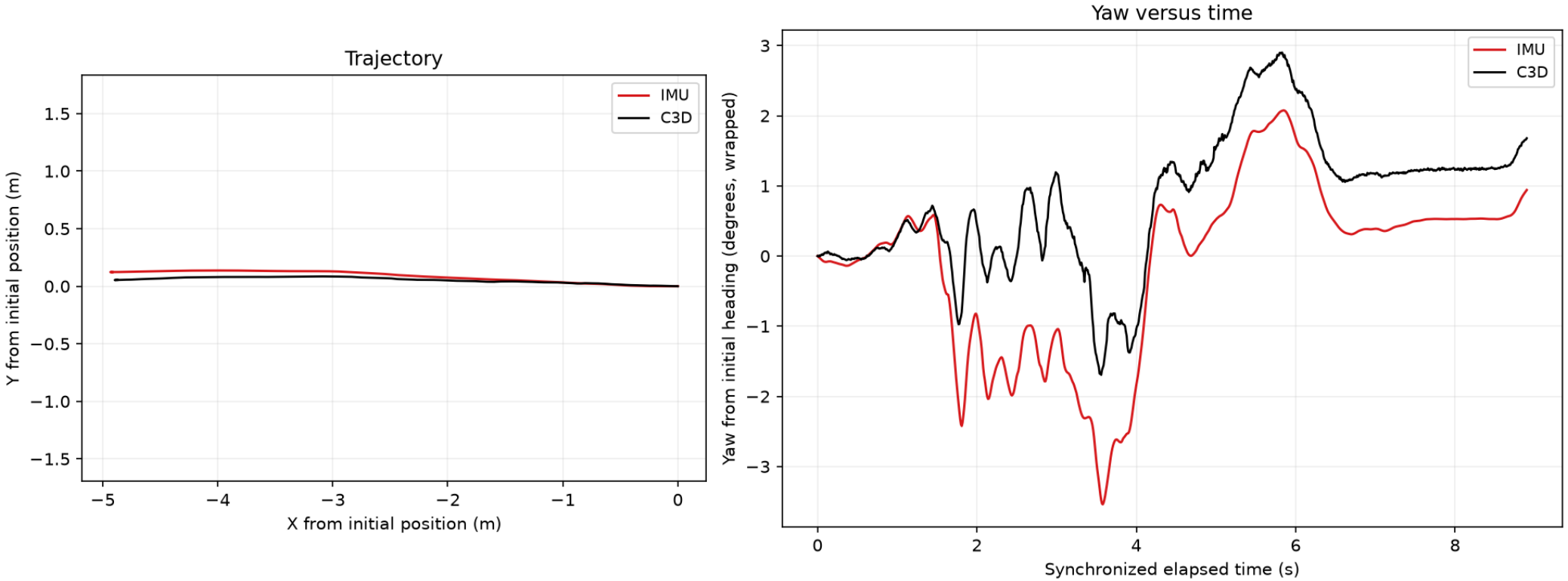
2026-09-24 | SL | SL_Trial3_No.5
Synchronized 2D RMSE: 110.2 mm



Supported-route metrics (mm): route RMSE 66.4; P95 120.7; sampled Hausdorff 189.6; max-run discrete Frechet 189.6; last-supported endpoint 165.9; absolute length error 101.3.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

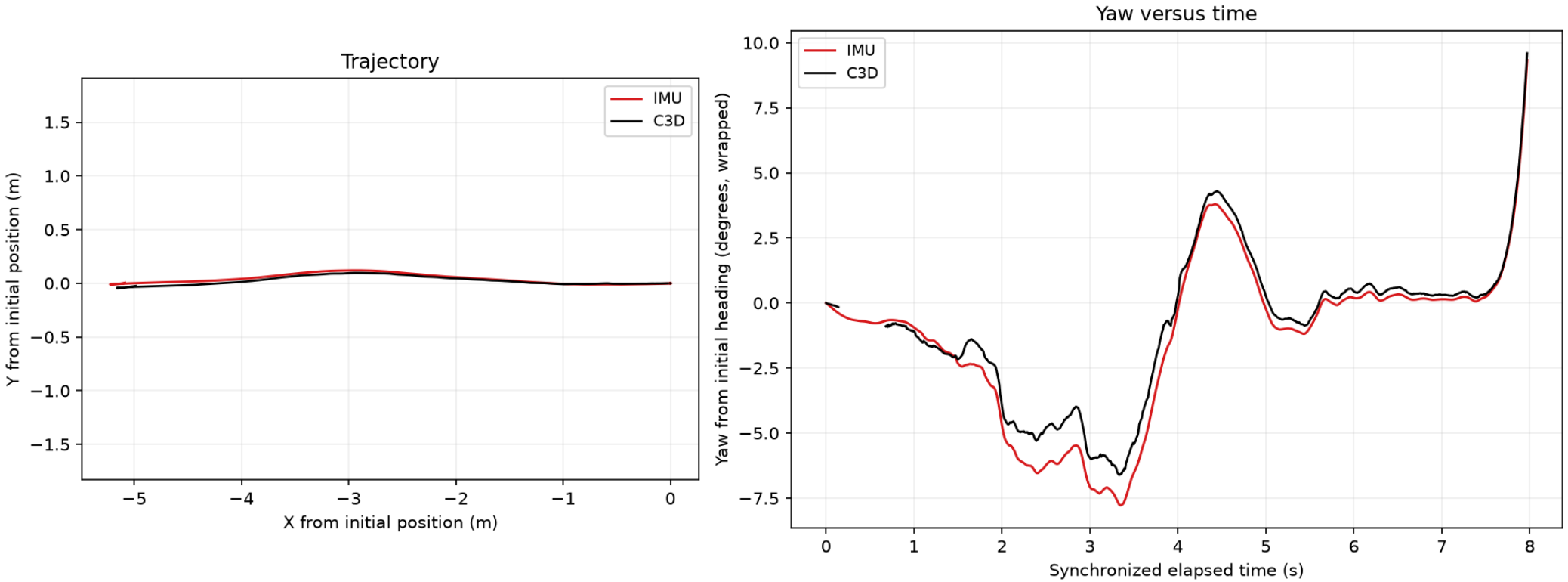
2026-09-24 | ST | ST_Trial1_No.10
Synchronized 2D RMSE: 61.3 mm



Supported-route metrics (mm): route RMSE 39.9; P95 66.5; sampled Hausdorff 76.9; max-run discrete Frechet 80.4; last-supported endpoint 80.4; absolute length error 35.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

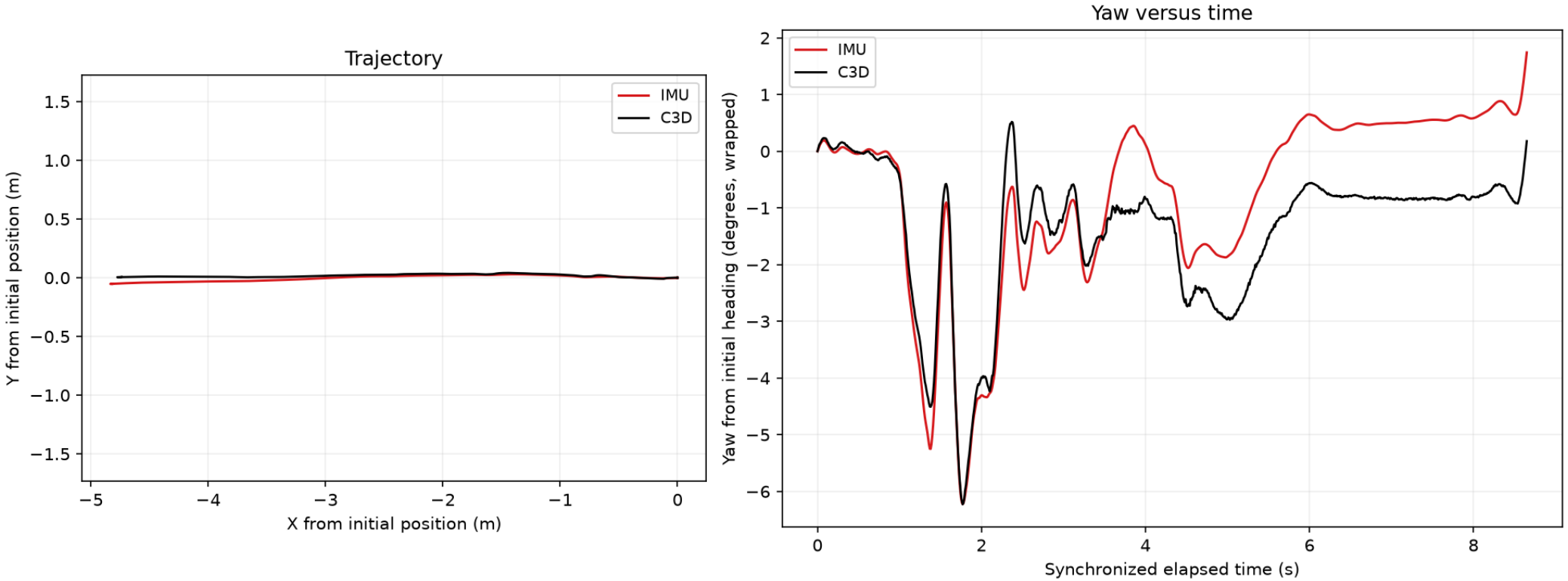
2026-09-24 | ST | ST_Trial1_No.5
Synchronized 2D RMSE: 50.4 mm



Supported-route metrics (mm): route RMSE 22.1; P95 35.8; sampled Hausdorff 70.6; max-run discrete Frechet 106.3; last-supported endpoint 106.3; absolute length error 37.4.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

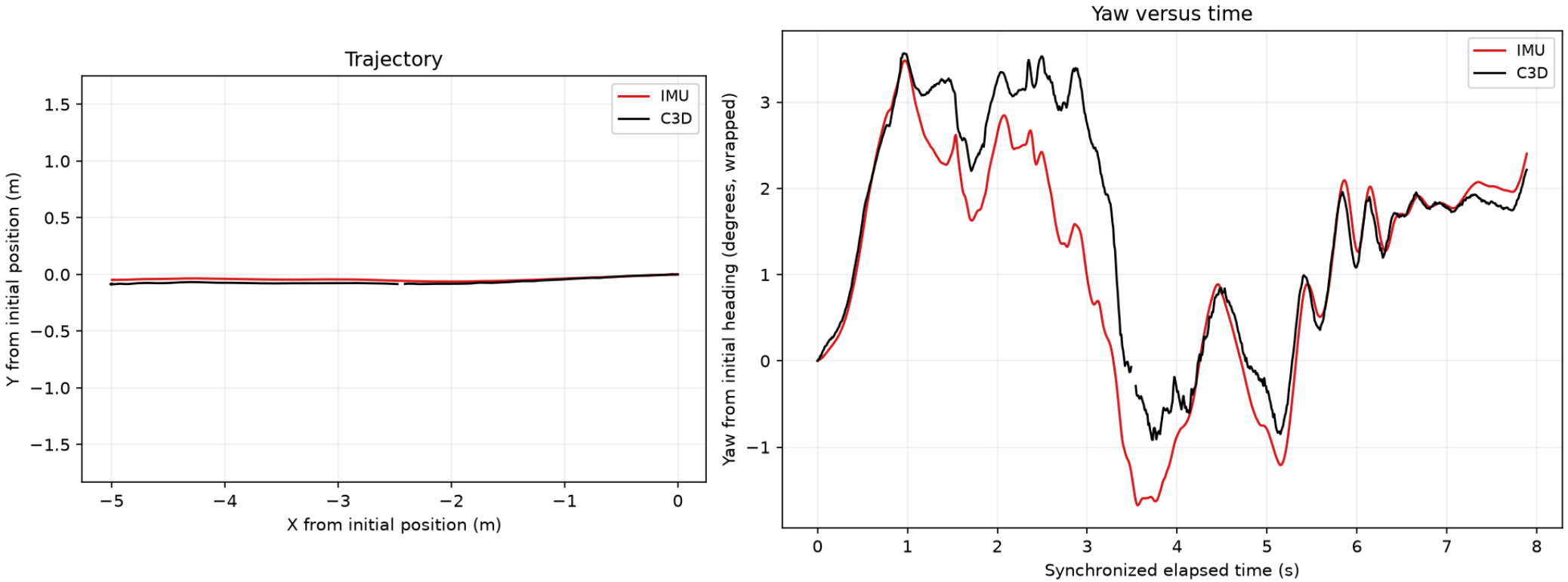
2026-09-24 | ST | ST_Trial2_No.10
Synchronized 2D RMSE: 54.1 mm



Supported-route metrics (mm): route RMSE 26.0; P95 50.9; sampled Hausdorff 78.8; max-run discrete Frechet 94.6; last-supported endpoint 94.6; absolute length error 56.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

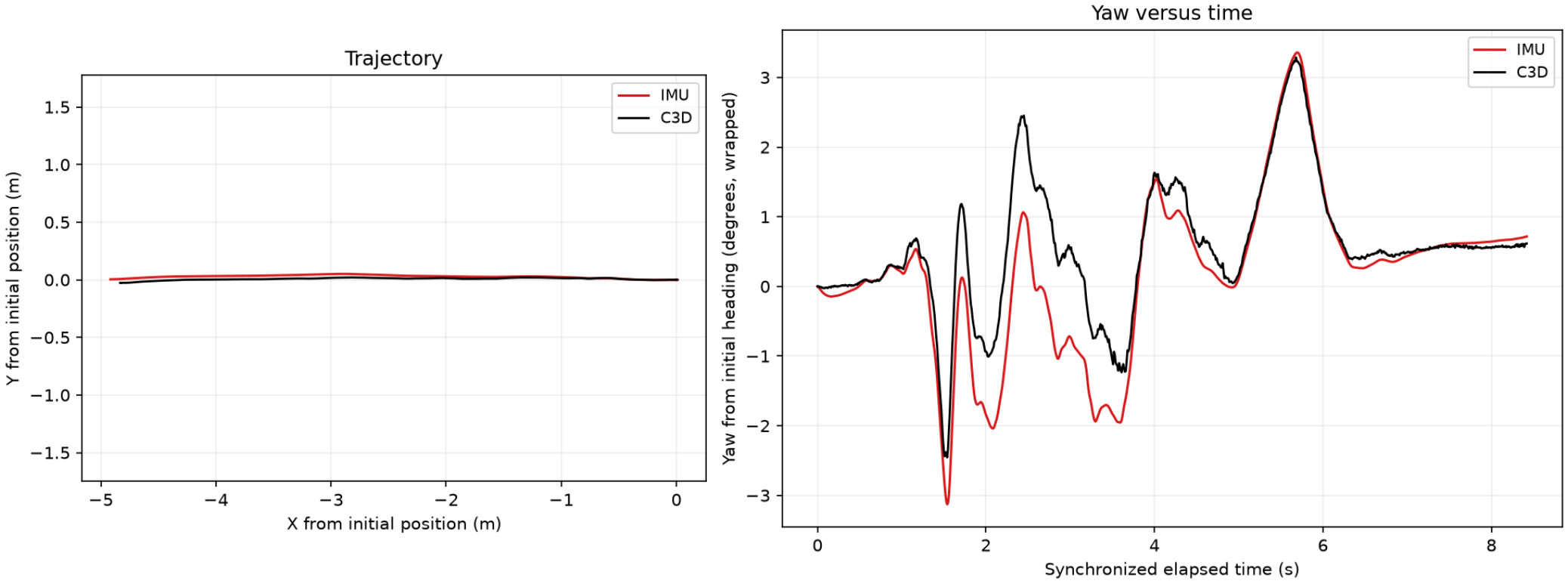
2026-09-24 | ST | ST_Trial2_No.5
Synchronized 2D RMSE: 55.5 mm



Supported-route metrics (mm): route RMSE 25.5; P95 36.5; sampled Hausdorff 38.5; max-run discrete Frechet 80.5; last-supported endpoint 38.1; absolute length error 24.4.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

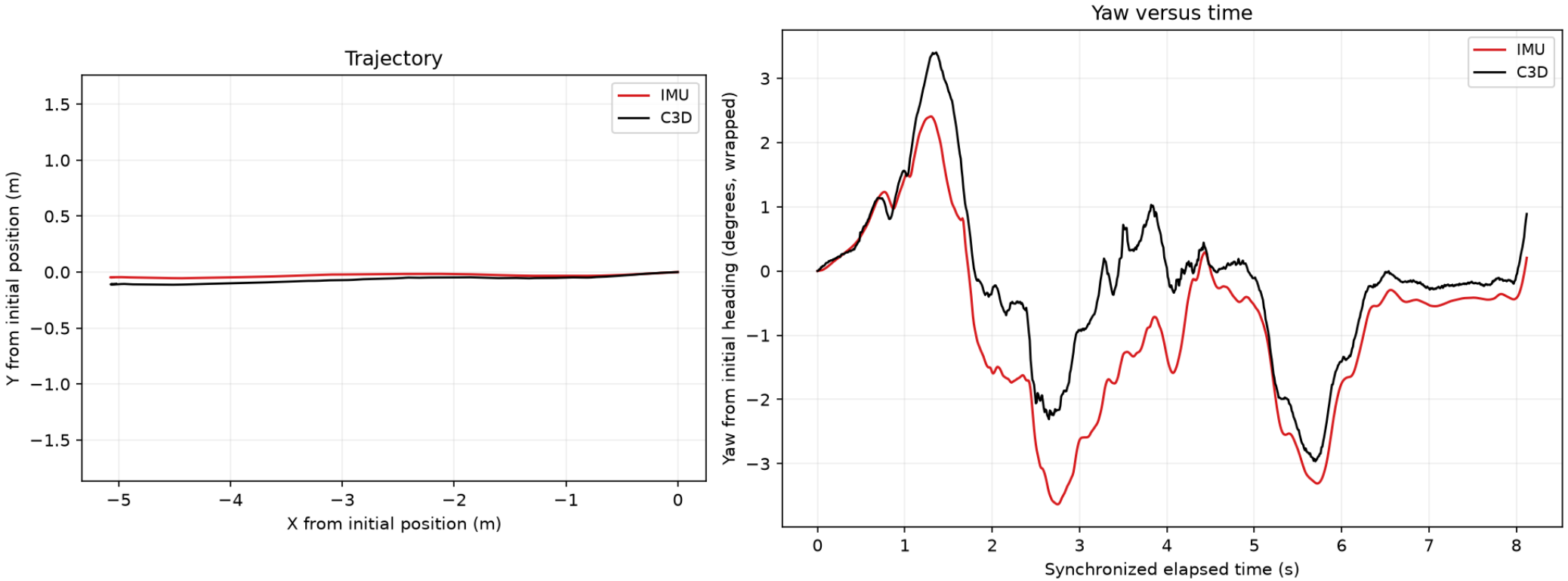
2026-09-24 | ST | ST_Trial3_No.10
Synchronized 2D RMSE: 56.2 mm



Supported-route metrics (mm): route RMSE 24.2; P95 33.7; sampled Hausdorff 88.0; max-run discrete Frechet 94.7; last-supported endpoint 94.7; absolute length error 77.6.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

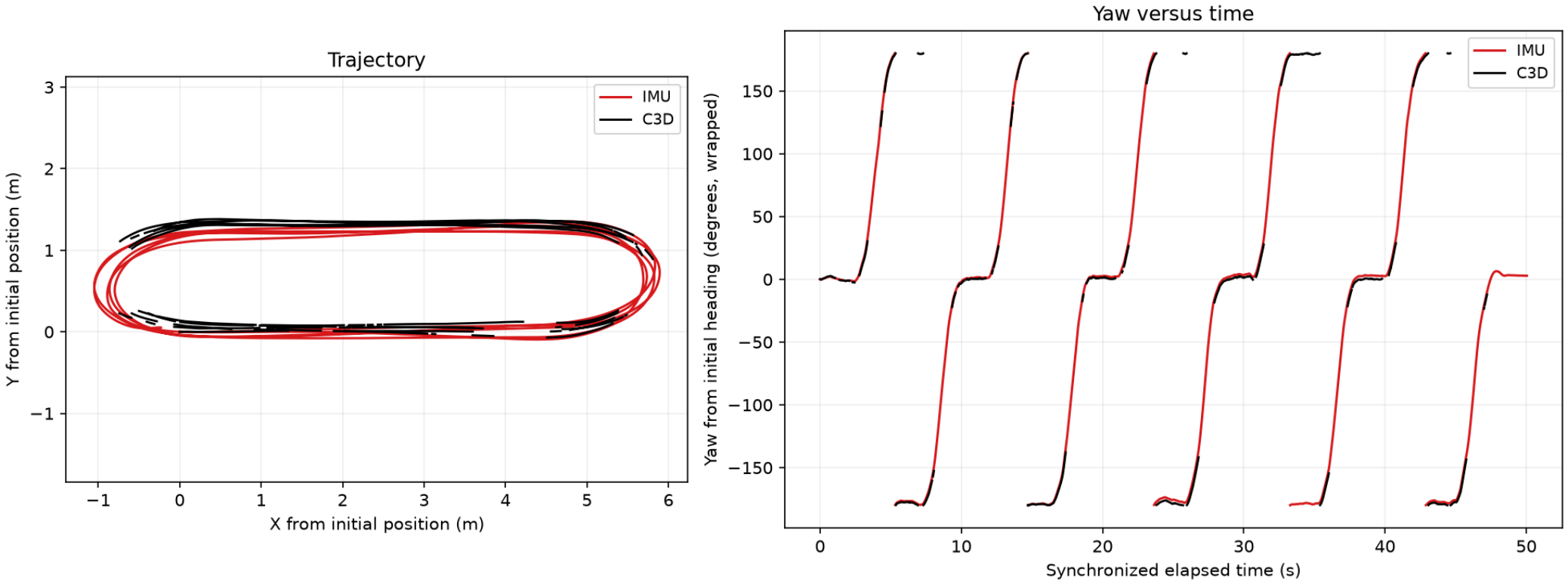
2026-09-24 | ST | ST_Trial3_No.5
Synchronized 2D RMSE: 73.2 mm



Supported-route metrics (mm): route RMSE 40.7; P95 60.9; sampled Hausdorff 62.8; max-run discrete Frechet 64.5; last-supported endpoint 64.5; absolute length error 19.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

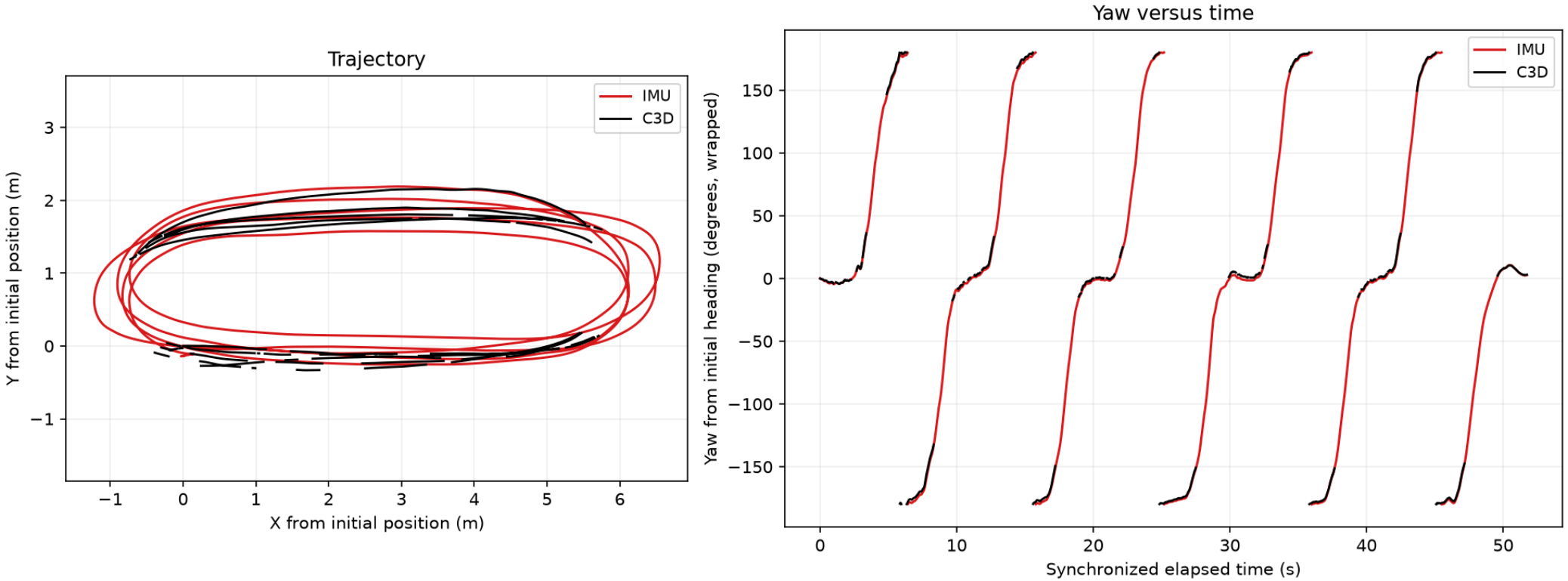
2026-09-25 | 10x5 | 10x5_Trial1_No.14
Synchronized 2D RMSE: 138.4 mm



Supported-route metrics (mm): route RMSE 59.1; P95 121.2; sampled Hausdorff 237.1; max-run discrete Frechet 238.8; last-supported endpoint 117.2; absolute length error 334.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

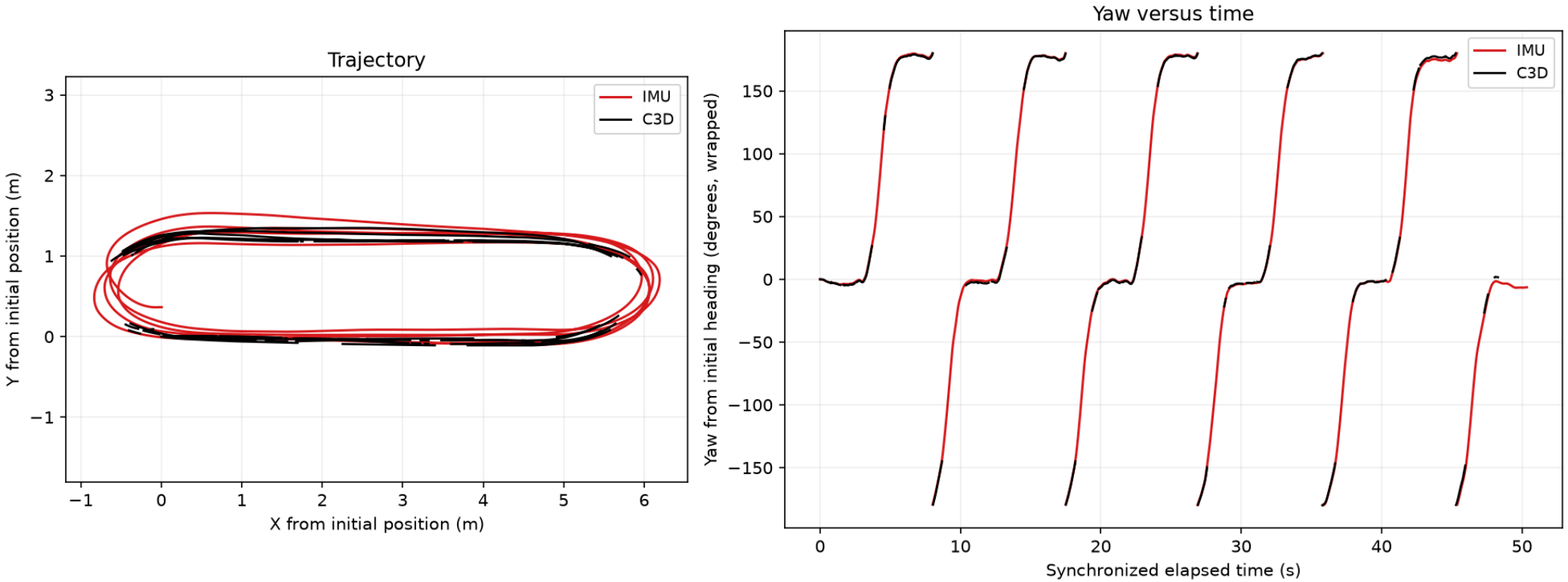
2026-09-25 | 10x5 | 10x5_Trial1_No.18
Synchronized 2D RMSE: 207.5 mm



Supported-route metrics (mm): route RMSE 79.8; P95 173.7; sampled Hausdorff 333.1; max-run discrete Frechet 345.0; last-supported endpoint 193.6; absolute length error 145.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

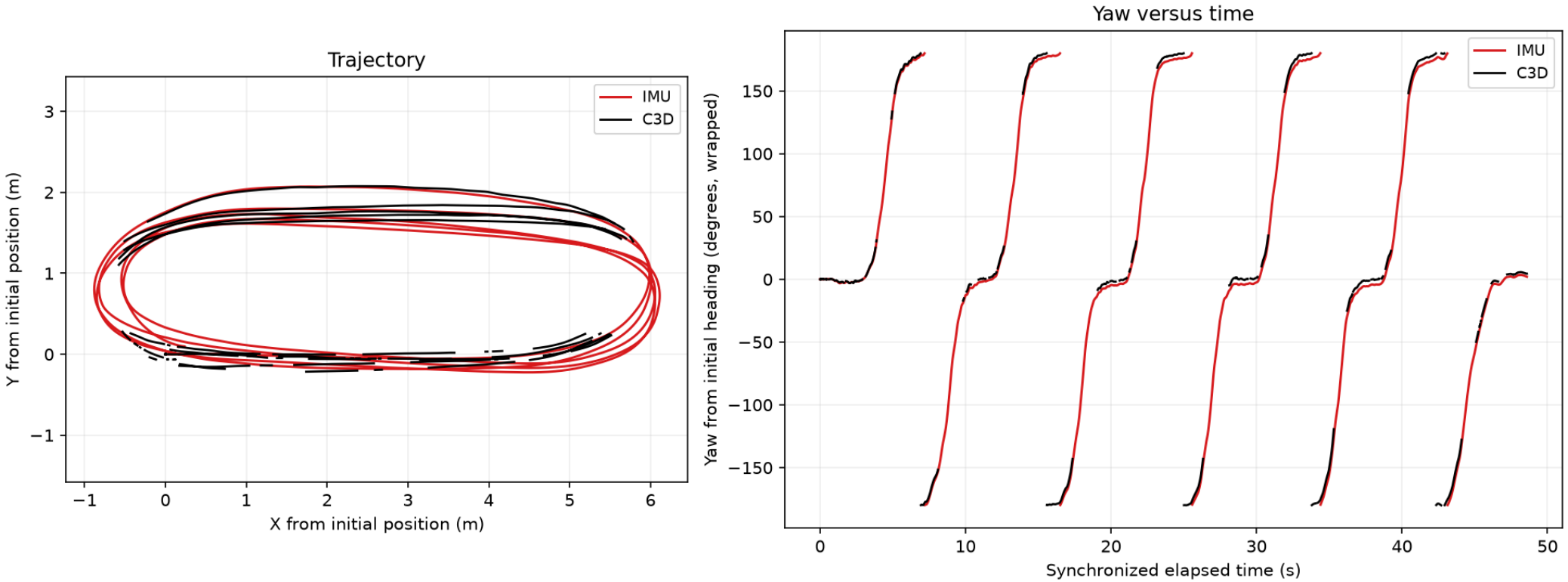
2026-09-25 | 10x5 | 10x5_Trial2_No.14
Synchronized 2D RMSE: 164.6 mm



Supported-route metrics (mm): route RMSE 49.5; P95 110.1; sampled Hausdorff 283.3; max-run discrete Frechet 304.0; last-supported endpoint 279.5; absolute length error 369.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

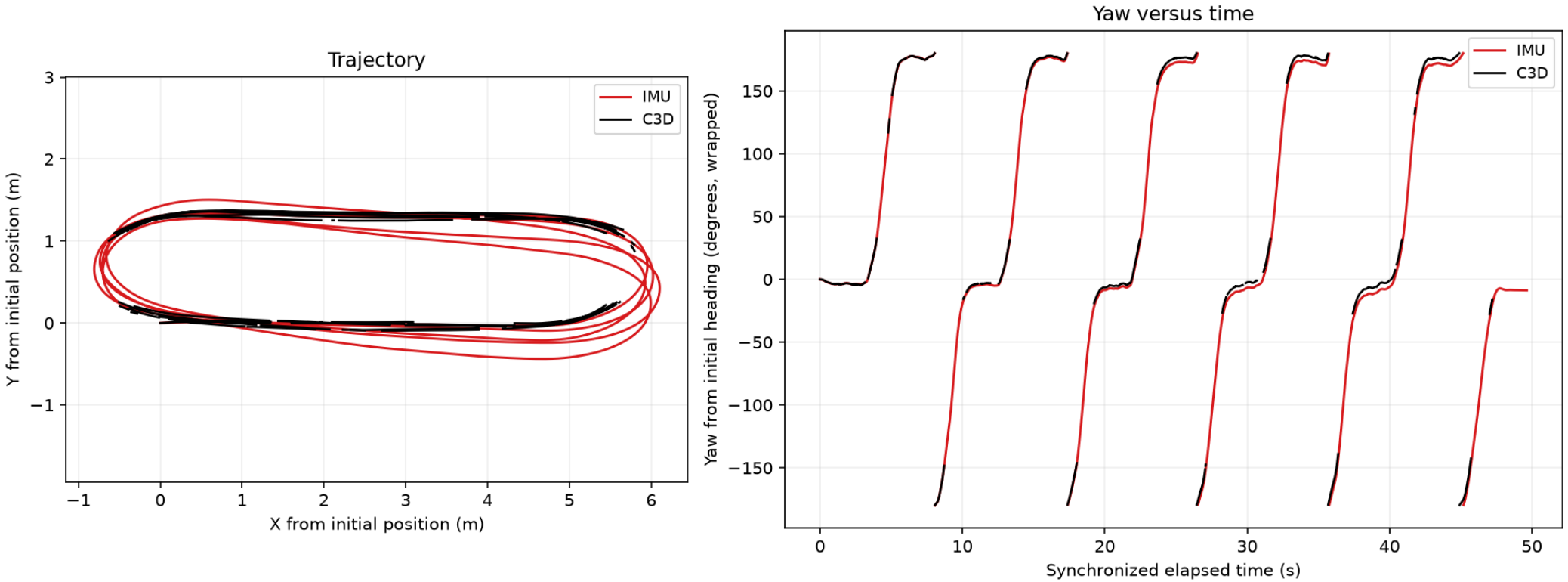
2026-09-25 | 10x5 | 10x5_Trial2_No.18
Synchronized 2D RMSE: 198.0 mm



Supported-route metrics (mm): route RMSE 72.3; P95 193.2; sampled Hausdorff 328.6; max-run discrete Frechet 392.6; last-supported endpoint 205.8; absolute length error 295.7.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

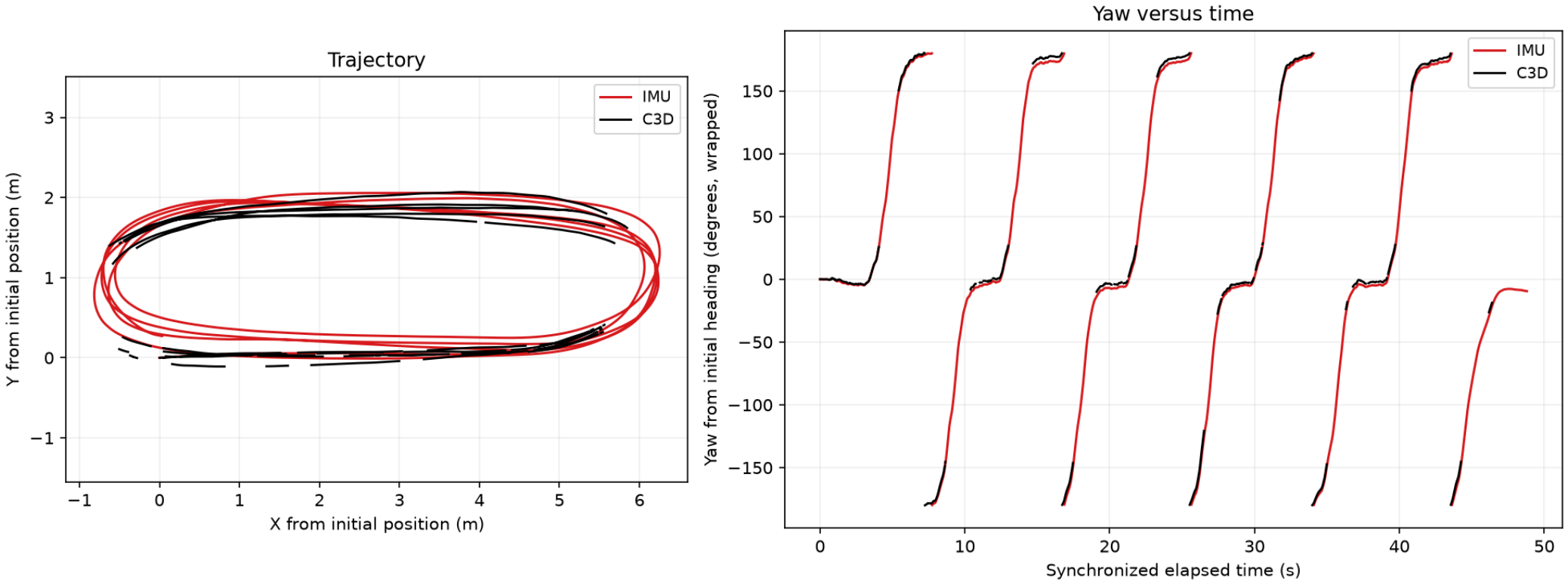
2026-09-25 | 10x5 | 10x5_Trial3_No.14
Synchronized 2D RMSE: 187.9 mm



Supported-route metrics (mm): route RMSE 95.0; P95 242.6; sampled Hausdorff 447.4; max-run discrete Frechet 525.9; last-supported endpoint 40.0; absolute length error 562.7.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

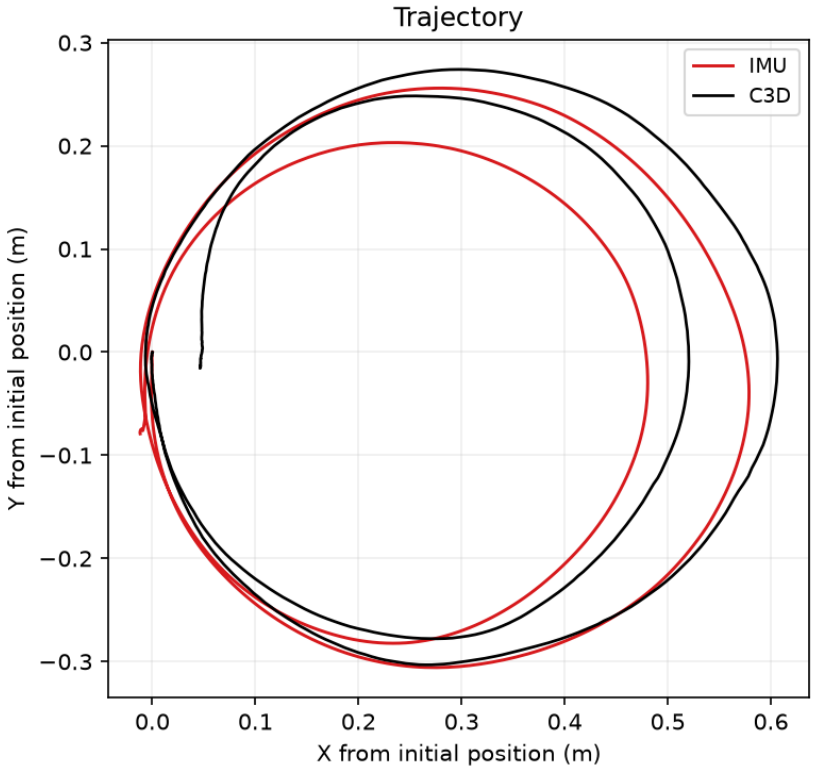
2026-09-25 | 10x5 | 10x5_Trial3_No.18
Synchronized 2D RMSE: 228.3 mm

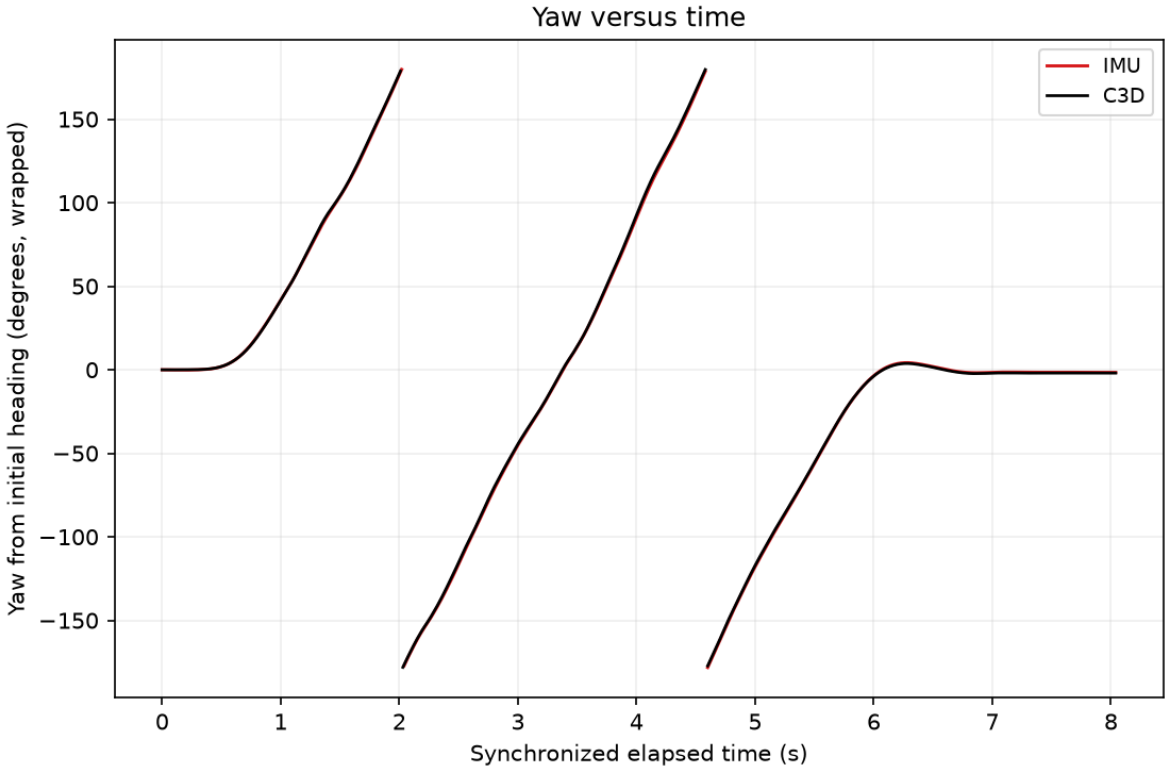


Supported-route metrics (mm): route RMSE 83.5; P95 176.4; sampled Hausdorff 399.7; max-run discrete Frechet 400.3; last-supported endpoint 360.0; absolute length error 408.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

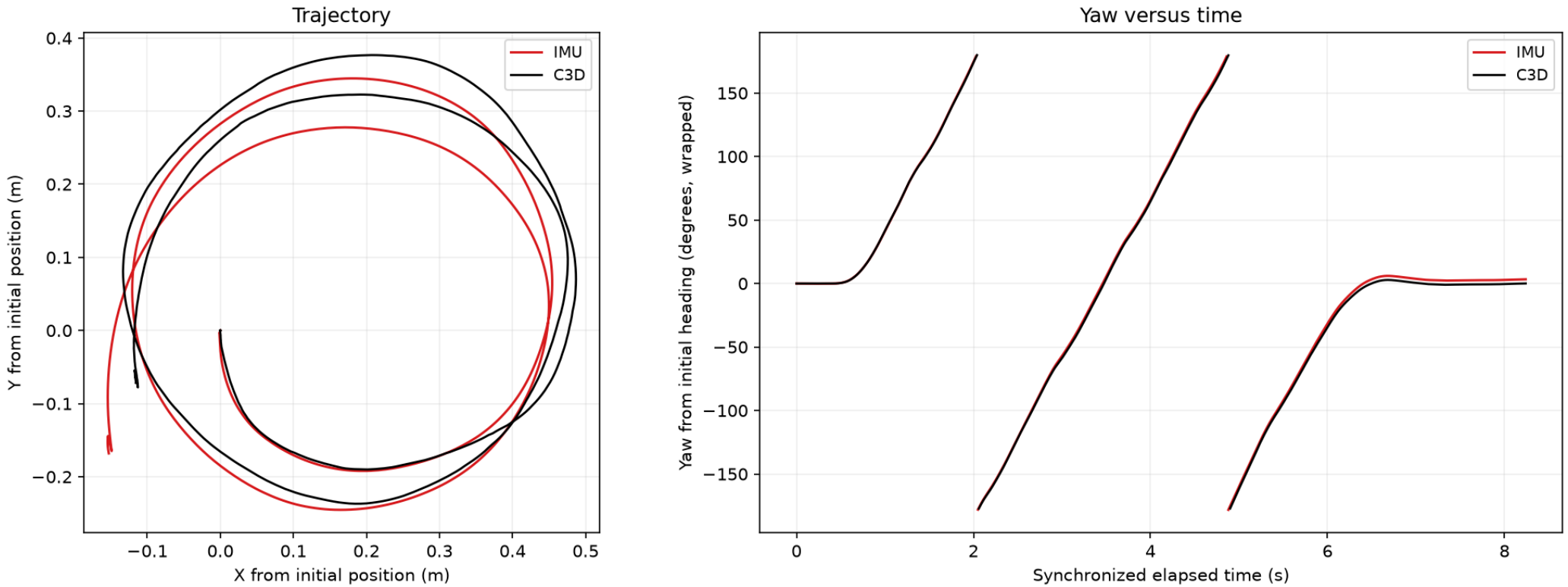
2026-09-25 | AR | AR_Trial1_No.14
Synchronized 2D RMSE: 52.2 mm





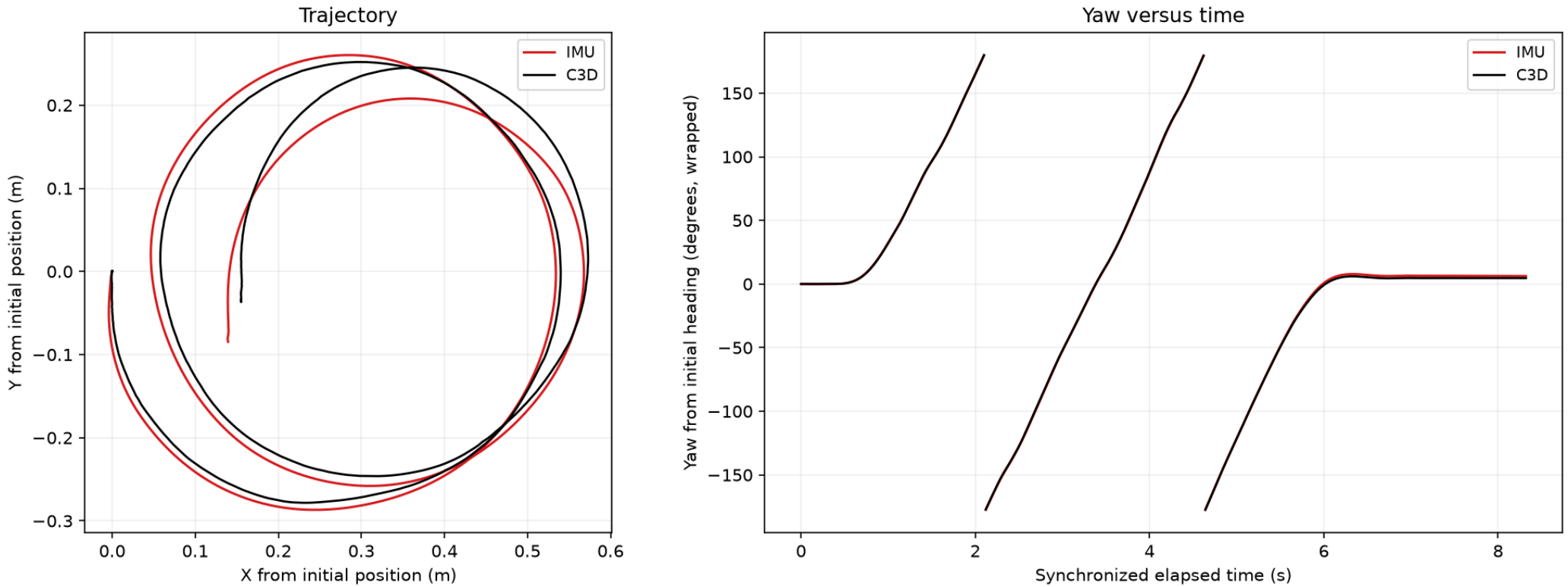
Supported-route metrics (mm): route RMSE 20.8; P95 45.4; sampled Hausdorff 56.9; max-run discrete Frechet 86.4; last-supported endpoint 86.4; absolute length error 96.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



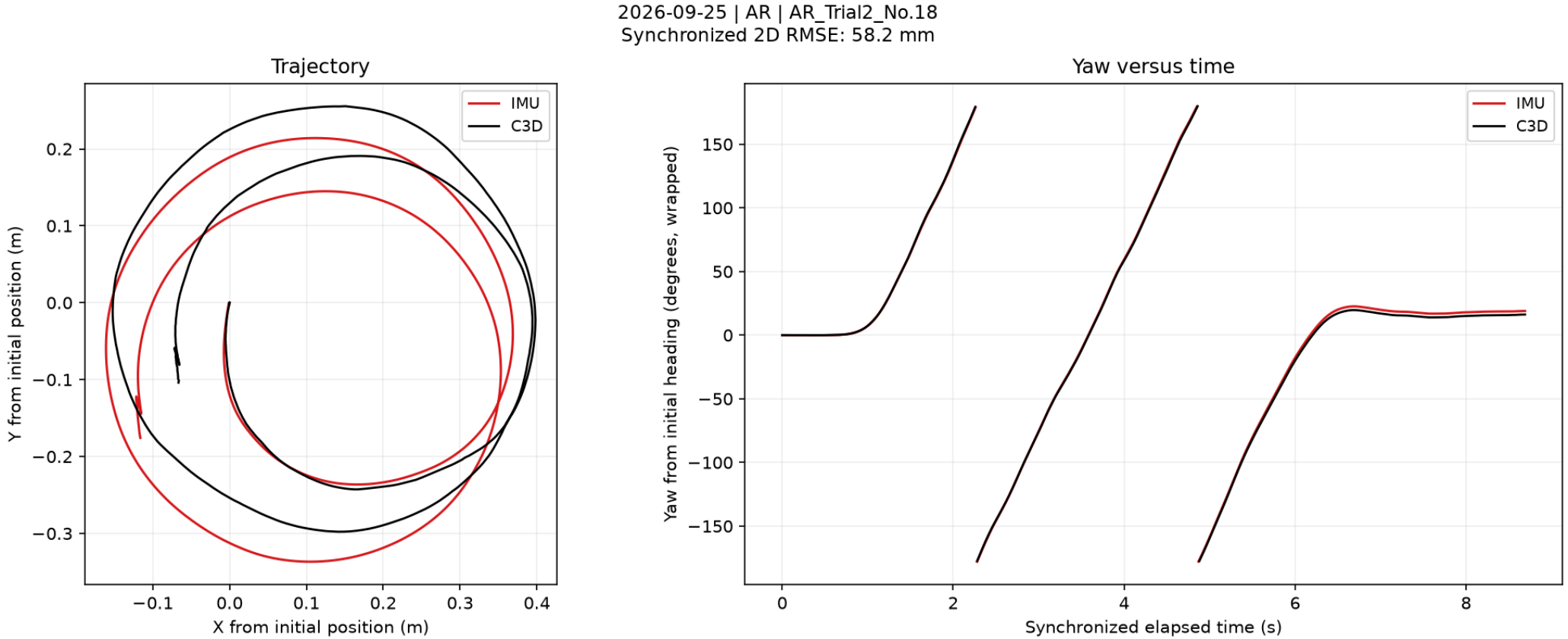
Supported-route metrics (mm): route RMSE 22.2; P95 46.5; sampled Hausdorff 98.7; max-run discrete Frechet 103.3; last-supported endpoint 103.3; absolute length error 97.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 11.8; P95 25.8; sampled Hausdorff 50.6; max-run discrete Frechet 50.6; last-supported endpoint 50.6; absolute length error 26.1.

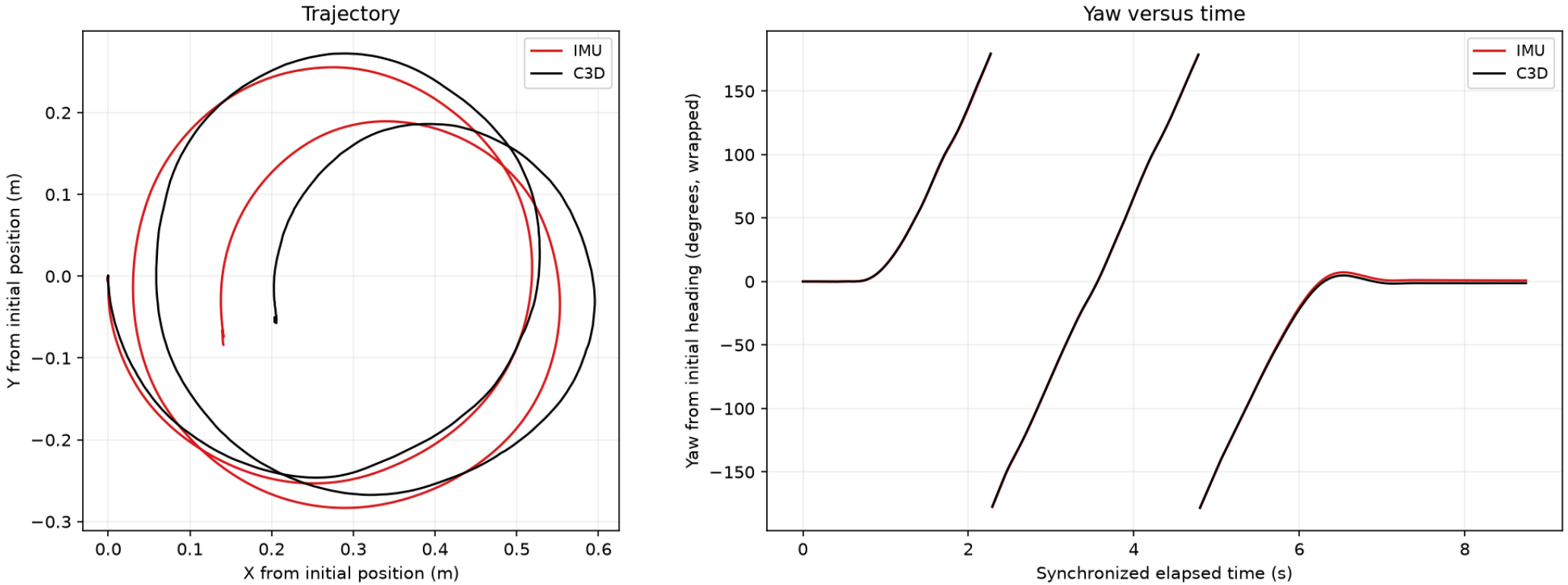
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 28.3; P95 51.9; sampled Hausdorff 60.1; max-run discrete Frechet 87.3; last-supported endpoint 87.3; absolute length error 68.7.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

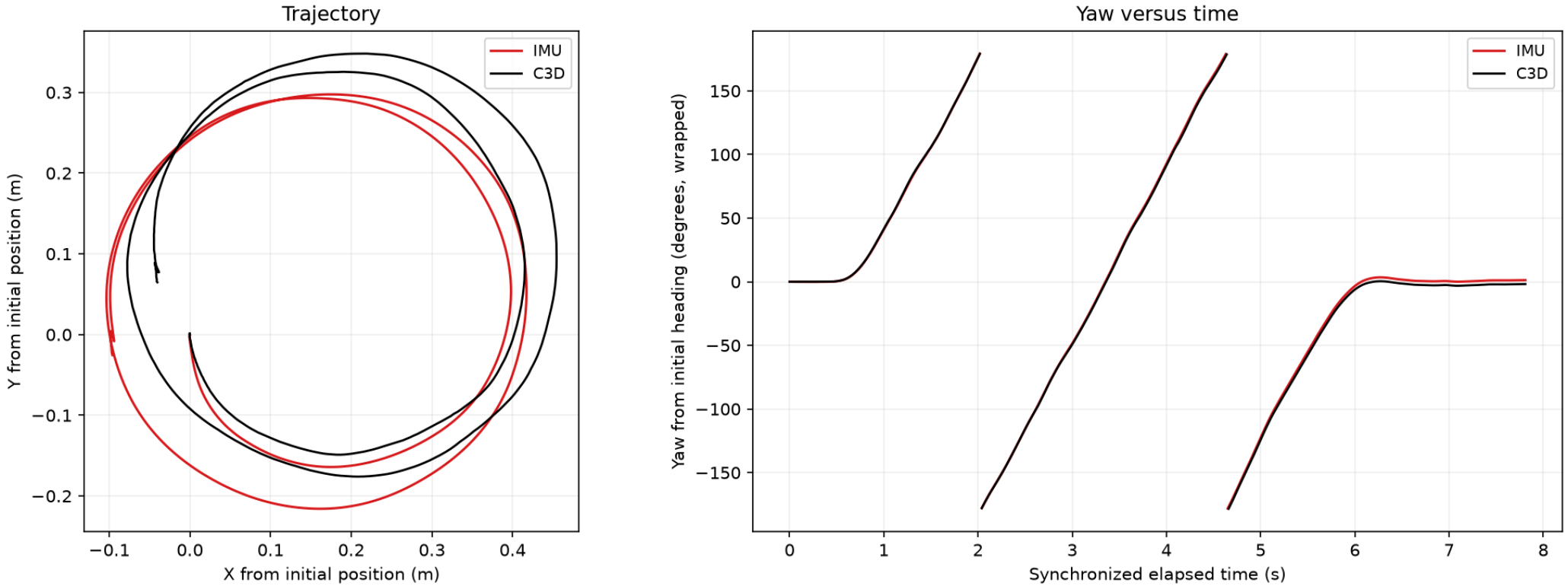
2026-09-25 | AR | AR_Trial3_No.14
Synchronized 2D RMSE: 46.0 mm



Supported-route metrics (mm): route RMSE 23.8; P95 57.7; sampled Hausdorff 66.7; max-run discrete Frechet 69.0; last-supported endpoint 69.0; absolute length error 54.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

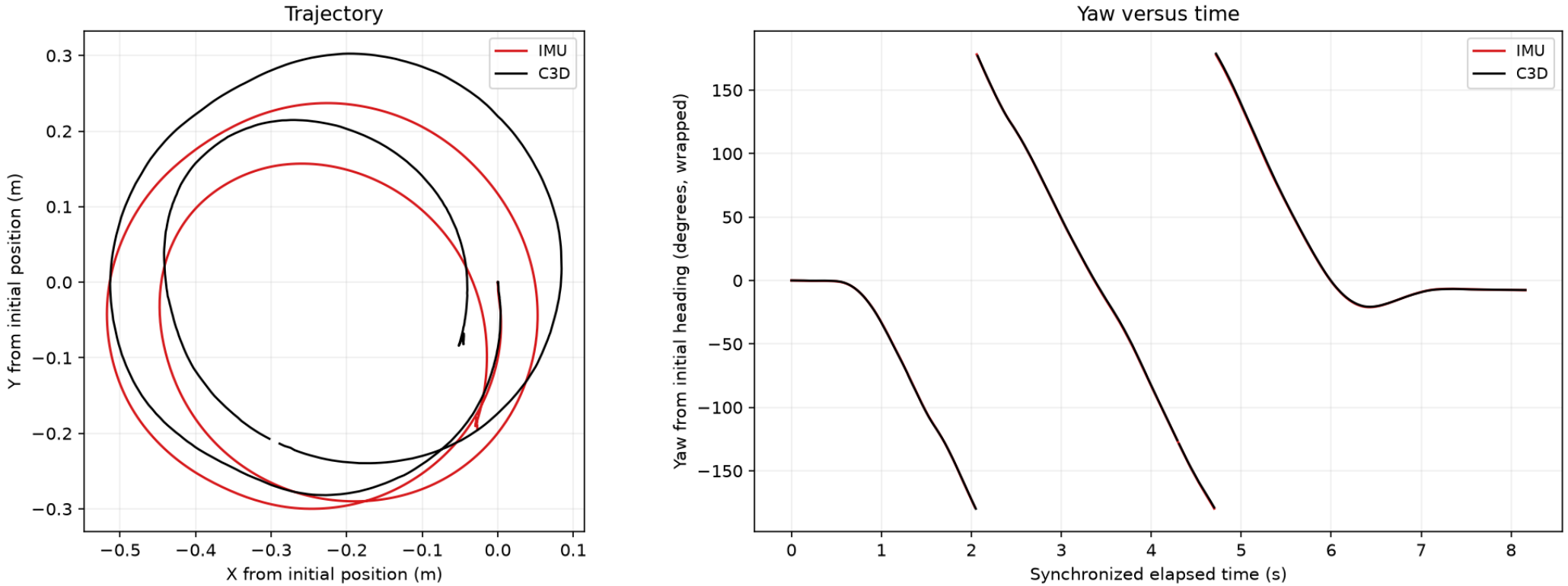
2026-09-25 | AR | AR_Trial3_No.18
Synchronized 2D RMSE: 64.6 mm



Supported-route metrics (mm): route RMSE 27.3; P95 55.7; sampled Hausdorff 58.8; max-run discrete Frechet 106.0; last-supported endpoint 106.0; absolute length error 33.3.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

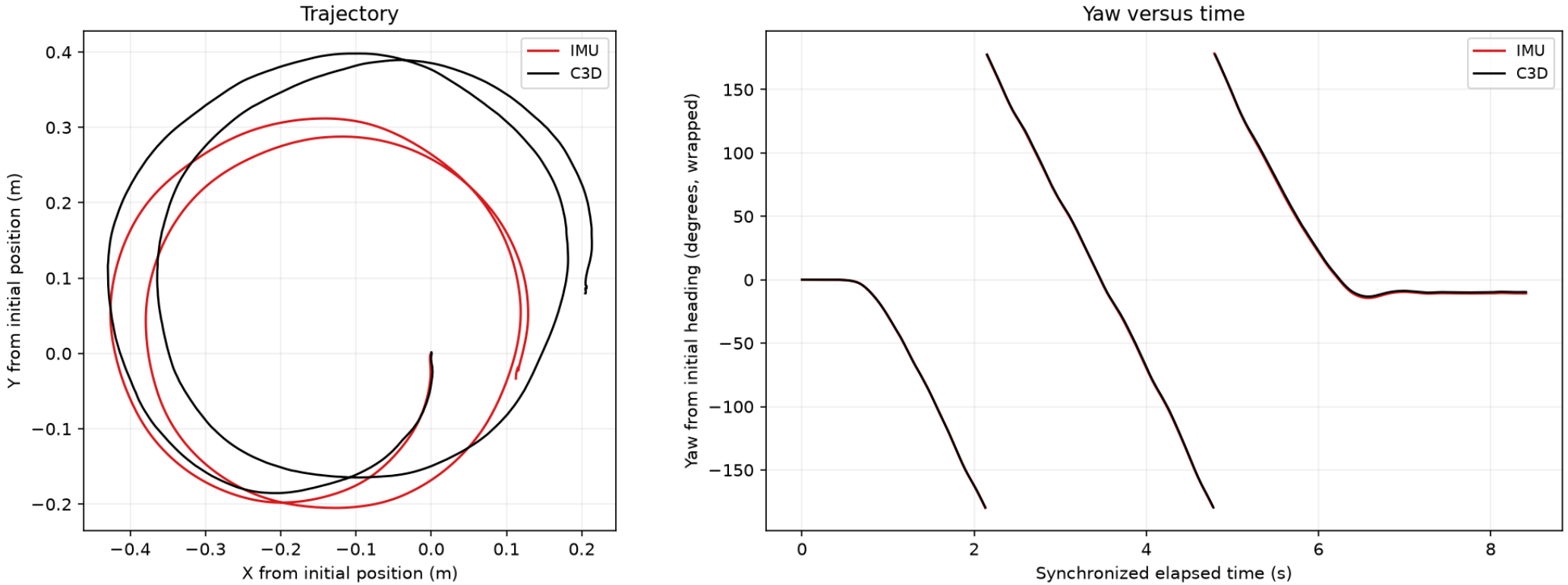
2026-09-25 | CR | CR_Trial1_No.14
Synchronized 2D RMSE: 72.1 mm



Supported-route metrics (mm): route RMSE 31.3; P95 63.3; sampled Hausdorff 73.2; max-run discrete Frechet 117.8; last-supported endpoint 117.8; absolute length error 22.8.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

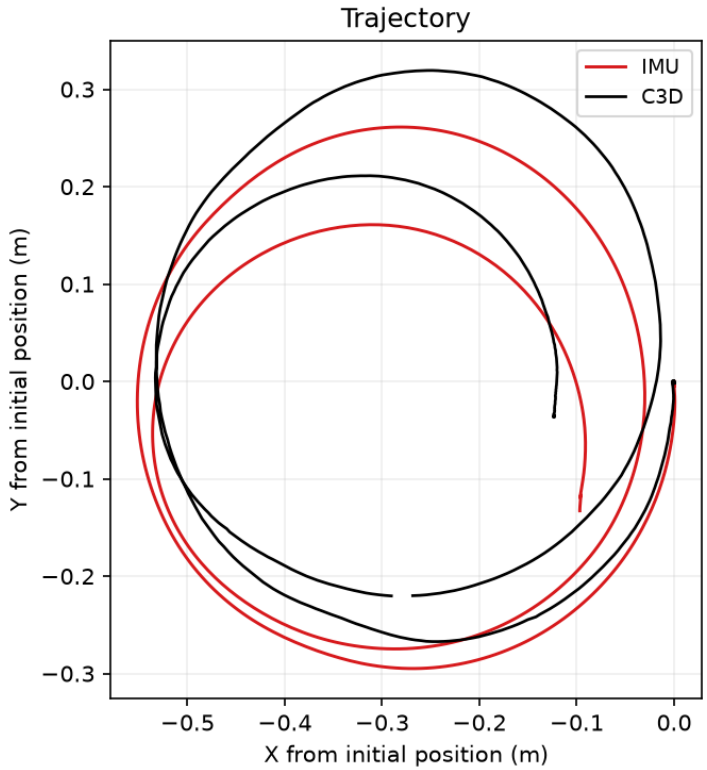
2026-09-25 | CR | CR_Trial1_No.18
Synchronized 2D RMSE: 93.6 mm

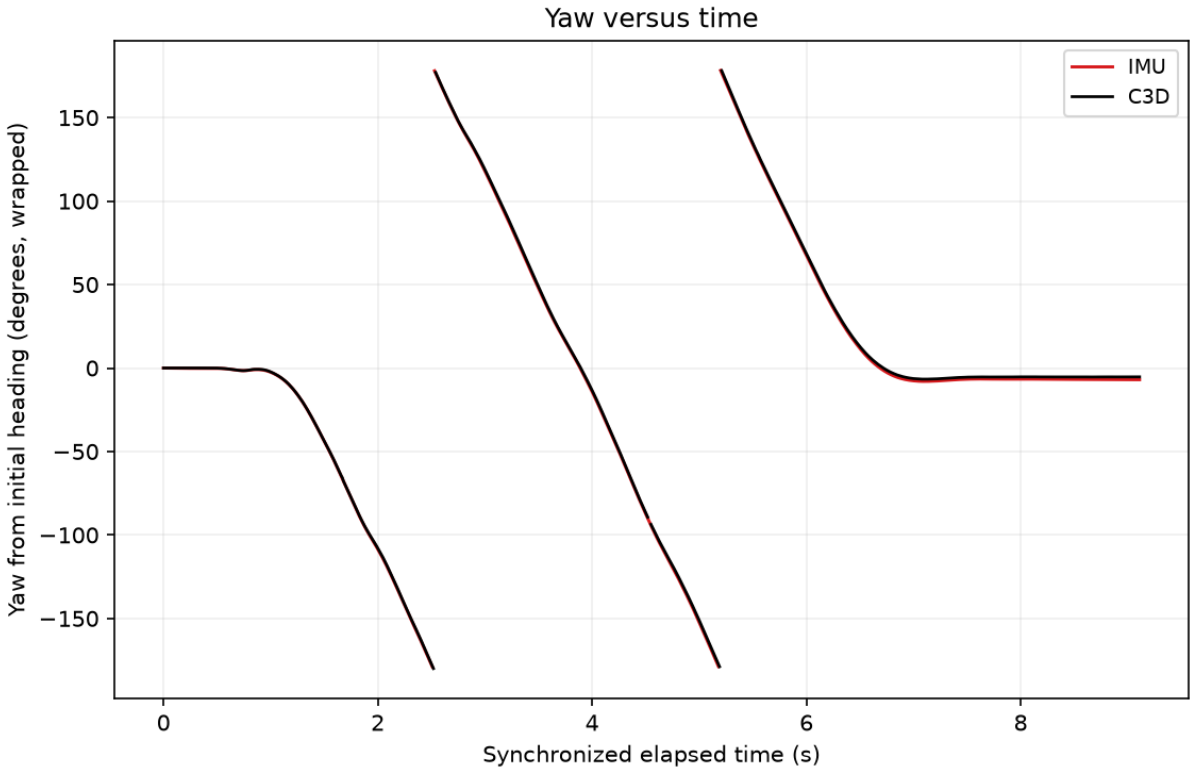


Supported-route metrics (mm): route RMSE 54.6; P95 106.8; sampled Hausdorff 127.2; max-run discrete Frechet 145.9; last-supported endpoint 145.9; absolute length error 273.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-25 | CR | CR_Trial2_No.14
Synchronized 2D RMSE: 63.8 mm

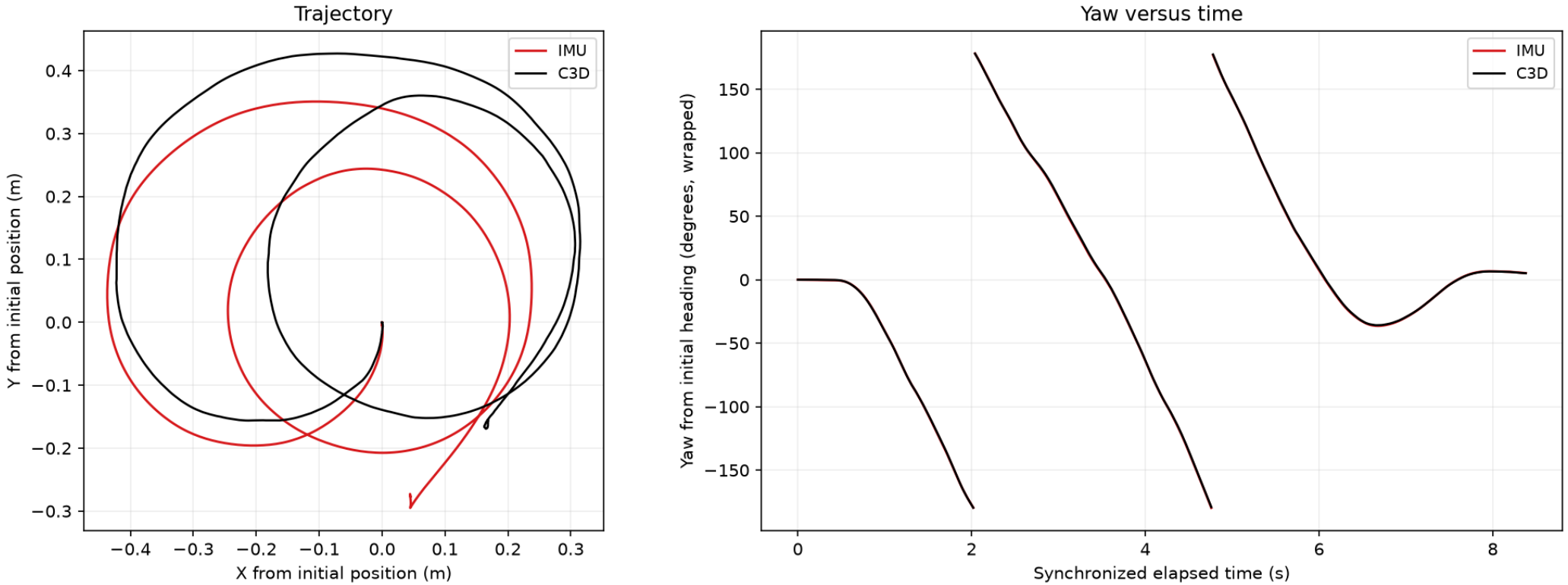




Supported-route metrics (mm): route RMSE 31.5; P95 59.0; sampled Hausdorff 64.0; max-run discrete Frechet 101.1; last-supported endpoint 101.1; absolute length error 64.6.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

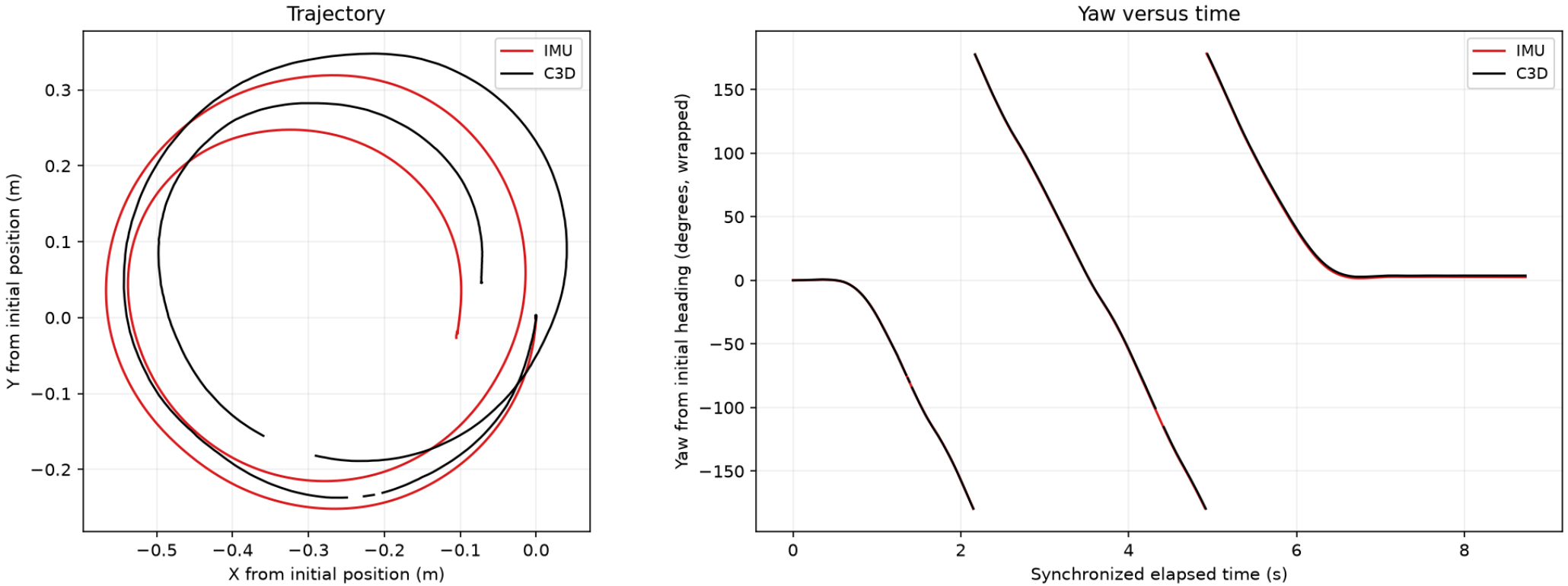
2026-09-25 | CR | CR_Trial2_No.18
Synchronized 2D RMSE: 117.2 mm



Supported-route metrics (mm): route RMSE 59.0; P95 104.7; sampled Hausdorff 148.8; max-run discrete Frechet 173.6; last-supported endpoint 170.7; absolute length error 140.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

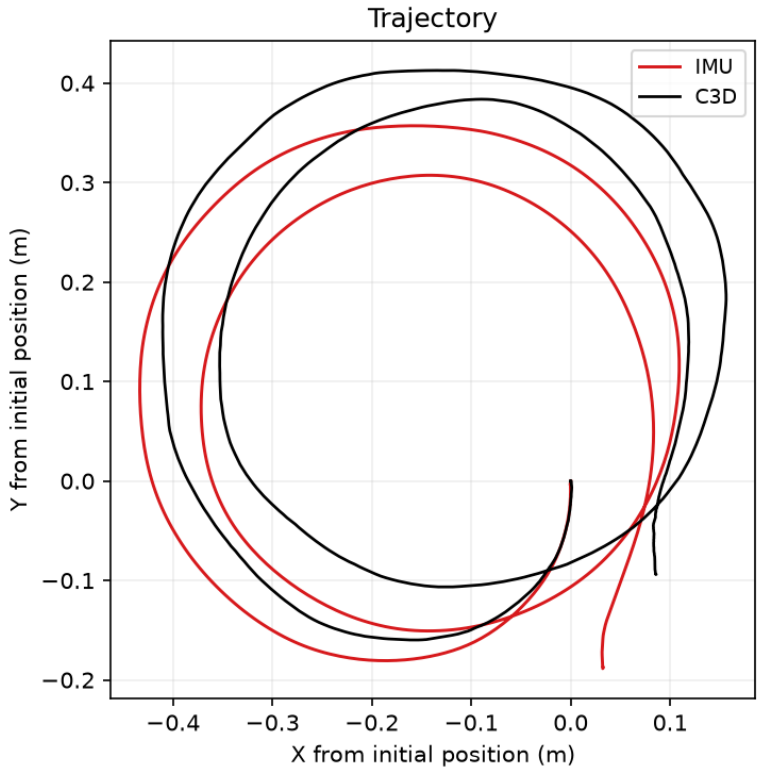
2026-09-25 | CR | CR_Trial3_No.14
Synchronized 2D RMSE: 53.6 mm

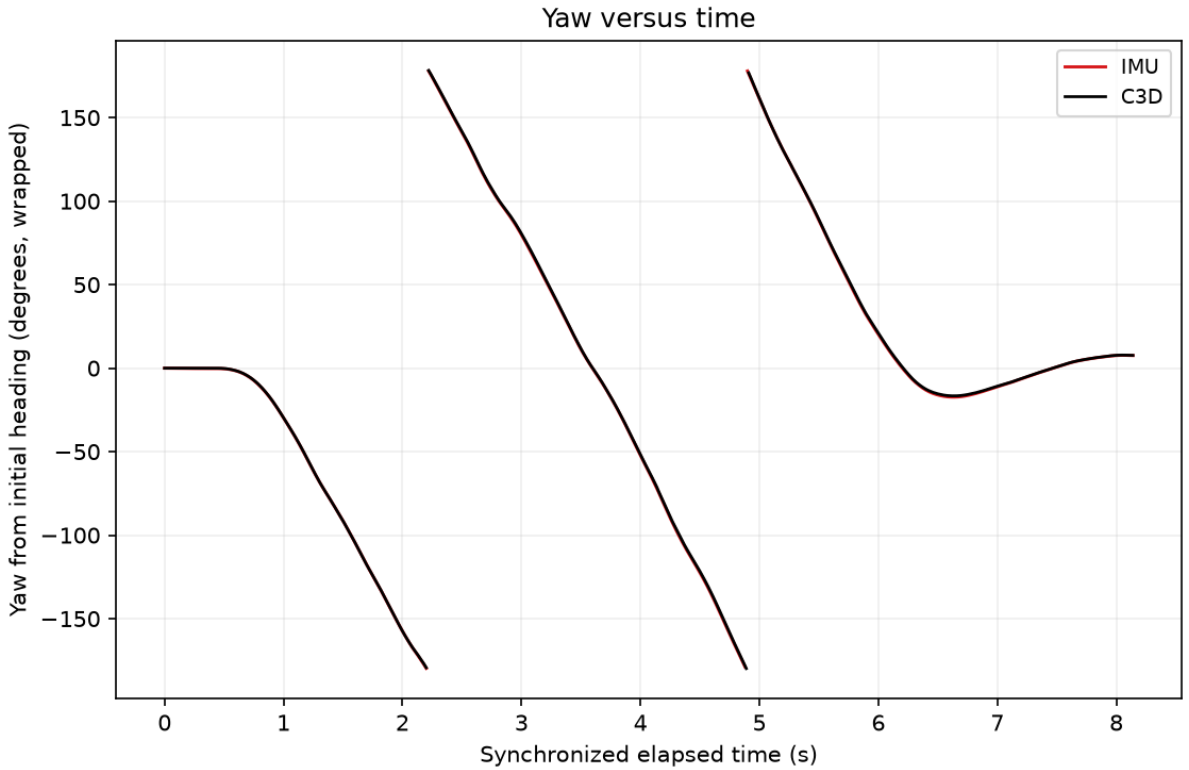


Supported-route metrics (mm): route RMSE 29.4; P95 54.9; sampled Hausdorff 79.6; max-run discrete Frechet 80.3; last-supported endpoint 80.3; absolute length error 8.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-25 | CR | CR_Trial3_No.18
Synchronized 2D RMSE: 74.8 mm

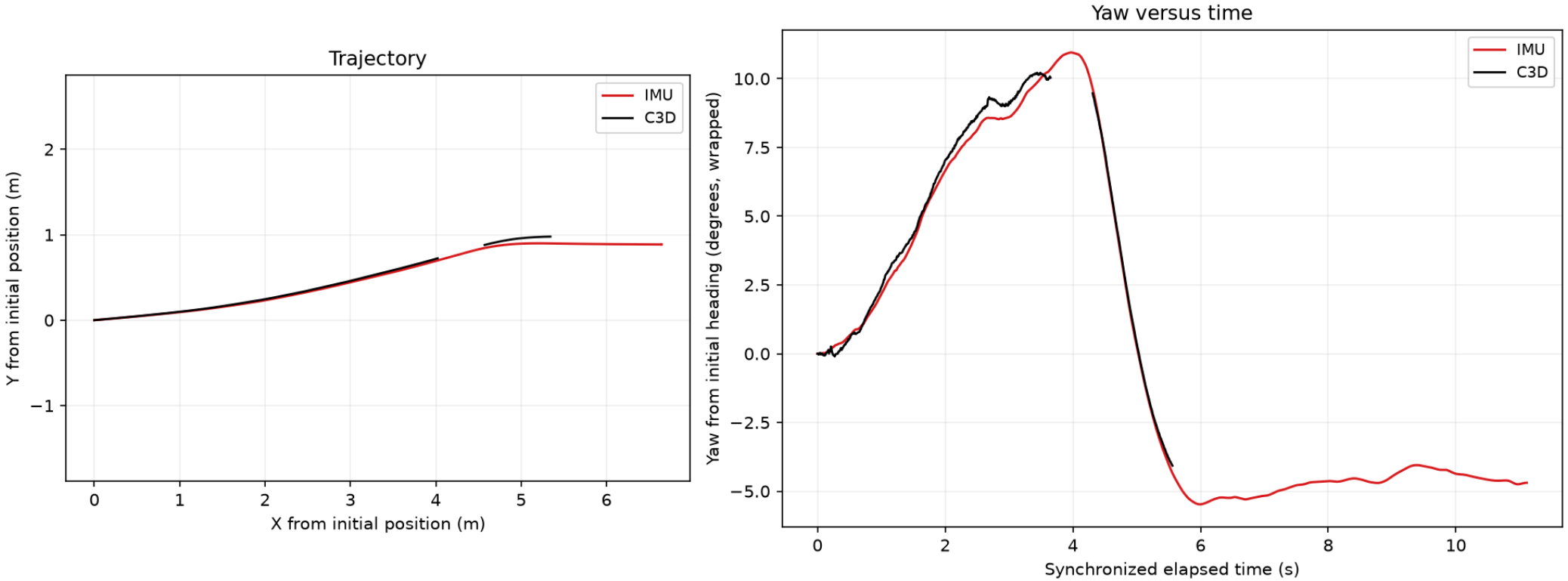




Supported-route metrics (mm): route RMSE 34.1; P95 71.8; sampled Hausdorff 105.8; max-run discrete Frechet 107.8; last-supported endpoint 107.8; absolute length error 76.1.

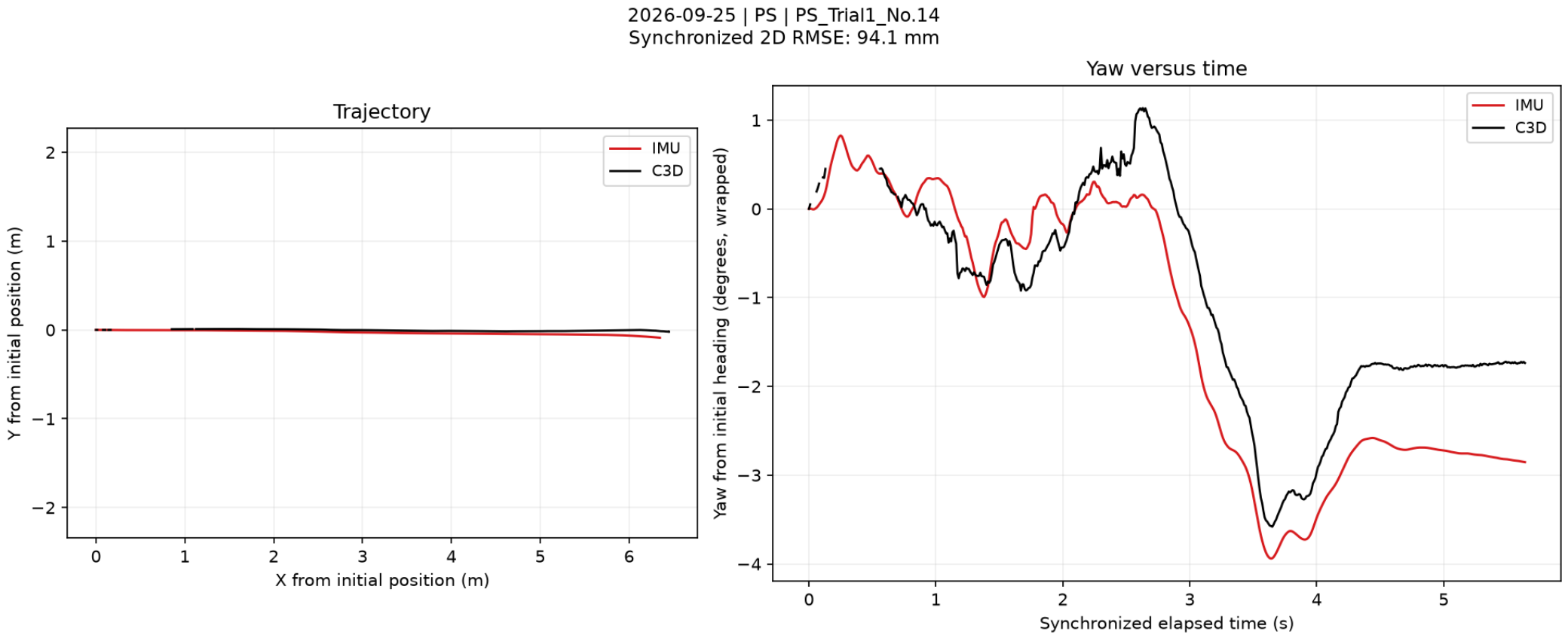
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-25 | OS | OS_Trial2_No.14
Synchronized 2D RMSE: 103.7 mm



Supported-route metrics (mm): route RMSE 32.1; P95 69.6; sampled Hausdorff 171.8; max-run discrete Frechet 171.8; last-supported endpoint 171.8; absolute length error 144.7.

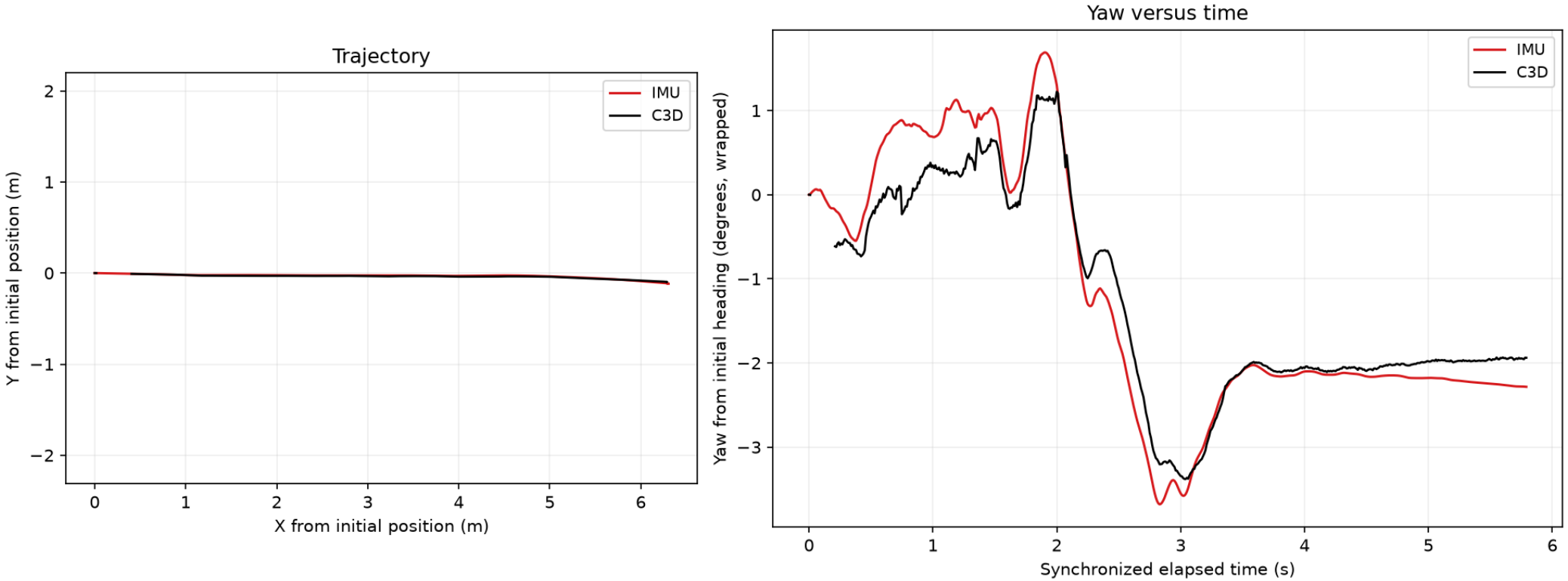
Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.



Supported-route metrics (mm): route RMSE 34.3; P95 69.9; sampled Hausdorff 120.8; max-run discrete Frechet 120.8; last-supported endpoint 120.8; absolute length error 102.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

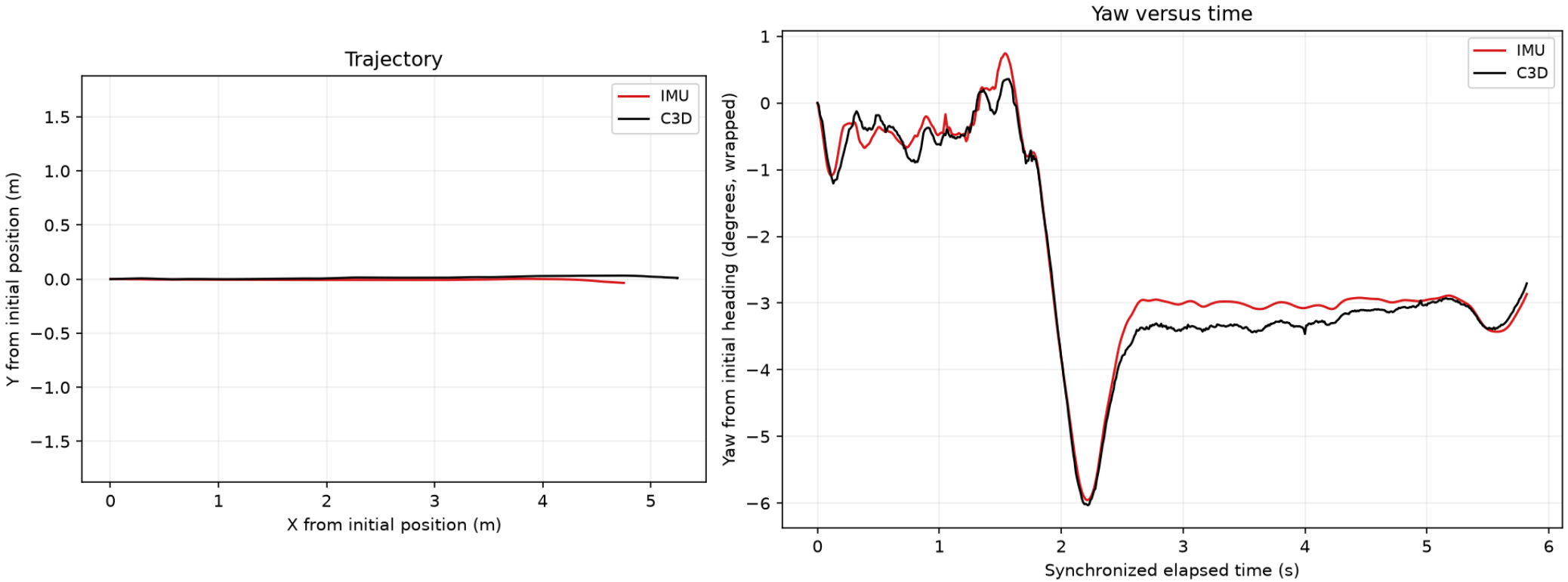
2026-09-25 | PS | PS_Trial2_No.14
Synchronized 2D RMSE: 20.7 mm



Supported-route metrics (mm): route RMSE 7.2; P95 11.5; sampled Hausdorff 25.1; max-run discrete Frechet 26.1; last-supported endpoint 26.1; absolute length error 26.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

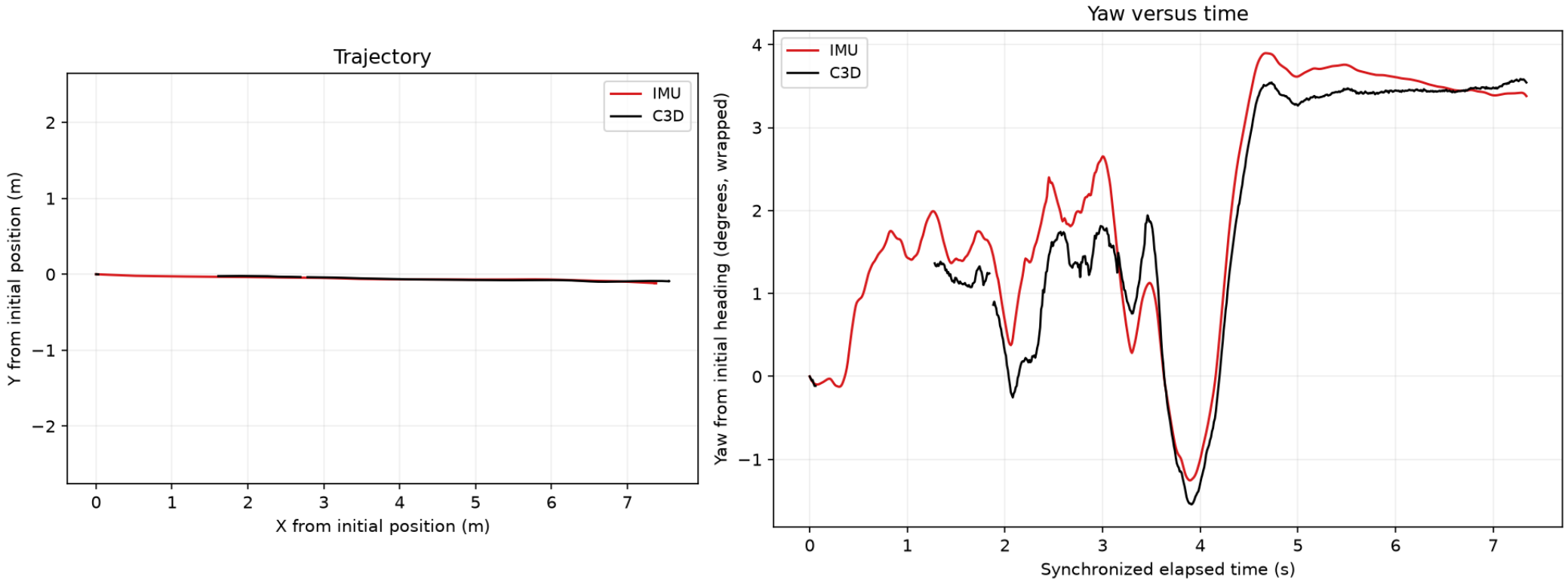
2026-09-25 | PS | PS_Trial2_No.18
Synchronized 2D RMSE: 404.2 mm



Supported-route metrics (mm): route RMSE 70.2; P95 67.4; sampled Hausdorff 499.7; max-run discrete Frechet 499.7; last-supported endpoint 499.7; absolute length error 506.3.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

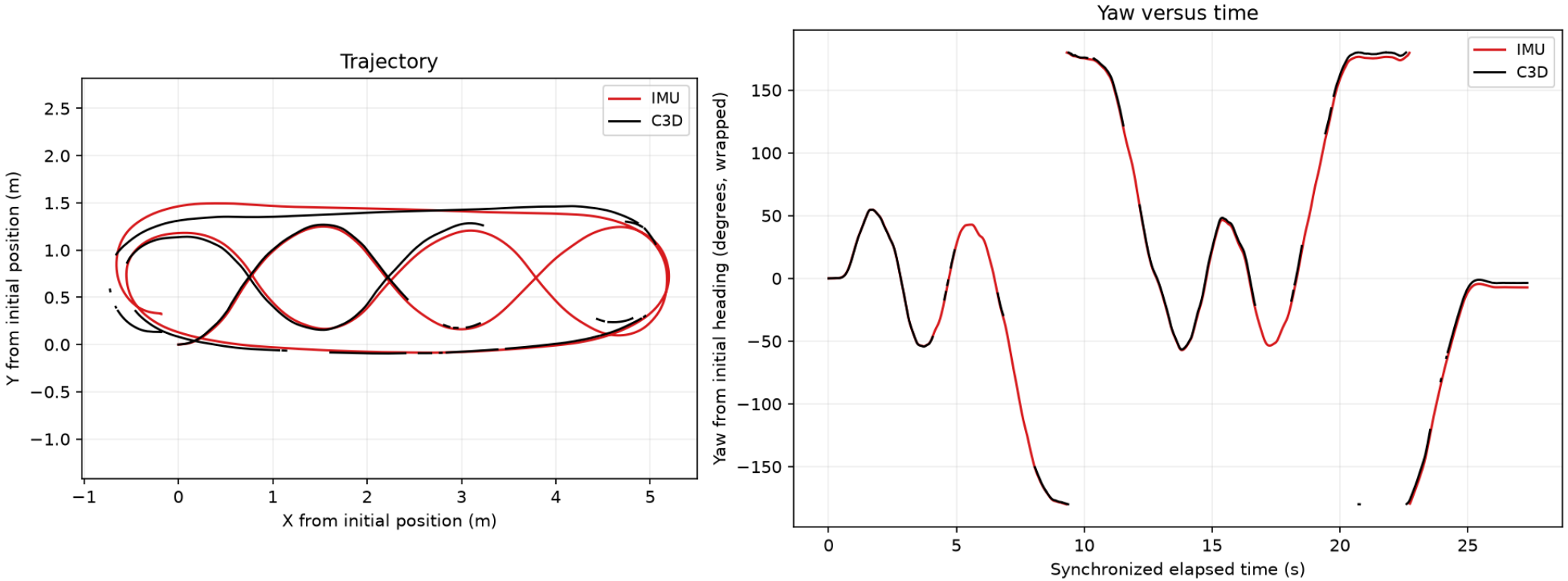
2026-09-25 | PS | PS_Trial3_No.14
Synchronized 2D RMSE: 141.9 mm



Supported-route metrics (mm): route RMSE 17.3; P95 20.1; sampled Hausdorff 176.4; max-run discrete Frechet 176.4; last-supported endpoint 176.4; absolute length error 191.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

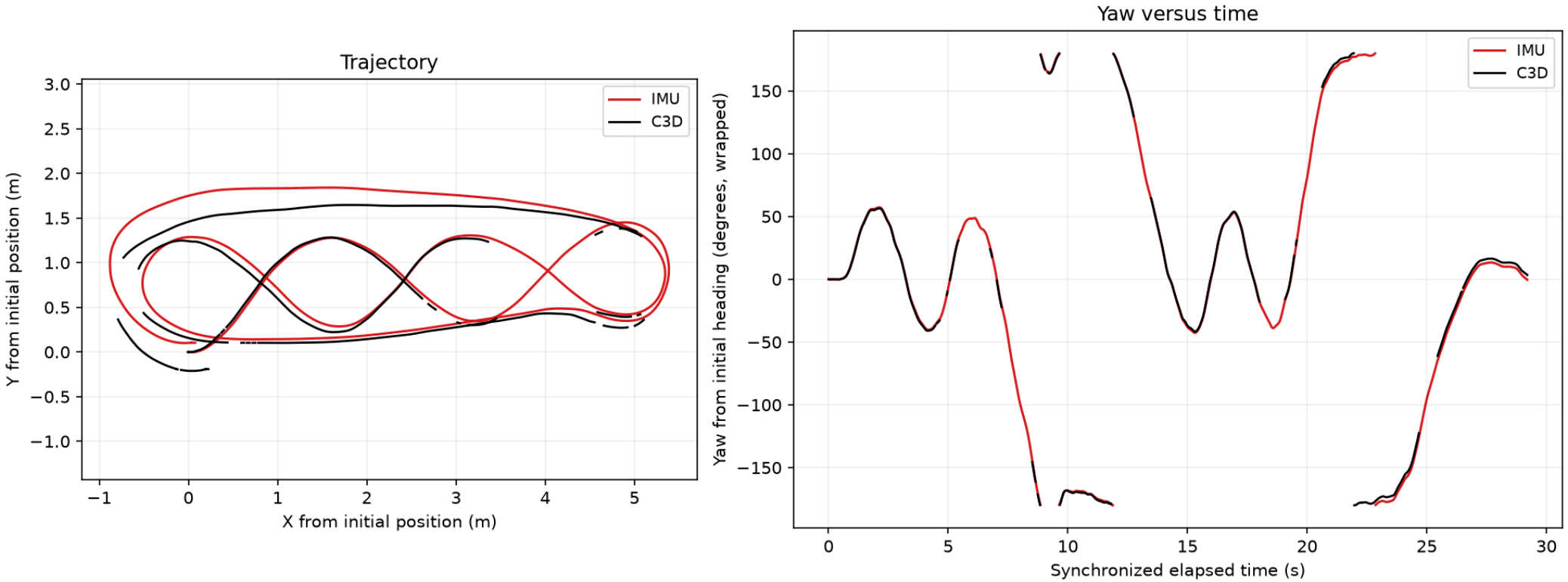
2026-09-25 | SL | SL_Trial1_No.14
Synchronized 2D RMSE: 111.4 mm



Supported-route metrics (mm): route RMSE 60.8; P95 140.4; sampled Hausdorff 188.3; max-run discrete Frechet 214.3; last-supported endpoint 189.4; absolute length error 150.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

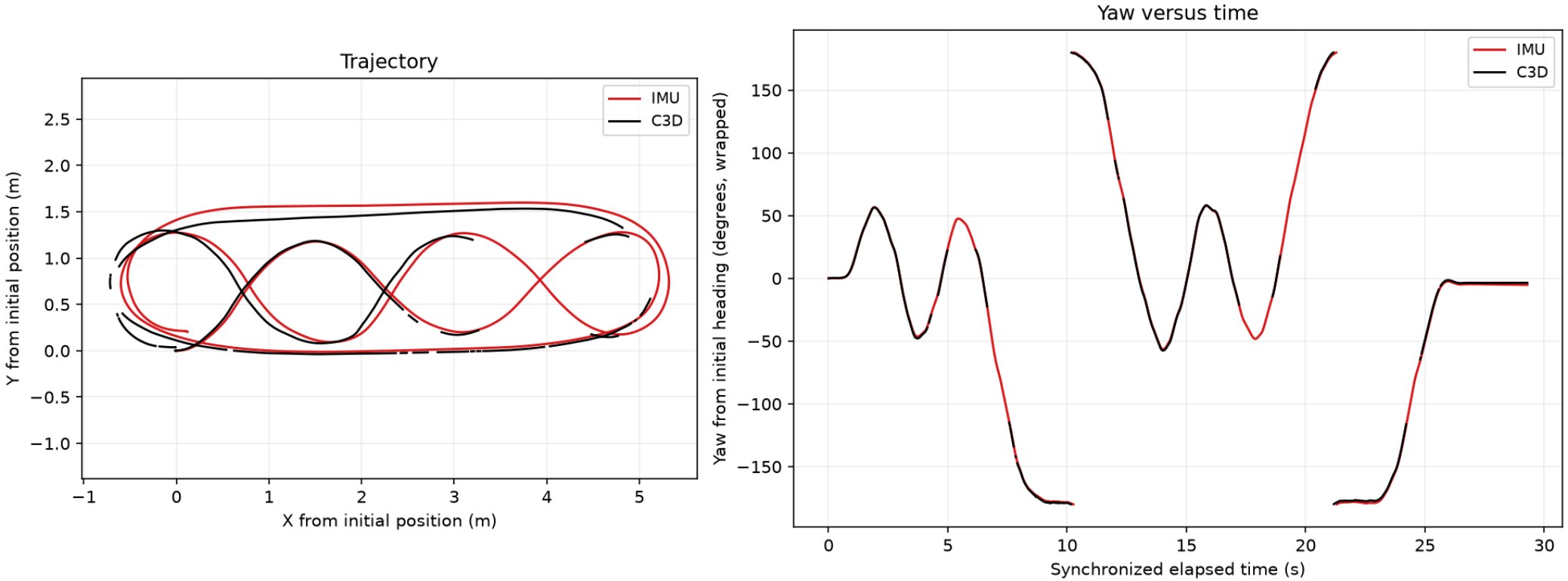
2026-09-25 | SL | SL_Trial1_No.18
Synchronized 2D RMSE: 174.3 mm



Supported-route metrics (mm): route RMSE 110.1; P95 259.5; sampled Hausdorff 329.8; max-run discrete Frechet 334.6; last-supported endpoint 330.6; absolute length error 160.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

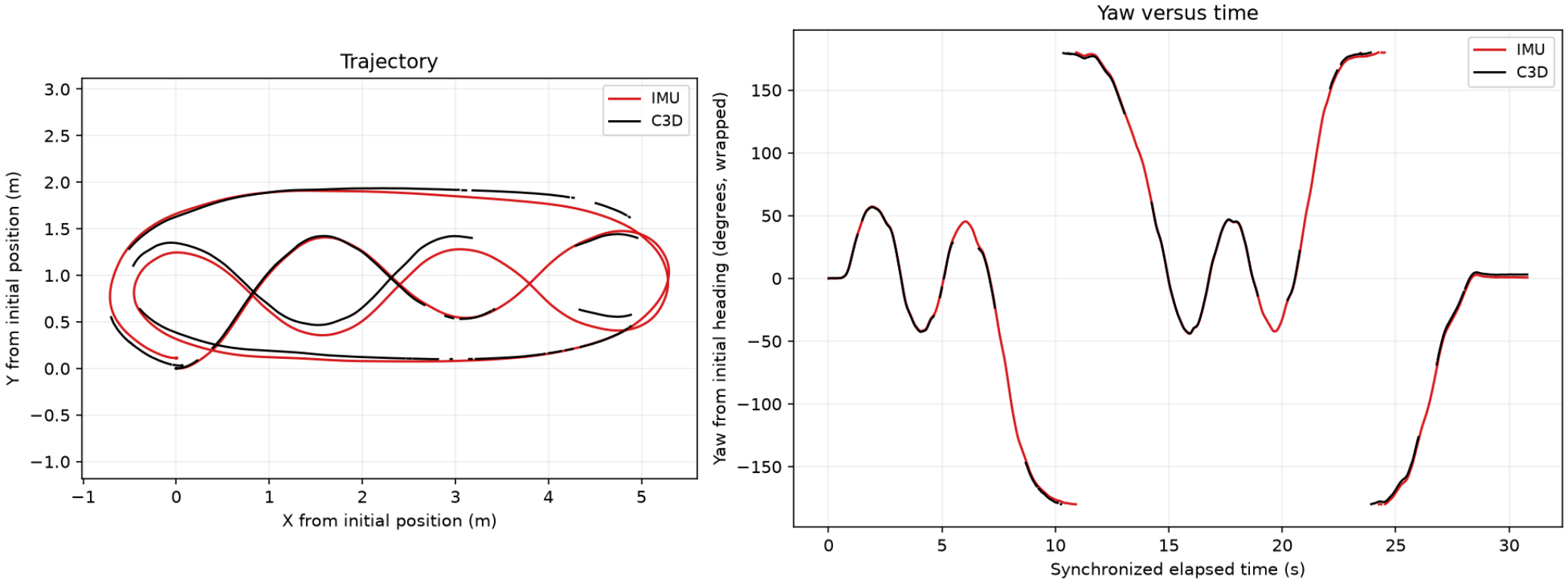
2026-09-25 | SL | SL_Trial2_No.14
Synchronized 2D RMSE: 145.5 mm



Supported-route metrics (mm): route RMSE 67.9; P95 136.9; sampled Hausdorff 196.8; max-run discrete Frechet 238.2; last-supported endpoint 210.0; absolute length error 83.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

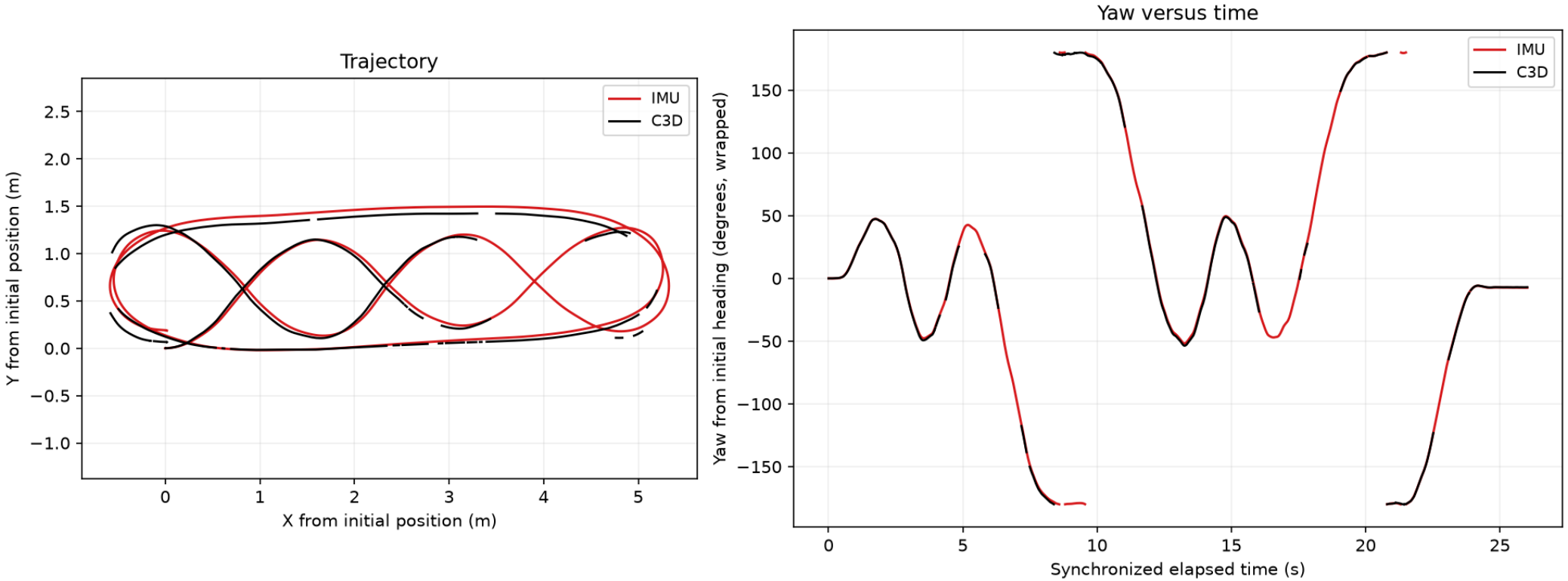
2026-09-25 | SL | SL_Trial2_No.18
Synchronized 2D RMSE: 88.7 mm



Supported-route metrics (mm): route RMSE 64.1; P95 127.5; sampled Hausdorff 157.0; max-run discrete Frechet 157.0; last-supported endpoint 94.8; absolute length error 22.1.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

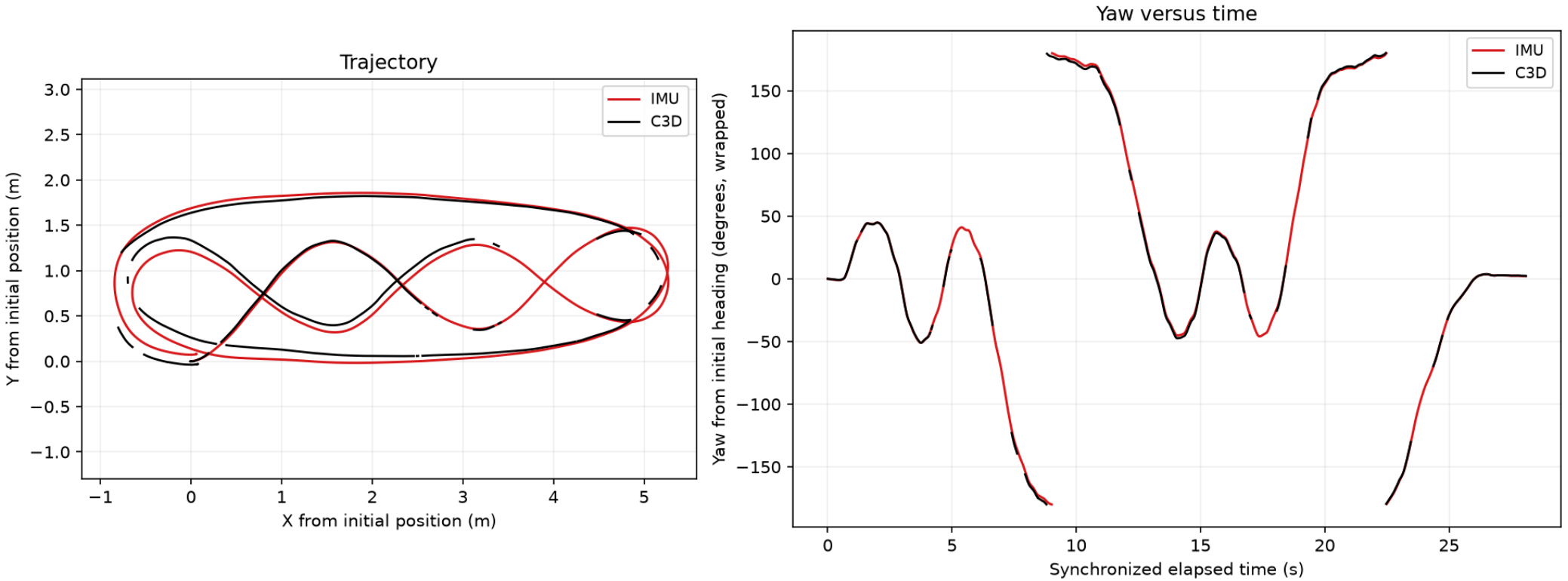
2026-09-25 | SL | SL_Trial3_No.14
Synchronized 2D RMSE: 80.7 mm



Supported-route metrics (mm): route RMSE 47.0; P95 79.5; sampled Hausdorff 127.7; max-run discrete Frechet 127.8; last-supported endpoint 121.0; absolute length error 141.5.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

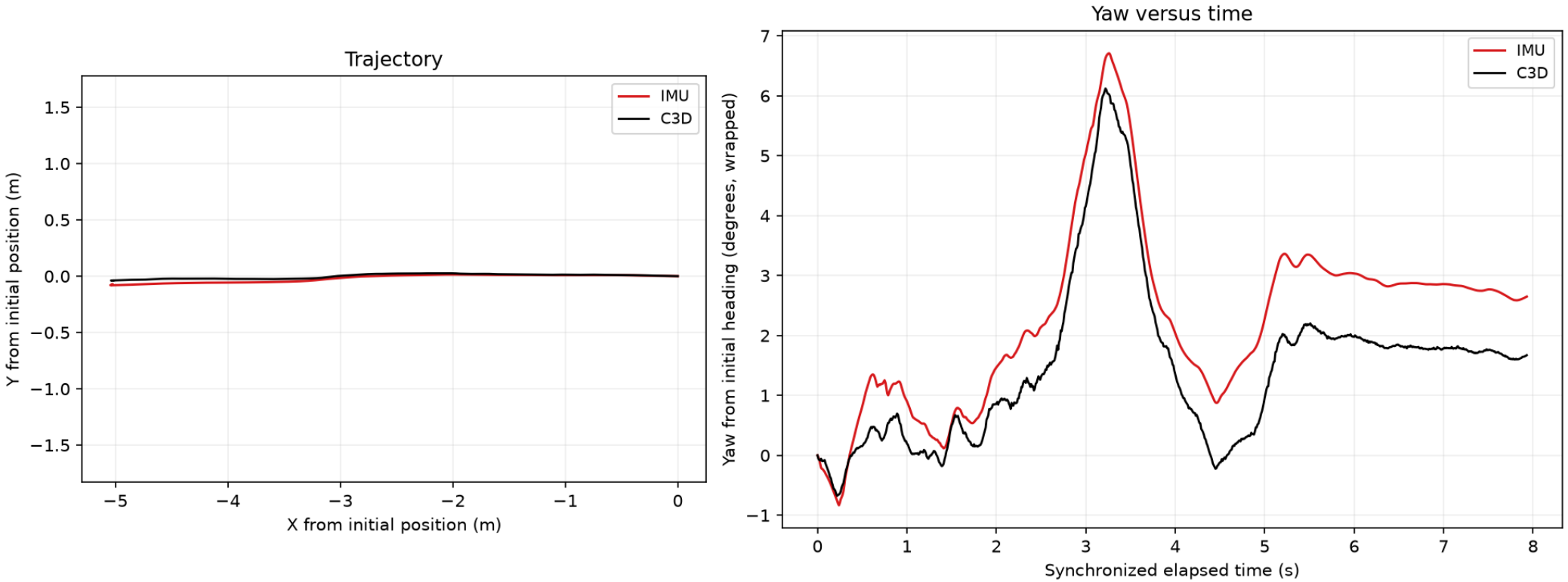
2026-09-25 | SL | SL_Trial3_No.18
Synchronized 2D RMSE: 90.9 mm



Supported-route metrics (mm): route RMSE 62.4; P95 118.8; sampled Hausdorff 156.5; max-run discrete Frechet 164.1; last-supported endpoint 112.0; absolute length error 92.6.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

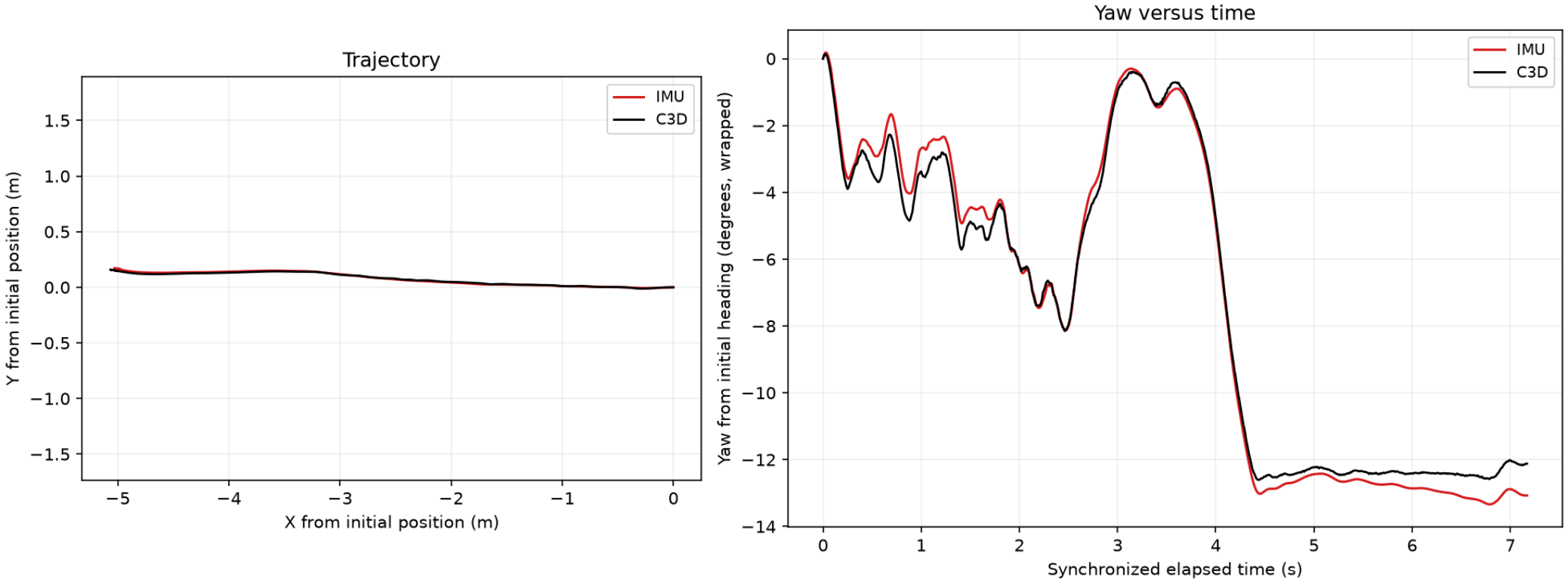
2026-09-25 | ST | ST_Trial1_No.14
Synchronized 2D RMSE: 51.9 mm



Supported-route metrics (mm): route RMSE 22.5; P95 41.8; sampled Hausdorff 43.4; max-run discrete Frechet 44.7; last-supported endpoint 34.5; absolute length error 9.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

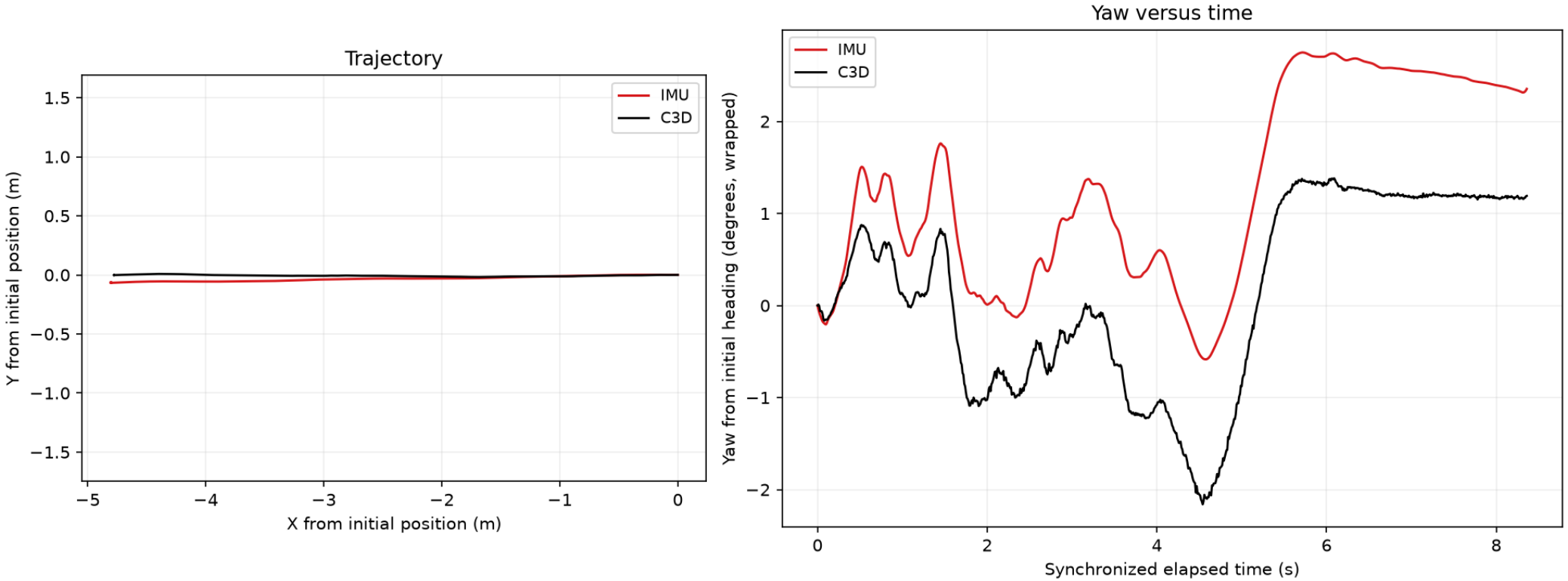
2026-09-25 | ST | ST_Trial1_No.18
Synchronized 2D RMSE: 48.9 mm



Supported-route metrics (mm): route RMSE 7.5; P95 14.2; sampled Hausdorff 35.0; max-run discrete Frechet 35.5; last-supported endpoint 34.5; absolute length error 47.3.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

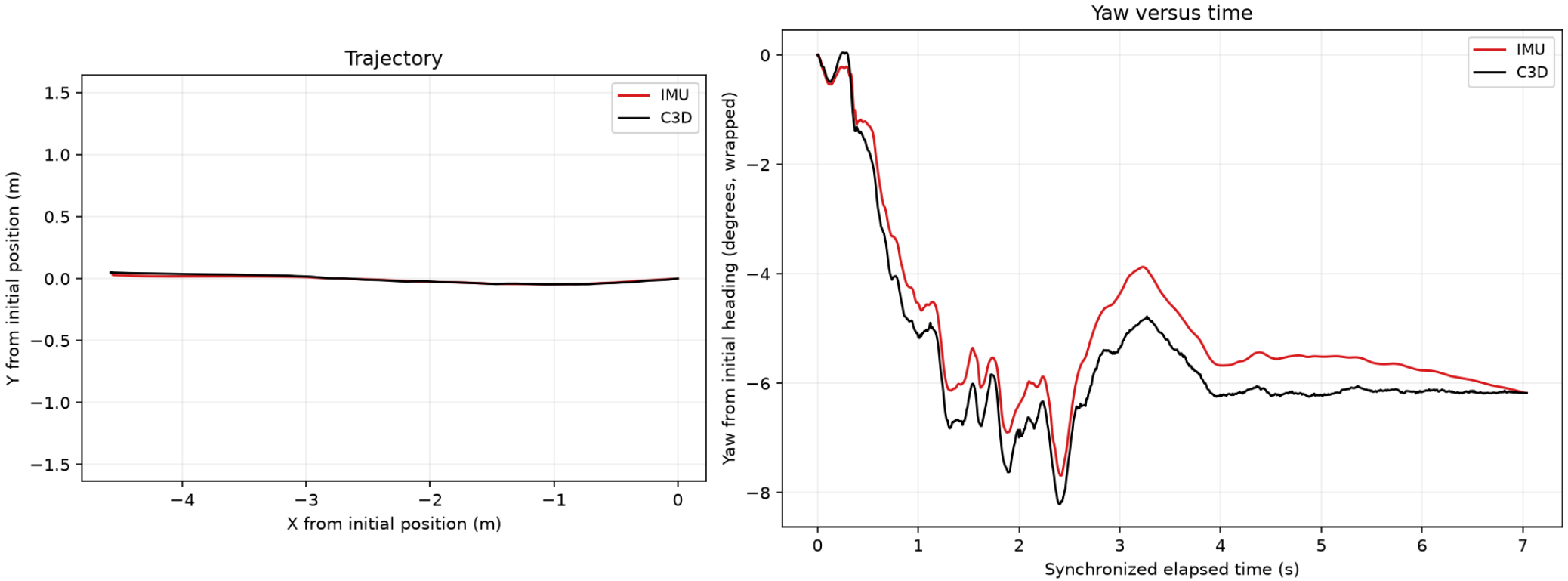
2026-09-25 | ST | ST_Trial2_No.14
Synchronized 2D RMSE: 78.1 mm



Supported-route metrics (mm): route RMSE 35.1; P95 63.2; sampled Hausdorff 69.7; max-run discrete Frechet 69.7; last-supported endpoint 65.7; absolute length error 28.2.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

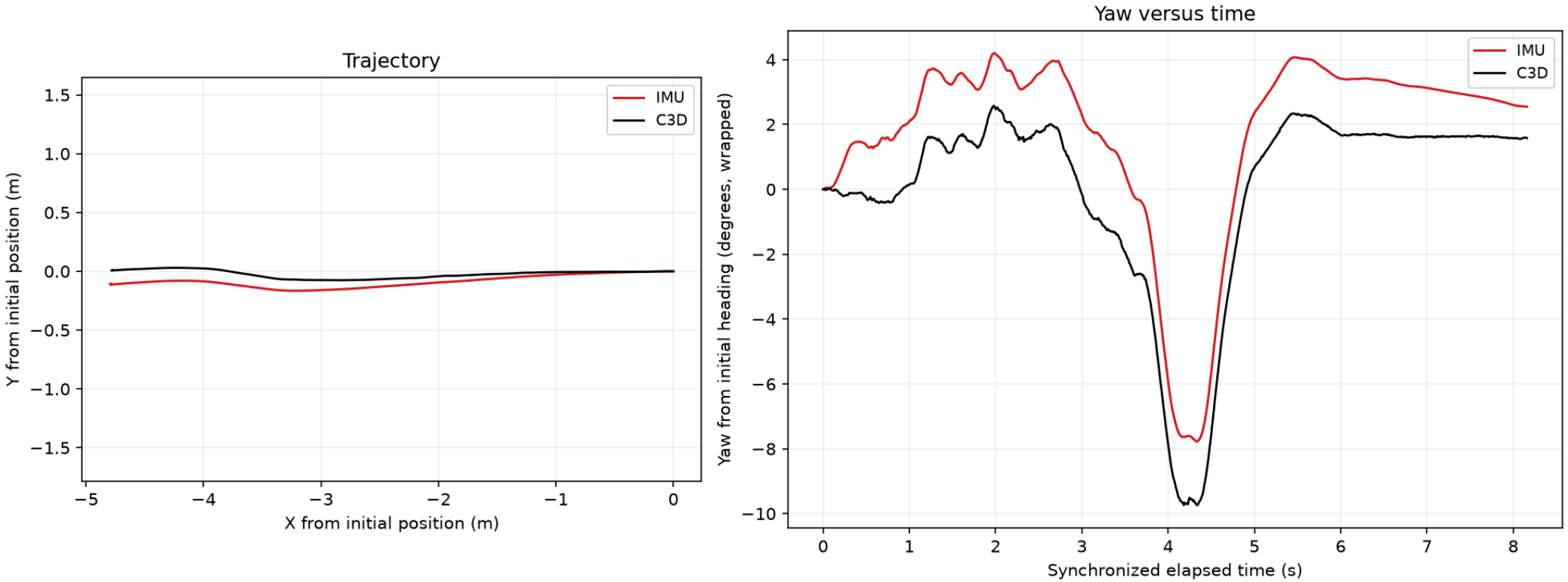
2026-09-25 | ST | ST_Trial2_No.18
Synchronized 2D RMSE: 30.6 mm



Supported-route metrics (mm): route RMSE 9.8; P95 21.5; sampled Hausdorff 22.3; max-run discrete Frechet 22.6; last-supported endpoint 17.0; absolute length error 26.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

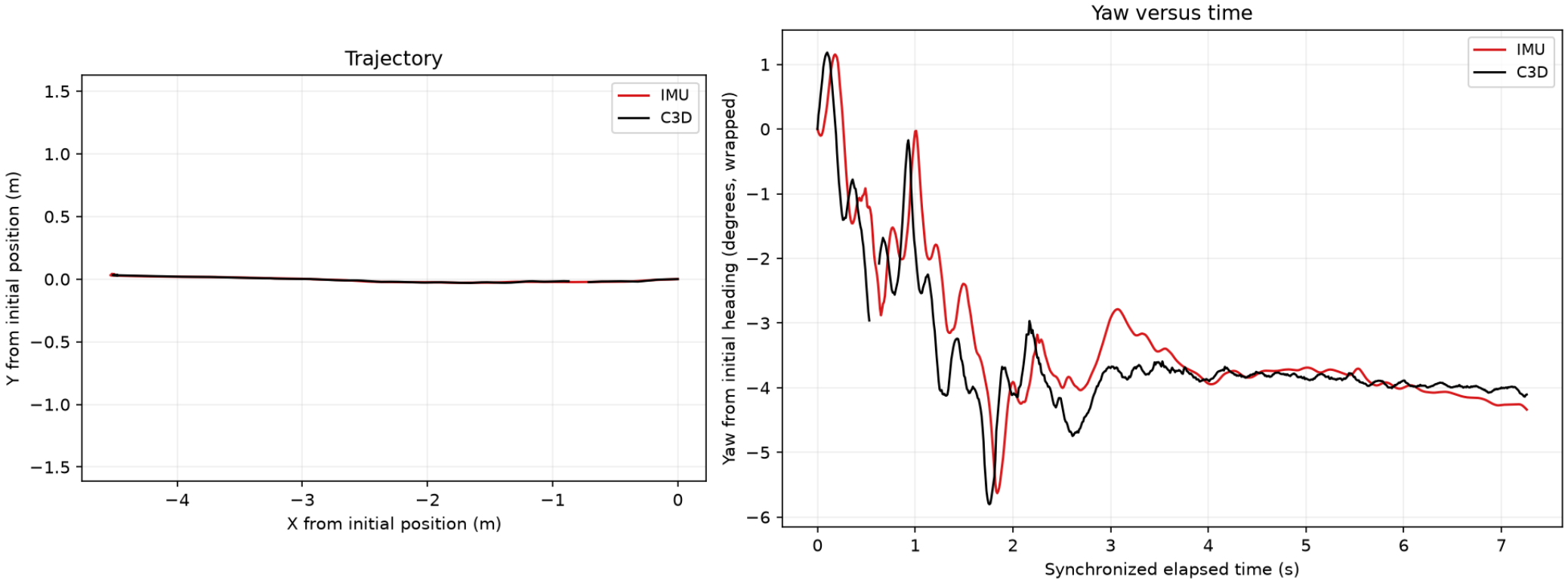
2026-09-25 | ST | ST_Trial3_No.14
Synchronized 2D RMSE: 99.2 mm



Supported-route metrics (mm): route RMSE 74.0; P95 115.1; sampled Hausdorff 119.8; max-run discrete Frechet 119.0; last-supported endpoint 117.7; absolute length error 15.0.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.

2026-09-25 | ST | ST_Trial3_No.18
Synchronized 2D RMSE: 23.4 mm



Supported-route metrics (mm): route RMSE 2.7; P95 5.2; sampled Hausdorff 17.2; max-run discrete Frechet 40.0; last-supported endpoint 40.0; absolute length error 10.9.

Common native observed support and initial pose; disconnected gaps are not bridged. Refit FAILED training-participant guard: diagnostic comparison only. Retained best100.702mm. No fitted spatial alignment or endpoint correction.